Bypass-LoRA: 低维循环注意力与 FFN 条件融合 v2

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Abstract

本文记录 Bypass-LoRA v2 截至 2026-11-02 的结果。研究对象是冻结的 Qwen3-0.6B-Base 旁的低维侧分支。V0 用预先锁定容差的检查验证了侧分支的实现语义（E-0001 修正后 85/85 项、E-0004 130/130 项通过）。V1 在 PIQA+BoolQ 上用不含记忆的 tokenwise proxy 比较八组接口（每组 2M tokens，seed 0）：最好的 concat 接口 C_dual（78.15%）只比加法 F_out（77.80%）高 0.35 pp，未达 1 pp 门槛，两个参数相近的 LoRA（79.20%、79.05%）更高，H-0002 记为 challenged（screening）。V2 研究只保留低维记忆时的单跳读取：十组 M 线配置都没有达到运行前锁定的门槛（正常记忆对清零与打乱记忆的 EM 优势最大 +0.6 pp，门槛 10 pp），训练后答案位置权重最高的槽位总是第一个写入的槽位，H-0003 记为 challenged（screening）。用户批准计划外的修复后，记忆路线重开。诊断显示上下文中编码的 value 只对第一条事实可解码；把每条事实单独编码后写入，并先在每个 episode 只有 1 条事实的任务上训练（D5 配方），16 条事实的单跳读取在三个 seed 上都达到门槛（EM 优势 41.1、47.8、46.2 pp），H-0012 记为 supported（screening）。同一批 adapter 在 V3 的链终点任务（T2）的一跳问题上没有达到门槛（EM 优势 0.9–3.0 pp）：解码器迁移了，寻址没有迁移。V3 的循环矩阵因此先在 T2 上训练共同起点，正在运行。V5 缩减版在官方 validation 上只开封一次：tokenwise C_up 对 28 层 rank1 LoRA 的三 seed 平均宏平均差为 −0.42 pp（95% 区间 [−1.02, +0.20]），按封存规则 H-0008 记为 challenged（confirmatory），这条路线停止。按 plan2 新增的残差旁路线（RB）完成了实现检查（397 项通过）与计时。除 V5 外，证据都是单 seed 或三个 seed 的筛选级证据。

1 Introduction

Bypass-LoRA v2 在冻结主干旁加一条低维循环侧分支：从注意力的 Q/K/V 取特征，压缩成低维状态，经记忆读写与循环更新，再通过 FFN 融合写回主干。计划把研究拆成几个阶段：V0 验证实现语义（H-0001）；V1 在没有记忆与循环的 tokenwise proxy 下比较融合接口，检验把信号注入 FFN 的 gate/up 是否优于加法写回（H-0002）；从 V2 起研究丢弃 support 主干 KV、只保留低维记忆时的单跳读取（H-0003），后续还有读算子选择性（H-0004）、多次读取（H-0005）与实用适配（H-0006）。V2 没有产生可用读取后，计划先缩减为 V5 的简单接口对比；用户随后批准了针对失败原因的修复（H-0009–H-0012），记忆路线与 V3/V4 因此重开。plan2 另加一条残差旁路线（RB）：侧分支在每个 token 处读取自身的因果历史，检验它的实用价值（H-0013）、对侧支历史的依赖（H-0014）、重复读取（H-0015）与 gate/up 融合（H-0016）。本文报告 V0–V2、记忆路线修复、V3 前置诊断、V5 缩减版的确认结果与 RB 线的实现检查。各假设的陈述、范围与当前状态如下。

- H-0001@1 (supported, active) V0 实现语义与恒等式正确：新侧分支实现（QKV tap、因果 KernelMatrix、低维累加循环、FFN 融合、M/F 协议）语义正确：出口全零时与未修改 Base 输出一致；两步梯度启动后 Q/K/V 压缩与 Rq/Ro 获得非零梯度且主干参数不更新；KernelMatrix 与 KernelSlot 的输出及 Q/K/V 梯度一致；Cg=PWg、Cu=PWu 约束下 C_dual 与 F_pre 的输出及对同一 P 的梯度一致；改动后缀不改变前缀输出；一次 prefill、分块 prefill 与逐 token decode 一致；padding 不写不读、reset/batch 重排不串状态；同 token 只写一次

且 S 不改变写次数；M 线 support 主干 KV 不进入 query forward。(scope: Qwen/Qwen3-0.6B-Base（锁定 revision）；FP32 小张量参考用严格阈值；GPU BF16 容差由同一 Base 的重复测量预先锁定，看到失败后不得放宽；检查围绕 plan §6.3 列出的风险，不要求重跑无关实验)

- H-0002@1 (challenged, paused) FFN gate/up 条件融合优于加法写回（V1 tokenwise proxy）: F 线（保留原 attention 上下文）、4 层 [6,13,20,27]、r32、无侧支记忆与循环的 tokenwise proxy Z=SiLU(UA) 下，把同一 Z 注入冻结 FFN 的 gate/up（C_gate/C_up/C_dual）比加法写回（F_out/F_pre）在非天花板 dev 切片上取得更好的任务准确率与更低 NLL；且该优势不能仅由线性退化组（Z=UA + C_dual，即共享 A 的 gate/up LoRA）或参数相近的 LoRA（4 层 all-7 rank10；28 层 all-7 rank2）解释。(scope: F 线；真实任务 train/dev（PIQA+BoolQ 官方 train 切分）或预先锁定的非天花板切片；每组 ≤2M 非 padding 输入 tokens、seed 0 单 seed 筛选（screening）；接口间不等参，只筛位置与信号，不据此宣称参数效率；无记忆 proxy 的负结果不否定有记忆 concat（V2 保留 C_dual 交互探针）)

- H-0003@1 (challenged, active) M 线单跳：learned memory 读取产生可用信号（V2）: M 线（问答阶段丢弃 support 主干 KV，仅保留声明的低维 memory）、单层 0-based index14、r128、S=1、T1 单跳（E=16 事实、4 关系、visible_fact_delimiter 写入、all_valid_question 读取）、仅答案 CE 从 Base 训练时，2×2×2 析因（特征 direct_hidden/qknorm_pre_rope × 读算子 KernelMatrix/SoftmaxSlot × 融合 F*/C*）中至少一组在新 dev 上正常 memory 的 EM 同时比 zero 与 shuffled memory 高至少 10 pp 且 NLL 更低，并有可用正对照与非退化读出。(scope: study_id concat_loop_v2 的 V2 数据（独立 phase/seed 命名空间，不复用旧验证集）；每组 ≤4M 输入 tokens、单 seed 筛选；门槛须在 V2 release 前锁定；是否继续有限循环探索的研究门槛，不是统计确认，也不替代旧 writer 的 80% gate)

- H-0004@1 (inconclusive, paused) 读算子选择性：SoftmaxSlot 读取比 KernelMatrix 更集中且更可用: 在相同正核 q/k 特征、相同写读位置与训练条件下，SoftmaxSlot（s0=sqrt(dk)、tau=1、g=1）比 KernelMatrix 产生更高的目标 key 权重、更少的有效读取项数、更低的归一化读误差与更高的单跳 EM；若 SoftmaxSlot 有效而 KernelMatrix 无效，瓶颈在读权重的选择性与混合，而非 memory 中的信息缺失。(scope: V2 析因内的读算子主效应与交互；单 seed 筛选；差异只称读算子差异，不归因于 "保存 KV 消除了矩阵压缩"；KernelSlot 只作恒等检查，不作训练 arm)

- H-0005@2 (untested, active) 固定记忆下演化 query 的多次读取有独立收益（V3→V4）: 固定 memory（同一 token 的各步共享 M、c）下，累加循环 S=2/4 的 hops2–3 EM 优于独立训练的 S1，且该收益不被固定 query 精确幅度对照（Z=S·eta·a·Ro）、固定总步长、同源 tokenwise 核、MAC/延迟匹配的更宽 S1、同核锚定更新与完整状态读出解释。(scope: M 线，单独编码写入协议：每条事实不带前文单独编码，在其分隔符处写入 key/value，问题侧不变；这是须声明的结构化支持输入条件，统计量与 E-0010 分开报告。单读配置沿用 H-0012 的 D5 配方：冻结 Qwen3-0.6B-Base，第 14 层，direct_hidden，SoftmaxSlot，F_out，r128，先在 E=1 上训练。任务：T2 链终点（E-0011 生成器），训练与 dev 只含 hops1–3，hops4/6/8 封存作外推。V3 各组样本预算相同，单 seed 筛选。确认（V4）需预注册：hops2–3 平均 EM 差 $\geq$ +3 pp，配对区间下界 > 0，3 个新 seed 方向都为正，hop1 差不低于 -1 pp。)

- H-0006@1 (untested, paused) 保留原 attention 的实用适配价值（V5）: F 线（4 层 r32，保留原 attention 上下文）的 PIQA+BoolQ 适配中，C* 多读相对参数匹配的强 LoRA（全 28 层 all-7）宏平均准确率提升至少 +1 pp 且每任务下降不超过 1 pp；并且侧支历史相对同参 tokenwise T0 与 zero/shuffle/permute 干预有可测贡献；同时报告延迟与状态成本。(scope: 官方 validation 封存为确认集，配置与分析锁定后只开一次；3 seeds × ≤12M、最多 3 epochs；

95% 配对区间 + Holm 校正；WikiText-103 保留 split 的 NLL 作能力保持；参数、状态与计算不同时等量，不宣称同时公平)

- H-0007@1 (untested, paused) 冻结 S-RE：已学 K/V 换用槽位读算子后更可用（旧资产诊断）: 冻结旧 S-RE（DPFP 实际 key 768 / value 128、原读位置、memory-only）的 writer 与低维 reader，仅把原生 ridge read 在所有正常读取位置同时替换为 SoftSlot 或 HardSlot（温度/增益只在独立 calibration 上按 NLL 选择），旧 1,000 个验证 episode 的 EM 高于原生读取（旧参考 723/1,000，NLL 1.642）。(scope: 需要旧 S-RE 推理资产（writer.pt 与 looplora.llm.sq2.load_inference）；只回答既有 learned K/V 在另一读算子下能否更有效使用，不证明训练后极限；复现旧验证只证明兼容性；冻结诊断上限 1 node·h)

- H-0008@2 (challenged, retired) 保留原 attention 的 tokenwise 简单接口的实用价值（V5 缩减）: F 线（保留原 attention 上下文）、4 层 [6,13,20,27] 的 tokenwise 接口 C*（C_up，r32，Z=SiLU(UA)，524,288 参数）在 PIQA+BoolQ 官方 validation 上，相对参数匹配的强 LoRA（全 28 层 all-7 rank1，630,784 参数）宏平均准确率高至少 1 pp，且每个任务的准确率差不低于 -1 pp。(scope: V1 的切分、模板与评估；3 个新 seeds；每 run 正好 3 epochs（约 4.46M tokens，不超过 12M）；训练目标同 V1（逐样本平均）；学习率按预先登记的端点扩展校准；官方 validation 封存为确认集，配置与分析锁定后只开一次；95% 配对 bootstrap 区间只反映给定这三个训练 seed 时的数据不确定性，不代表训练过程的分布；+1 pp 与每任务 -1 pp 是点估计门槛，supported 不表示以 95% 置信保证收益至少 +1 pp，也不是逐任务非劣检验；次要比较做 Holm 校正；WikiText-103 保留 split 的 NLL 作能力保持；参数接近但不相等，不宣称参数效率或同时等计算；LoRA 结论只针对 rank1 与本合同的短校准 LR 规程；不含侧支历史与多次读取（记忆路线已停止，D-0008）)

- H-0009@1 (untested, active) 记忆路线修复：改变训练配方后单跳读取可用: 在 V2 的 M 线架构（冻结 Qwen3-0.6B-Base，第 14 层，r128，S=1，分隔符写入，问题位置读取）下，V2 的失败（EV-0070、EV-0072）主要来自训练动态而非容量：至少一种改动后的训练配方——更少事实或事实数课程、预先吸收答案先验的独立通路、地址权重辅助监督、或去中心化余弦寻址加可学温度——能让 T1 E=16 的单跳读取在 dev 上达到 E-0010@2 锁定的 H3 门槛（EM_normal − max(EM_zero, EM_shuffled) ≥ 10 pp，NLL_normal < min(NLL_zero, NLL_shuffled)，目标事实平均地址权重 ≥ 0.125）。(scope: M 线 T1（hop 1）；E=16 沿用 V2 数据，E ∈ {1,4,8} 为新登记的数据版本；单 seed 筛选，每 arm 与 V2 相同的步数与 batch；额外监督（目标事实索引）单独标注，结论只在其条件下陈述；不读 confirmation；不改 V2 的结论与 E-0010 判定)

- H-0010@1 (challenged, paused) E=16 上下文中 value 解码器先行：干净读取下可学成，之后寻址可从答案 CE 发现: 在 E=16 上下文里，若训练时把记忆限制为只含目标槽位（teacher-forced 单槽读取，用到目标索引，属额外监督），value→ 答案的解码器能在 6000 步内达到 teacher-forced 读取下 EM_normal − max(EM_zero, EM_shuffled) ≥ 50 pp；从该 checkpoint 恢复全部槽位、只用答案 CE 再训练 6000 步后，单跳读取在 E=16 dev 上达到 E-0010@2 的 H3 门槛。(scope: M 线 T1 hop 1，E=16 沿用 V2 数据（ATT-0039-1）；冻结 Qwen3-0.6B-Base，第 14 层 r128 S=1，direct_hidden、SoftmaxSlot、F_out，LR 3e-3；单 seed 筛选；额外监督与协议改动单独标注；不读 confirmation；不改 V2 与 E-0014 的结论)

- H-0011@1 (challenged, paused) 槽位隐变量的边际似然训练避免塌缩并让寻址可从答案发现: 把训练目标从向量空间混合读取改为槽位隐变量的边际似然 L = −log Σ_i w_i·p(answer | 只读 v_i)，读取只在最后问题位置、寻址权重来自正常 softmax、不用目标索引，6000 步后最后位置不再塌缩到固定槽位，并在 E=16 dev 上按硬读取（argmax 槽位）协议达到 H3 门槛（EM 差 ≥ 10 pp、NLL 条件、最后位置目标权重 ≥ 0.125）。(scope: M 线 T1 hop 1，E=16 沿用 V2 数据（ATT-0039-1）；冻结 Qwen3-0.6B-Base，第 14 层 r128 S=1，direct_hidden、

SoftmaxSlot、F_out，LR 3e-3；单 seed 筛选；额外监督与协议改动单独标注；不读 confirmation；不改 V2 与 E-0014 的结论；硬读取与只读最后位置是新登记的读取协议，与 E-0010 的全问题位置统计量分开报告)

- H-0012@1 (supported, active) 每条事实单独编码后写入，消除 value 的上下文迁移: 支持中的每条事实单独编码（格式同 E=1）后在其分隔符处写入，读取协议不变，只用答案 CE：从零训练或从 R1 checkpoint 续训 6000 步后，E=16 单跳读取在 dev 上达到 H3 门槛，且最后位置不塌缩到固定槽位。(scope: M 线 T1 hop 1，E=16 沿用 V2 数据（ATT-0039-1）；冻结 Qwen3-0.6B-Base，第 14 层 r128 S=1，direct_hidden、SoftmaxSlot、F_out，LR 3e-3；单 seed 筛选；额外监督与协议改动单独标注；不读 confirmation；不改 V2 与 E-0014 的结论；单独编码是须声明的结构化支持输入条件（写入时不看其他事实），以 E-0015 规则 5 成立为前提)

- H-0013@1 (untested, active) 残差旁路低维因果 linear attention 的实用价值（R_res+KernelMatrix S1，plan2 主线）：保留原 attention 的 F 线上，在冻结 Qwen3-0.6B-Base 的四层 [6,13,20,27] 各加一个 r32 块输出残差旁路 Y=B(X)+Z·P_R（R_res），Z 由真实因果 KernelMatrix 低维 linear attention 单次读取（S1）产生；它在 PIQA+BoolQ 上的宏平均准确率比参数接近的标准 LoRA 至少高 1 pp，且每个任务的差不低于 −1 pp。(scope: RB 线。筛选（RB1）：dev 每任务 1,000 条、单 seed，对照 Base、同参数 T0 与同四层 all-7 LoRA r7（630,784 参数，比 R_res 的 663,552 少 4.94%）。确认（RB5，尚未建契约）：新登记并封存的确认集、3 个新 seed，对照全 28 层 all-7 LoRA r1（630,784）。固定 qknorm_pre_rope、inclusive 先写一次再读、η=0.25、g=1；参数不完全相等，不宣称参数效率；同时报告延迟与状态字节；+1 pp 与每任务 −1 pp 是点估计门槛。)

- H-0014@1 (untested, active) 残差旁路的收益依赖侧支自身的因果历史（同参数 T0 与冻结历史干预）：在 R_res+KernelMatrix 上，同参数、同深度、没有侧支历史的同源逐 token 核 T0 的宏平均至少低 1 pp；并且保留原 attention 与其 KV、只改侧支历史的冻结干预（episode 间交换历史状态、历史内打乱 key/value 配对）各使宏平均至少下降 1 pp。(scope: RB 线、F 线。筛选：RB1-02 对 RB1-01（S1）及 RB1 的冻结干预（dev）；确认：RB5 的同深度 T0（RB5-08）与确认集上的干预。原 Q/K/V 特征本身含上下文，清空侧支历史不等于模型失去历史；干预可能带来分布偏移，单次 zero/shuffle 下降不作机制证明；T0 也改变了读函数，因此同时报告窗口、strict 与当前 token 读权重。不成立时按 plan2 §11.4 只称动态/非线性适配。)

- H-0015@1 (untested, active) 同一残差旁路内的低维重复读取有超出幅度、步长与逐 token 深度的增量（S_F*∈{2,4}）：R_res+KernelMatrix 的累加式循环 S_F*（2 或 4，由 RB1 dev 锁定）比 S1 的宏平均至少高 1 pp；该增量不能由重复同一读取的输出放大（S4 固定第 1 步 query）、累积总步长（S4 且 η=0.25/4）或同深度 T0 解释。(scope: RB 线、F 线；S_F* 只由 F 线 train/dev 决定，不移植 M 线的 S_M*。筛选：RB1-03/04 对 RB1-02，RB1-05/06 为幅度与步长对照；确认：RB5-02 对 RB5-01 与 RB5-08。RB1 的 T0 只有 S1，不能排除 S4 的逐 token 深度解释；S4 若只等于固定 query 对照，记为暂无迭代查询证据（plan2 §17.2）；不宣称因果逐跳推理；报告读取次数、MAC 与实测延迟。)

- H-0016@1 (untested, active) 同一 KernelMatrix 状态下 FFN gate/up 条件融合（C*_RB）优于块输出残差写回: 在相同的四层 r32 KernelMatrix 状态与相同 S 下，把 Z 送入原 FFN gate/up 的条件融合 C*_RB（C_dual 或 C_up，按 plan2 §7.4 在 RB1 选定）比 R_res 的宏平均至少高 1 pp；确认时该收益在匹配 C*_RB 参数的 LoRA、更宽的 R_res 与 C*_RB 自己的同深度 T0 对照下仍成立。(scope: RB 线、F 线。筛选：RB1-08（C_dual）与 RB1-09（C_up）对 RB1-02；确认：RB5-03/04 对 RB5-01/02，RB5-06/07/09 作参数与历史对照。接口参数不同（C_dual 1,318,912，C_up 925,696，R_res 663,552），不宣称参数效率。V1

proxy 的结论（EV-0048、D-0004）不改写；concat 失败不影响残差主线（plan2 §6A R1.3）。)

- H-0017@1 (untested, active) RB 线新实现正确：因果前缀 KernelMatrix、T0、四层 R_res/concat 与侧支历史干预: 向量化因果前缀 KernelMatrix 与已审核的逐 token 参考扫描在输出和梯度上一致；四层 R_res 与 concat 在零出口时与 Base 一致，R_res 只在本块原 FFN 之后加 $Z \cdot P_R$ 且原块每层只执行一次；四层两步梯度启动成立；T0 与侧支历史干预符合定义且不泄漏未来 token；参数计数与 plan2 §15 一致。(scope: RB0（E0）；FP32 小张量用运行前锁定的严格容差，GPU BF16 沿用 E-0001/E-0004 锁定的容差；只检查本线的新路径，已有 V0 与记忆路径测试作回归；不产生任务性能结论。)

- H-0018@1 (untested, candidate) RB1 无信号时一次同布局容量或特征探针（r64/r128 或 raw/activation taps）: 若 RB1 的残差与 concat 组都没有信号，而 LoRA 与 T0 正对照可学、实现检查通过，则一次同布局探针（同四层、同 KernelMatrix 与 S1，只把 r 改为 64 或 128；或在 r32 下比较 qknorm_pre_rope、raw 与 SiLU(raw) taps）能让 R_res 达到 Base+1 pp 以上，且不比同布局的近预算 LoRA 低 1 pp 以上。(scope: 只在 RB1 规则 9 触发时释放，一次为限；样本预算取 plan2 §14.3 的 8M 容量/特征储备（例如三臂各 2M）；r 探针配匹配参数的同四层 LoRA；四层 r32 的负结果只界定该配置，不推广到所有低维残差旁路（plan2 §6A R1.3）。)

2 Related work

目前的记录中还没有核查过的文献条目，因此这里不对相关工作作任何断言。与本工作最直接相关的是同一计划此前的 LoopLora 实现（UnbearableFate/LoopLora @ a255e81）。本研究只把它作为只读的数据重叠核对来源：E-0005 的新数据与其 registry 重叠为 0。本研究没有复现它的结果；冻结 S-RE 诊断因缺少推理资产而暂停（D-0001）。V1 中的 LoRA 只作为参数相近的对照。系统的文献核查留待后续阶段。

- none recorded

3 Method

所有实验都使用 Qwen/Qwen3-0.6B-Base（锁定 revision da87bfb6），BF16 冻结主干、FP32 新参数、eager attention，硬件为 GH200 MIG 2g.24gb；所有工作负载都在 PBS 作业中运行。E-0001 与 E-0004 是不训练的实现检查，容差在检查前由同一 Base 的重复测量锁定（BF16 atol 0.50002，FP32 2e-5），失败后不放宽。E-0002 从 PIQA 与 BoolQ 的官方 train 中切出 train、calibration 与 dev（dev 每任务 1,000 条，官方 validation 封存不读），在三个代表结构上用 0.2M tokens 比较 1e-4 与 3e-4，按 calibration NLL 选学习率，三个家族都选中 3e-4。E-0003 用 seed 0 训练八组：proxy $Z = \mathrm{SiLU}(UA)$（4 层 [6,13,20,27]，r32）分别接 F_out、F_pre、C_gate、C_up、C_dual；线性退化组 $Z = UA$ 接 C_dual；以及 4 层 rank10 与 28 层 rank2 的 LoRA。每组在累计 2M 非 padding 输入 tokens 的那一步停止（13,077 步，batch 2），warmup 5% 后 cosine；主结果用最终 checkpoint，并要求重载后复评逐条一致。主指标是 dev 上 PIQA 与 BoolQ 准确率的宏平均。F*、C*、“有效”与“concat 胜出”的规则都在第一个正式 run 之前写进契约，按点估计判定；配对 bootstrap 区间（10,000 次，按任务分层）只作描述。E-0005 生成 V2 的单跳合成任务（每个 episode 16 条事实、4 种关系、单 token 答案），并在 dev 上评估 Base 的 M 线（只有问题）与 F 线（事实在上下文中）。E-0006 实现 M 线训练与诊断流水线并做冒烟。E-0007 在 calibration 切片上诊断 Base 在 F 线上失败的原因。V2 沿用 M 线协议：support 只写不读，问题阶段丢弃 support 的主干 KV，只保留单层（index 14）、r128、S=1 的低维记忆。E-0008 先训练事实在上下文中的 F 线正对照；

E-0009 按家族在 0.2M tokens 上校准学习率（KernelMatrix 1e-2，SoftmaxSlot 3e-3）。E-0010 用 seed 0 训练十组，每组 6,000 步、4,000,470 tokens，按运行前锁定的门槛判定：正常记忆的 EM 同时比清零与打乱记忆高至少 10 pp、NLL 更低、目标权重至少 0.125。它还报告 oracle 与读取诊断，以及 10,000 次配对 bootstrap 区间（只作描述）。E-0012 只做测量，不设自动判定：提取与探针正对照（L0）；冻结第 14 层特征上的线性探针与以目标事实为标签的双线性寻址探针（L1）；训练后与随机初始化投影上的探针（L2）；计划 §8.3 的独立 clean-value 与 read-value ridge 探针（L3）；读取槽位、key 几何与融合幅度（L4）。E-0011 为 V3 生成 T2 链终点数据：每个 episode 有 12 条关系事实组成若干条链，另加 4 条干扰事实；问题给出起点，答案是沿链走到的终点，问题不给出跳数。train 与 dev 只含 1–3 跳，4/6/8 跳的外推切片封存。

记忆路线重开后（D-0012），E-0014 在 V2-03 配置（第 14 层、r128、S=1、direct_hidden、SoftmaxSlot、F_out，LR 3e-3，6,000 步）上做六个单项改动：每个 episode 1 条事实（R1，value 通路正对照）、4 条事实（R2）、事实数课程 1→4→8→16（R3）、先单独训练并冻结的先验通路（R4）、最后问题位置的地址辅助 CE（R5，额外监督）、去中心化余弦寻址（R6），判定沿用 E-0010@2 的门槛。E-0015 只在 E-0014 的 checkpoint 上评估，分离 value 解码侧与寻址侧，其中包括每条事实单独编码后的 oracle 对照。E-0016 检验顾问轮次 1（AD-0001、AD-0002）的三条路线：解码器先行（D1 用 teacher-forced 单槽读取；D2a、D2b 再只用答案 CE 学寻址）、槽位边际似然（D3）、单独编码写入（D4 从零训练；D5 从 R1 的 E=1 checkpoint 续训）。单独编码写入指每条事实不带前文单独过主干，在分隔符处写入 key 与 value，问题侧不变。E-0017 用 seed 1、2 复现 D5。E-0018 不训练，检查 D5 adapter 在 T2 上的迁移，并扫描评估时的读取步数 $S_{\text{eval}} \in \{1, 2, 4, 8\}$。E-0022 是 V3 的训练矩阵：先从 D5 checkpoint 在 T2 一跳问题上训练共同起点（6,000 步，须过 H3 门槛），再从同一起点训练十组，每组 6,000 步，包括累加读取 S=1/2/4、固定总步长、固定 query 的幅度对照、不读记忆的逐 token 深度、按运算量匹配的更宽 S1（r=152）等。主指标是 T2 dev 上 2–3 跳的 EM，进入 V4 的规则在运行前写定。

E-0013（V5 缩减版）在 F 线四层 [6,13,20,27] 上比较 tokenwise C_up（r32）、F_out（r64，两者都是 524,288 个参数）与全 28 层 all-7 rank1 LoRA（630,784 个参数），seed 1–3，各训练 3 个 epoch，每个 arm 按 calibration NLL 选学习率。配置与分析锁定后，PIQA+BoolQ 官方 validation 只开封一次。主比较是 C_up 减 LoRA 的三 seed 平均宏平均差，95% 区间来自配对、按任务分层的 bootstrap（10,000 次）；封存规则规定上界低于 +1 pp 记为 challenged。两个次要比较用 Holm 校正，另报 WikiText NLL 与延迟。

RB 线（D-0025）把侧分支放在四层的残差旁路上：每个 token 用因果前缀 KernelMatrix 读取侧分支自己的历史，读出写回块输出残差（R_res）或接 FFN gate/up；T0 是参数相同但不读历史的对照。确认集 A2 从 V1 train 中每任务分层切出 1,000 条后封存，RB 的训练数据不含这些行。E-0019（RB0）做实现检查与计时，E-0020（RBc）校准四个训练家族的学习率，E-0021（RB1）是首轮筛选。RB0 的 BF16 比较仍用锁定容差 0.50002；跨 GEMM 形状改变（逐 token decode、batch 扩展）的 BF16 比较，若未改动的 reference 给出相同的超出量，只作诊断记录，由同一比较在 FP32 下按逐张量 $\max|a - r| \le 10^{-5} + 10^{-4} \max|r|$ 判定（D-0027）。各实验的契约摘要如下。

E-0001 V0 实现与不变量检查

- question: 新侧分支实现能否通过 plan §6.3 的针对性检查（函数保持、两步梯度启动、冻结输入 Jacobian、同核恒等、约束融合恒等、因果、cache、reset/padding、所有权、read/write 计数、M 线信息访问、导出重载），从而可以进入训练？
- intervention: 在 src/ 实现 FeatureTap（qknorm_pre_rope 为主，另有 raw_qkv、direct_hidden；K/V 在 repeat_kv 之前压缩，不重复执行原投影）、LowDimProjection、MemoryBackend（kernel_matrix、kernel_slot_reference、soft_slot；M、c 用 FP32 累积，空记忆显式输出零）、LatentReader（累加与锚定更新，readout=increment 或 full，eta=0.25 与 g=1 固定）、Fusion（F_out、F_pre、F_attnres、C_gate、C_up、C_dual，出口零初始化）、Protocol（M 线 support write-only、query read-only、support 主干 KV 不进 query；F 线 inclusive/strict

因果写入）与 Accounting；原 attention 与原 FFN 每层每次前向只执行一次；用 pytest 在 Miyabi-G 作业中检查

- comparator: 未修改的 Qwen3-0.6B-Base 前向（同输入、同 dtype）；KernelSlot 参考读取；用同一 P 构造的 F_pre；一次 prefill 对分块 prefill 与逐 token decode；固定 query 时的精确幅度关系 Z=S·eta·a·Ro

- metric: plan §6.3 每项检查的 pass/fail；输出与梯度的最大绝对/相对误差对比预先锁定的容差（FP32 严格阈值；BF16 容差由同一 Base 的重复测量锁定，并在检查运行前写入配置）；各参数组的梯度范数；主干参数更新量（应为 0）；write/read/原块调用计数

- scope: Qwen/Qwen3-0.6B-Base（锁定 model 与 tokenizer revision）；F 线 layers [6,13,20,27]、M 线 layer [14]；r ∈ {32,128}；S ∈ {1,2,4}；FP32 小张量参考与真实模型（BF16 主干 + FP32 分支）两种设置；shared→untied 复制恒等推迟到 V3 之前

- contract: E-0001@1, sealed as DOC-0003@1 (research/experiments/E-0001.md)

E-0002 V1 任务切片与家族学习率校准

- question: PIQA+BoolQ（从官方 train 切出 train/dev/calibration，纯文本固定模板）在 4 层 r32 设置下是否处于可学习且未饱和的区间？LoRA、tokenwise 加法、tokenwise concat 三个家族在 calibration 上各取哪个学习率？

- intervention: Base 只评估不训练；三个家族各两个学习率 {1e-4, 3e-4}，每个 run 0.2M 非 padding 输入 tokens，seed 0，相同数据顺序：LoRA（4 层 [6,13,20,27] all-7 rank10）、tokenwise 加法（proxy Z=SiLU(UA) + F_out）、tokenwise concat（proxy Z=SiLU(UA) + C_dual）

- comparator: Base；同一家族的两个学习率之间

- metric: calibration NLL（选择依据，运行前固定）与准确率；Base 与 LoRA 在 dev 上的 PIQA、BoolQ 准确率（地板/天花板检查）；实测吞吐（tokens/s）、每步时间、加载/评估/保存时间与 node·h

- scope: F 线 context-retained；max length 512；AdamW betas (0.9, 0.999)、eps 1e-8、矩阵 weight decay 0.01；warmup 占实际优化步数 5% 后 cosine；global-norm clip 1.0 并记录 pre-clip 范数；dropout 0；PAD 与 EOS 同 ID 时按实际 padding 位置屏蔽；模板、标签 token、候选打分与多 token 标签归一化规则在本实验内固定并写入 release；官方 validation 不开封

- contract: E-0002@1, sealed as DOC-0019@1 (research/experiments/E-0002.md)

E-0003 V1 FFN 融合接口筛选（八组）

- question: tokenwise proxy 下，哪种接口能把低维状态转成任务收益：加法写回（F_out/F_pre）还是 FFN gate/up 条件融合（C_gate/C_up/C_dual）？这种优势是否超出线性退化组与参数相近的 LoRA？

- intervention: V1-01 至 V1-05：proxy Z=SiLU(UA)（4 层 [6,13,20,27]，r32）分别接 F_out、F_pre、C_gate、C_up、C_dual；V1-06：线性 Z=UA + C_dual

- comparator: V1-07：同 4 层 all-7 LoRA rank10（约 0.92M 参数）；V1-08：全 28 层 all-7 LoRA rank2（约 1.26M 参数）；V1-06 线性退化组；Base 只评估

- metric: dev 上 PIQA、BoolQ 准确率及其宏平均（主指标）与 dev NLL；stored、trainable 与实际有梯度的参数数；实测吞吐、wall time 与 node·h；所有 arm 在相同 dev 样本上评估
- scope: F 线；沿用 V1 切片与学习率校准锁定的模板、切分、各家族学习率与优化配方；seed 0 单 seed 筛选（screening），不作确认；max length 512；checkpoint 选择规则在运行前固定且所有 arm 相同；接口间不等参，不宣称参数效率；官方 validation 不开封
- contract: E-0003@2, sealed as DOC-0040@2 (research/experiments/E-0003.md)

E-0004 记忆路径就绪检查（RV-0002 跟进）

- question: V2 用到的记忆路径在分块缓存与支持写入容量两处是否正确：FP32 下 chunk3 的记忆状态是否与整段前向一致；支持写入容量保护是否对所有 backend 一致生效？
- intervention: 按 RV-0002 两条 low 跟进：为 chunk3 比较 cache.memory[14] 的 M、c（slot backend 还有 slots）与整段前向的 FP32 状态，并记录 cache.chunk3.state.max_abs_err；在 visible_fact_delimiter 的 support 阶段，对所有 backend（含 kernel_matrix）执行 counts <= slot_capacity
- comparator: 整段前向的记忆状态；slot backend 已有的容量保护行为
- metric: 检查通过数；chunk3 状态 max_abs_err（FP32 atol/rtol 2e-5，来自已锁定的 tolerances.json）；chunk1 状态误差只作诊断；kernel_matrix 在 slot_capacity=1 时按预期报错
- scope: 单层 index14、r128（V2 默认）与 r32；KernelMatrix、KernelSlot、SoftmaxSlot；GH200 MIG 2g.24gb；容差不改
- contract: E-0004@1, sealed as DOC-0047@1 (research/experiments/E-0004.md)

E-0005 M 线合成任务生成器与 Base 上下限（concat_loop_v2）

- question: 新的 concat_loop_v2 合成关系任务能否按 plan §8.1、§13.1–13.2 的协议确定性生成、与旧 LoopLora 数据不重叠，并且 Base 在 M 线（只有问题）下接近先验基线、在 F 线（支持事实在上下文中）下明显高于先验基线，从而适合 V2 的单跳记忆读取研究？
- intervention: 实现生成器、episode registry 与切分；生成 V2 的 T1 单跳任务（E=16 个支持事实、4 种关系、固定公开渲染模板、单 token 答案）：train、calibration 2,048、dev 1,000，以及只登记身份与哈希的封存 confirmation 2,000；多跳（训练/dev hops 1–3，封存外推 hops 4/6/8）只实现接口与身份；Base 只在 dev 上评估 M 线与 F 线
- comparator: 旧 LoopLora registry（UnbearableFate/LoopLora @ a255e81，只读）；先验基线（按 train 答案频率的最佳常数答案）；M 线 Base 对 F 线 Base
- metric: 两次生成哈希一致；各 split 的 episode 身份互不相交；与旧 registry 重叠为 0；答案都是单 token；各轴统计（关系、事实数、干扰项、文本长度、名字池与 tokenization 长度、支持顺序）；每条事实恰好一个写入分隔符位置；Base dev 的完整词表 EM 与答案 NLL（M 线、F 线）
- scope: study_id=concat_loop_v2，phase V2；seed 由 derive_seed(20261029, "concat_loop_v2", phase, purpose) 派生，derive_seed 与 T-0002 的定义相同；write_scope=visible_fact_delimiter，read_scope=all_valid_question；不训练；封存切片不做任何模型评估；GH200 MIG 2g.24gb
- contract: E-0005@1, sealed as DOC-0048@1 (research/experiments/E-0005.md)

E-0006 V2 M 线训练与诊断流水线（实现与冒烟）

- question: 按 plan §8.1 的 M 线协议（support write-only、query read-only、单层 index14、r128、S=1）实现的训练与评估流水线，以及 §8.3、§8.4 规定的诊断，是否在小规模冒烟中行为正确（初始化时与 Base 相同、梯度按预期流动、因果对照与 oracle 开关按定义改变读出）?
- intervention: 实现 M 线训练/评估入口：support 阶段只写、问题阶段从同一只读快照读取、按 episode owner 索引、单 token 答案 CE、完整词表 EM 与 NLL；§8.3 诊断（寻址 top-1、margin、目标权重、熵、n_eff，读误差，输出范数，数值统计）；zero / shuffled / permuted memory 因果对照；§8.4 oracle 开关（真实 key 替换 query、直接提供目标 value、zero memory、其他 episode 的 memory，只作离线评估）
- comparator: Base（出口零初始化时应与之完全相同）；正对照小任务上 normal memory 对 zero memory
- metric: 检查通过数；初始化时与 Base 的 logits 差；各控制与 oracle 开关的读出变化；冒烟步时、显存与吞吐（为记忆家族校准与 V2 估算成本）
- scope: 特征 direct_hidden 与 qknorm_pre_rope；读算子 KernelMatrix 与 SoftmaxSlot（g=1，s0=sqrt(dk)，tau=1）；融合 F_out 与 C_dual；write_scope=visible_fact_delimiter，read_scope=all_val GH200 MIG 2g.24gb；只做冒烟，不做正式训练
- contract: E-0006@1, sealed as DOC-0053@1 (research/experiments/E-0006.md)

E-0007 F 线 Base 难度诊断（concat_loop_v2）

- question: E-0005 中 Base 的 F 线 EM 接近先验（dev 1.3%，先验 1.5%）；协调者只读核查已有预测，dev 上 F 线 argmax 有 94.4% 是问题主语，M 线为 76.7%。Base 是在上下文中找到了正确实体、只是被复制问题主语的倾向压过，还是在 E=16 时根本没有找到？失败是否来自提示格式或评估实现？
- intervention: 在 calibration 切片上沿用 E-0005 的 Base 提示与评分路径，逐条输出完整词表 top-5、正确答案的秩与概率、问题主语的概率与 top-1 类别；计算限定候选准确率；F 线另做事实数梯度 E ∈ {1,2,4,8,16}：保留被问事实，其余事实按派生 seed 确定性抽取并保持原相对顺序，用生成器原模板重新渲染
- comparator: E-0005 的 Base dev 结果（M 1.1%、F 1.3%、先验 1.5%）；同一 calibration 上的 M 线；按候选数计算的随机基线；E=1 作为最容易的正对照
- metric: 完整词表 EM 与答案 NLL；正确答案的秩与概率；问题主语复制率；top-1 类别分布；限定候选准确率：(a) 支持中出现的全部不同名字，(b) 同一集合去掉问题主语，(c) 与被问事实同关系的事实的不同尾实体；按 E、被问事实位置、关系、support_order 与 answer_frequency_stratum 分层
- scope: 只用 Base（Qwen3-0.6B-Base @ da87bfb608c14b7cf20ba1ce41287e8de496c0cd），不训练；只用 calibration 切片（runs/ATT-0039-1/data/calibration.jsonl，sha256 c08753365165216b5a9f4c3d629 dev 不再评估，confirmation 不读取；E 变体只是诊断视图，不是新切分；不改 E-0005 的协议、模板与数据；diagnostic 证据，不支持也不挑战 H-0003
- contract: E-0007@2, sealed as DOC-0063@2 (research/experiments/E-0007.md)

E-0008 V2 F 线训练后正对照（事实在上下文中）

- question: 在与 V2 M 线相同的数据、样本顺序与 4M 输入 tokens 预算下，事实放在上下文中、只用答案 CE 训练的标准 LoRA，能否在 E=16 的单跳任务上明显超过先验？即任务本身在该规模与预算下是否可学，从而 M 线结果可以解读。
- intervention: LoRA（4 层 [6,13,20,27]，all-7 模块，rank 10，与 V1-07 相同布局）接在冻结 Base 上；F 线提示 = support_text + question_text（与 E-0005/E-0007 的 F 线评估相同），只在答案位置计算 CE；学习率在 {1e-4, 3e-4, 1e-3} 上各跑 0.2M tokens，按 calibration 答案 NLL 选择；选中后跑完整的 4M tokens
- comparator: E-0005/E-0007 的未训练 Base F 线（calibration EM 1.42%，E=16）；按 train 答案频率的最佳常数答案（先验基线）
- metric: calibration 与 dev 上的完整词表 EM 与答案 NLL；答案不等于问题主语的子集上的 EM；问题主语复制率；学习曲线；吞吐、wall time 与 node·h
- scope: concat_loop_v2 V2 数据（E-0005，train 16,384 / calibration 2,048 / dev 1,000）；seed 0；F 线结果单独登记，不替代 M 线；confirmation 不读取；只作为 V2 的正对照与第二步的放行门槛
- contract: E-0008@1, sealed as DOC-0086@1 (research/experiments/E-0008.md)

E-0009 V2 记忆家族学习率校准与正式 M 线训练程序

- question: 正式 M 线训练程序（4M token 日程、对照评估、checkpoint 重载）能否正确运行？在 V2 共同配方下，KernelMatrix 与 SoftmaxSlot 两个读算子各应使用哪个学习率？
- intervention: 实现正式 M 线训练程序（复用 E-0006 的 collate、train_forward、evaluate_batch 与对照，加入 C_up）；代表配置 qknorm_pre_rope + F_out（层 14，r128，S=1）分别接 kernel_matrix 与 soft_slot，学习率 {3e-4, 1e-3, 3e-3} 各跑 0.2M tokens
- comparator: 同一 run 中 zero memory 与 shuffled memory 的 calibration 答案 NLL（诊断）；E-0006 的冒烟结果
- metric: calibration 答案 NLL（normal memory，选择依据）；zero 与 shuffled 的 NLL 与 EM（诊断）；梯度范数与裁剪比例；吞吐、显存、wall time 与 node·h
- scope: concat_loop_v2 V2 数据；seed 0；V2 共同训练配方（E-0008 Design 段）；校准只选学习率，不作 H-0003 的证据；选中的学习率用于 E-0010 中同一读算子的全部 arm（包括 C_dual 探针）
- contract: E-0009@1, sealed as DOC-0087@1 (research/experiments/E-0009.md)

E-0010 V2 M 线单跳记忆读取：2×2×2 析因与 C_dual 探针

- question: M 线（问答阶段丢弃 support 主干 KV，只保留低维 memory）下，哪种特征 × 读算子 × 融合组合能让 learned memory 读取产生可用信号（normal memory 明显优于 zero 与 shuffled）? SoftmaxSlot 是否比 KernelMatrix 读得更集中、更可用？
- intervention: 十组：特征 direct_hidden / qknorm_pre_rope × 读算子 kernel_matrix / soft_slot × 融合 F*=F_out / C*=C_up（V2-01 至 V2-08），以及 qknorm_pre_rope × {kernel_matrix, soft_slot} × C_dual 交互探针（V2-09、V2-10）；层 14，r128，S=1，只用答案 CE，每组 4M 输入 tokens

- comparator: 同一 checkpoint 的 zero memory、shuffled memory 与 permuted association；E-0008 的 F 线训练后正对照；key/value oracle（诊断）；先验基线
- metric: dev 上 normal/zero/shuffled/permuted 的完整词表 EM 与答案 NLL（主指标为 normal 对 zero 与 shuffled 的 EM 差与 NLL 差）；读取诊断（目标权重、n_eff、熵、读误差）；答案 $\neq$ 问题主语子集上的 EM；学习曲线；参数数、吞吐、wall time 与 node·h
- scope: concat_loop_v2 V2 数据（E-0005），seed 0 单 seed 筛选，不是确认；V2 共同训练配方（E-0008 Design 段）；各读算子的学习率来自 E-0009；判定规则在任何 V2 run 之前锁定；M 线结果与 F 线正对照分开登记；不替代旧 writer 的 80% gate；confirmation 不读取
- contract: E-0010@2, sealed as DOC-0083@2 (research/experiments/E-0010.md)

E-0011 V3 T2 链终点任务生成器、登记与难度检查

- question: 能否按 plan §9.1、§13.1–13.2 确定性地生成 V3 的 T2 链终点任务（训练/dev 只含 hops 1–3，封存 hops 4/6/8 与未见名字），与 V2 数据和旧 registry 不重叠，结构捷径留有余量，并且 M 线 Base 接近先验、F 线 LoRA 小 pilot 能学会 hops 1–3？
- intervention: 新模块 src/v3_data.py（不改 src/v2_data.py）：phase 参数化的 T2 链终点生成器、episode registry 与切分；生成 phase V3 的 train/calibration/dev，并登记封存切片的身份与哈希；计算无模型的结构捷径基线；Base 在 dev 上评估 M 线与 F 线；F 线 LoRA 小 pilot（复用合并后的 src/v2_fline.py，0.2M tokens）
- comparator: 先验基线（train 最佳常数答案）；无模型结构捷径基线（随机 r 终点猜测、忽略关系类型走到底、最长 r 链终点、支持中最后出现的 r 终点）；V2 T1 数据（ATT-0039-1）与旧 LoopLora registry 的内容哈希
- metric: 两次生成哈希一致；身份与语义内容在切片内外互不重复，与 V2 和旧 registry 重叠为 0；单 token 答案与答案边界检查；各轴统计（hop、链长构成、r 链数、干扰事实位置、支持顺序、名字频率层、tokens）；每个 hop 的结构捷径 EM；M 线 Base 与 F 线 Base 的 dev 完整词表 EM/NLL（分 hop）；LoRA pilot 的 calibration EM/NLL（分 hop）；node·h
- scope: study_id=concat_loop_v2，phase V3；derive_seed(20261029, "concat_loop_v2", "V3", purpose)；write_scope=visible_fact_delimiter；E=16；只用 V2 的 seen 名字池（unseen 只用于封存切片）；封存切片不做任何模型评估；不训练 M 线；GH200 MIG 2g.24gb
- contract: E-0011@2, sealed as DOC-0097@2 (research/experiments/E-0011.md)

E-0012 V2 记忆失败层次定位（表示探针与寻址几何）

- question: E-0010 十组都没有读出 episode 记忆（value oracle 也无效）。在 M 线的各层次上，答案相关信息的线性可解码性在哪里下降：冻结第 14 层特征（写入位置的尾实体、问题特征对目标事实的寻址）、训练后的 key/value/query 投影、学到的目标 value（clean-value probe）、实际读取向量（read-value probe），以及读取的槽位与融合幅度？测量结果供协调者选择 plan §8.4 允许的唯一探针，或以预期价值有限为由停止扩大记忆路线。
- intervention: 不训练 M 线：L0 提取与探针正对照；L1 冻结 Base 第 14 层特征（direct_hidden；qknorm_pre_rope 的 q_norm/k_norm 与 v_proj 输出）上的线性探针与双线性寻址探针（秩 128/256/512/满秩）；L2 E-0010 十个 checkpoint 与随机初始化投影的 key/value/query 探针；L3 plan §8.3 的独立 clean-value 与 read-value 探针（src/v2_probe.py 路径）；L4 读取槽位分布、key 几何与融合输出幅度

- comparator: 随机水平（名字 1/92，关系 1/4，寻址 1/16）与按 train 频率的常数基线；L0 正对照；随机初始化投影对训练后投影；E-0010 的 oracle 与读取诊断
- metric: calibration 上各探针的 top-1 准确率（按层次、特征集、位置、目标与秩，附正确数与样本数）；clean_to_clean、read_to_read、clean_to_read；读取 argmax 槽位分布；episode 内 key 两两余弦与目标 margin；问题位置融合输出范数与第 14 层残差范数之比；node·h
- scope: V2 T1 数据（ATT-0039-1）train、calibration 与 dev，seed 0；不读 confirmation；不训练 M 线，不改 E-0010 结论；只作诊断测量
- contract: E-0012@1, sealed as DOC-0117@1 (research/experiments/E-0012.md)

E-0013 V5 缩减版：tokenwise 简单接口对强 LoRA 的实用确认（PIQA+BoolQ 官方 validation）

- question: 记忆路线停止（D-0008）后，按 plan §14.2 把 V5 收缩为简单接口对比：F 线 4 层 [6,13,20,27] 的 tokenwise C*（C_up，r32）在 PIQA+BoolQ 官方 validation 上，宏平均准确率是否比全 28 层 all-7 rank1 LoRA 高至少 1 pp，且每个任务不低于 -1 pp？与 C* 等参数的加法接口 F*（F_out，r64）和两者相比如何？各自的代价（参数、可合并性、训练时间、推理延迟、WikiText-103 NLL）是多少？
- intervention: 三个 arm 各用 3 个新 seed（1,2,3）独立训练：C_up r32 tokenwise SiLU（524,288 参数）；F_out r64 tokenwise SiLU（524,288）；LoRA 全 28 层 all-7 rank1（630,784，alpha=rank）。沿用 V1 的切分、模板、候选打分与 max_length 512；每 run 正好 3 epochs（约 4.46M 非 padding 输入 tokens），有效 batch 2；学习率按预登记网格逐 arm 在 calibration NLL 上选择；只用最终 checkpoint。未训练 Base 作参照
- comparator: 主比较：LoRA28 all-7 rank1（强基线，参数多约 20%）；次要：F_out r64（与 C_up 等参数的加法接口）；参照：未训练 Base
- metric: 官方 validation 上 PIQA、BoolQ 准确率与宏平均（主），seed 平均差、各 seed 差与配对分层 bootstrap 95% 区间；次要比较的 Holm 校正；dev 准确率与 NLL（只作监控）；WikiText-103 test NLL 相对 Base 的变化；可训练参数、可合并性、训练时间、峰值显存、推理延迟；node·h
- scope: F 线（保留原 attention）、Qwen3-0.6B-Base、V1 的 train/dev/calibration 切分与模板；官方 validation 封存，配置、训练规则与分析脚本锁定后只开封一次；3 个训练 seed；参数接近但不相等，不宣称同时等参数、等计算或等状态；不含侧支历史、记忆与多次读取；V1 已用这些任务的 train/dev 选择接口，不称零调参迁移；LoRA 结论只针对 rank1 与本合同的短校准 LR 规程，不代表所有合理的 LoRA 配置或完整训练长度上的最优调参；区间以三个训练 seed 为条件
- contract: E-0013@3, sealed as DOC-0128@3 (research/experiments/E-0013.md)

E-0014 记忆路线修复筛选：事实数、课程、先验吸收、寻址监督与去中心化余弦寻址

- question: V2 的失败是否主要来自训练动态？以 V2-03（direct_hidden、SoftmaxSlot、F_out）为基础，分别改变 (a) 每个 episode 的事实数（E=1、E=4），(b) 事实数课程 1→4→8→16，(c) 先训练并冻结一条吸收答案先验的通路，(d) 对最后问题位置的地址权重加辅助交叉熵（额外监督），(e) 去中心化余弦寻址加可学温度。哪些能在 E=16 dev 上达到 H3 门槛？E=1 能否说明 value 的写入、读取、解码通路本身可用？

- intervention: 六个 arm，seed 0，M 线第 14 层 r128 S=1，direct_hidden 特征，SoftmaxSlot，F_out 融合，LR 3e-3（E-0009），步数与 batch 同 V2（6000 步，每步 8 个 episode），只用最终 checkpoint：R1 E=1；R2 E=4；R3 课程 E 1→4→8→16；R4 先验通路（第 14 层 tokenwise F_out r32，先在零记忆下单独训练 1000 步后冻结）+ E=16；R5 E=16 + 地址辅助 CE（λ=1，额外监督）；R6 E=16 + 去中心化余弦寻址与可学温度

- comparator: V2-03（E-0010，ATT-0082-1）作参照；每个 arm 内的 zero、shuffled、permuted、oracle_key、oracle_value 记忆对照；R1 作 value 通路正对照

- metric: E=16 dev：EM 与 NLL（normal/zero/shuffled/permuted/oracle），目标事实平均地址权重，n_eff，最后问题位置的 argmax 槽位分布，融合增量与残差范数之比；R1/R2/R3 另报各自 E 的 dev；训练曲线；node·h

- scope: M 线 T1 hop 1；E=16 沿用 V2 数据（ATT-0039-1），E ∈ {1,4,8} 为新登记的数据版本（新 seed 命名空间）；单 seed 筛选；额外监督只在 R5，结论单独陈述；不读 confirmation；不改 V2 结论与 E-0010 判定

- contract: E-0014@1, sealed as DOC-0135@1 (research/experiments/E-0014.md)

E-0015 记忆路线诊断：value/解码侧与寻址侧的分离（E-0014 checkpoint 上只做评估）

- question: E-0014 多事实配方的失败落在哪一侧？训练出的 value 是否仍携带尾实体信息？非末位问题位置的混合读取是否干扰解码？E=1 学到的 value 映射能否迁移到多事实上下文？

- intervention: 只做评估，不训练：E-0014 的 R1（ATT-0122-1）、R2（ATT-0123-1）、R5（ATT-0126-1）和 V2-03（ATT-0082-1）的最终 checkpoint，在 E=1、E=4、E=16 calibration 上加新的读取替换对照（last_only、oracle_last、oracle_last_only、uniform、last_only_shuffled、wrong_oracle）、增量幅度扫描、单独编码替换（每条事实单独编码后放回原槽位：iso_kv、iso_v、iso_oracle_value）、按目标位置分层，以及线性探针（训练出的 value、冻结特征、最后位置的融合增量；上下文编码与单独编码各一次）

- comparator: 同一 checkpoint、同一 split 的 normal 读取；已有的 zero、shuffled、oracle_value 对照；冻结特征探针作参照

- metric: 每个 checkpoint×split× 对照的 EM 与 NLL（calibration，按目标位置分层）；探针在 calibration 上的准确率（train 前 4,096 条上拟合）及其基线；重新加载后 dev 上的 normal 预测须与 E-0014 记录逐例一致

- scope: M 线 T1 hop 1；诊断只用 train 与 calibration，dev 只用于加载一致性检查，不读 confirmation；数据为 ATT-0039-1（E=16）与 ATT-0112-1（E=1/4）；单 seed 的已有 checkpoint；只产出 diagnostic 证据，不改变任何假设状态

- contract: E-0015@4, sealed as DOC-0157@4 (research/experiments/E-0015.md)

E-0016 记忆路线训练筛选：解码器先行、槽位边际似然、单独编码写入（顾问轮次 1 的三条路线）

- question: 在 E=16 上下文里，value→ 答案的解码器能否在干净读取下学成？之后寻址能否只从答案 CE 发现？把训练目标改成槽位隐变量的边际似然能否避免塌缩？每条事实单独编码后写入能否消除上下文迁移并让单跳读取可用？

- intervention: seed 0，E=16（ATT-0039-1），V2-03 基础配置（direct_hidden、SoftmaxSlot、F_out、第 14 层 r128 S=1、LR 3e-3），每段 6000 步 ×8 episode。D1：teacher-forced 单槽读取（记忆只留目标槽位，额外监督）；D2a/D2b：从 D1 续训，恢复全部槽位，只用答案 CE（D2a 只训练寻址参数，D2b 全部训练）；D3：槽位隐变量边际似然，只在最后问题位置读取，不用标签；D4：每条事实单独编码后写入，从零训练；D5：单独编码写入，从 R1（ATT-0122-1）续训。D4/D5 以 E-0015 规则 5 成立为前提
- comparator: V2-03（ATT-0082-1）与 E-0014 各 arm；每个 arm 内的 zero、shuffled、permuted、oracle 对照；D1 另有 teacher-forced 的 shuffled 与错误槽位对照；D3 另报软读取
- metric: E=16 dev：EM 与 NLL（各对照），目标权重（全问题位置平均与最后位置），最后位置 argmax 分布，按目标位置分层；D1 的 teacher-forced 门槛；D3 的硬读取门槛；训练曲线；node·h
- scope: M 线 T1 hop 1；单 seed 筛选；D2a/D2b 的配方在 D1 用了目标索引，结论注明；D3 的硬读取与只读最后位置、D4/D5 的单独编码写入都是新登记的协议，与 E-0010 的统计量分开报告；不读 confirmation；不改 V2 与 E-0014 的结论
- contract: E-0016@1, sealed as DOC-0160@1 (research/experiments/E-0016.md)

E-0017 记忆路线：E=1 预训练加单独编码写入（D5 配方）的 seed 复现

- question: D5 在 seed 0 上达到单独编码写入协议下的 H3（EV-0095）。换成 seed 1、2，从各自重新训练的 R1（E=1）出发走完整个配方，是否同样达到？
- intervention: 主流程 P1、P2（s=1、2）：阶段 A 按 E-0014 的 R1 配方，用 seed s 在 E=1 上训练 6000 步；阶段 B 按 E-0016 的 D5 配方，从阶段 A 的最终 adapter 续训 6000 步，单独编码写入，E=16。次要 C1、C2：从 R1 seed 0（ATT-0122-1）出发，续训 seed 取 1、2，其余同 D5，用来区分阶段 A 与续训的随机性
- comparator: 每个 run 内单独编码写入下的 zero、shuffled、permuted、oracle_key、oracle_value 对照；E-0016 的 D5（seed 0，EV-0095）与 D4（EV-0096）
- metric: E=16 dev（1000）单独编码写入下的 EM 与 NLL、目标权重（全问题位置平均与最后位置）、n_eff，按目标位置分层；阶段 A 在 E=1 dev 的 normal、zero、shuffled；E=16 calibration 与上下文编码下的描述性对照；学习曲线；node·h
- scope: M 线 T1 hop 1；单独编码写入协议：每条事实不带前文编码后写入，问题侧不变，与 E-0010 的统计量分开报告；配方含 E=1 预训练阶段，只用答案 CE，不用地址标签；screening；不读 confirmation
- contract: E-0017@1, sealed as DOC-0180@1 (research/experiments/E-0017.md)

E-0018 V3 前置：单独编码写入下 T2 链终点任务的读取迁移、无训练深度扫描与循环实现检查

- question: D5 配方训练的单读 adapter（seed 0、1、2）在 T2 链终点任务的 dev 上，按单独编码写入，hop1 能否读到目标事实并答对？不训练时把读取次数扩展到 S_eval=2/4/8，hops2–3 的 EM 与每步对路径上各跳目标事实的地址权重怎样变化？固定记忆、累加更新的 S>1 循环在单独编码写入与 T2 数据上实现是否正确，训练步耗时多少？

- intervention: E0：在单独编码写入的 M 线代码中接入 E-0011 的 T2 链终点数据（phase V3，只用 train/calibration/dev）与 S>1 累加循环（同一问题位置各步共享固定记忆 M，z 按 plan §3.5 累加），新选项默认关闭，关闭时输出与梯度同 master；S=1 时与现有 D5 评估路径逐例一致。E1 只评估：ATT-0159-1（seed 0）、ATT-0172-1（seed 1）、ATT-0173-1（seed 2）的最终 adapter，在 T2 dev 1000 上，S_eval ∈ {1,2,4,8}。计时：T2 上 S=1/2/4 的训练步冒烟各约 100 步，不作证据

- comparator: E-0011 的先验与结构捷径基线及 M 线 Base；每个 adapter 内的 zero、shuffled、permuted 对照，hop1 另报 oracle_key、oracle_value；S_eval=1 作各 adapter 的参考

- metric: 分 hop（1、2、3）的 EM 与 NLL（各对照）；每步对路径上第 j 跳目标事实的地址权重、argmax 槽位分布与 n_eff；各步 query 余弦、状态与读出范数；训练步耗时、峰值显存；node·h

- scope: M 线，单独编码写入协议（须声明的结构化支持输入，统计量与 E-0010 分开报告），H-0012 的 D5 配置；T2 链终点 phase V3 的已发布切片，封存切片（confirmation、ood_h4/6/8、ood_unseen）不做任何评估；只评估已有 adapter，不训练（计时冒烟除外）；diagnostic；不读 confirmation

- contract: E-0018@1, sealed as DOC-0202@1 (research/experiments/E-0018.md)

E-0019 RB0：残差旁路线实现与计时（因果前缀 KernelMatrix、T0、四层 R_res/concat、侧支历史干预）

- question: plan2 主线需要的新 F 线路径是否正确、训练成本是否可承受：向量化因果前缀 KernelMatrix 与已审核的逐 token 参考扫描在输出与梯度上是否一致？T0、固定 query、四层 R_res 与 concat 的零出口、残差拓扑、两步梯度启动、因果、cache、padding 与侧支历史干预是否按定义工作？各 RB1 arm 的实测步时与 node·h 是多少？

- intervention: 新增并测试（默认行为不变）：KernelMatrix 向量化因果前缀实现（impl=prefix）；T0 分支（branch=kernel_tokenwise，与 KernelMatrix 分支同参数）；AdapterConfig 的 fixed_query 与 impl；F 线推理期侧支历史干预（iv_off、iv_window1/8、iv_strict、iv_swap、iv_permute、当前 token 读权重）；RB 训练入口（arm 表与参数断言、residual_la_v21 种子、2M tokens/3 epochs 日程、LR 校准模式）；latency 脚本；按 D-0025 的方案 A2 生成 RB 确认集切分清单（从 V1 train 每任务切 1,000 条；不做任何模型评估）。不训练正式模型

- comparator: 已审核的逐 token 参考扫描 process_sequence 与 KernelSlotReference；未修改的 Base（零出口）；E-0001/E-0004 已有的不变量测试作回归；V1 tokenwise arm 的实测 0.183–0.194 s/step（research/notes/T-0003-tables.md）作计时参照

- metric: 前向输出与对 $\bar{q}/\bar{k}/\bar{v}$/Rq/Ro 的梯度的最大绝对、相对误差；四层零出口 logits 与 Base 的差；拓扑、因果、cache、padding、重排、写读计数、导出重载与干预恒等各项通过/失败；故意破坏的实现被拒绝；stored/trainable/有梯度参数；≥100 个稳定步的 s/step、tokens/s、峰值显存；按 plan2 §14.4 推算的每个 RB1 arm 与 RB5 run 的 node·h

- scope: F 线（context-retained）、冻结 Qwen3-0.6B-Base、四层 [6,13,20,27]、r32（另核对 r44、r64 参数计数）、qknorm_pre_rope；实现与计时检查，不产生任务性能结论；FP32 容差在运行前写进测试，GPU BF16 容差沿用 E-0001/E-0004 的锁定值，不看结果后放宽；计时作业在 2g.24gb 上 fail-fast，其它作业的 MIG guard 只记录（D-0019）

- contract: E-0019@1, sealed as DOC-0203@1 (research/experiments/E-0019.md)

E-0020 RBc：残差旁路线四个训练家族的学习率校准（LoRA4 r7、T0、R_res+KernelMatrix、C_dual+KernelMatrix）

- question: 在 RB1 的数据、模板与优化配方下，四个训练家族（同四层 all-7 LoRA rank7；R_res+T0 S1；R_res+KernelMatrix S1；C_dual+KernelMatrix S1）的学习率按预登记规则应锁定为多少？
- intervention: 每家族 seed 0 短训练，到首次累计达到 200,000 非 padding 输入 tokens 的完整步为止（同 V1、E-0013）；初始网格：LoRA 与 T0 为 {1e-4, 3e-4, 1e-3}，R_res+KernelMatrix 与 C_dual+KernelMatrix 为 {3e-4, 1e-3, 3e-3}；初始网格的原始最小 NLL 在上端或下端时追加一个端点（LoRA/T0：3e-3 或 3e-5；两个 KernelMatrix 家族：1e-2 或 1e-4），只扩一次
- comparator: 同家族候选互比。plan2 §6.4 的建议点（LoRA/逐 token {1e-4,3e-4}，记忆 {3e-4,3e-3}）都在初始网格内；E-0002（V1 三族 3e-4，停在网格上端）、E-0013 校准（三个 arm 都是 1e-3）与 E-0009（M 线 KernelMatrix 1e-2）只作先验参照，不沿用
- metric: RB calibration 子集上的 NLL（V1 定义：各任务 gold CE 按 token 求和后取任务均值，再取两任务算术平均）；非有限值；每个候选的步数、tokens、步时与 node·h
- scope: F 线；RB 数据（V1 切分 split map SHA256 47cf5c777d07863b8def6903487697eafc970b0cd5b5d80babb 模板、max_length 512；calibration 用完整的每任务 2,048 条；train 为 V1 train 去掉 RB 确认集行，D-0025 方案 A2）；优化配方同 RB1；初始化与训练顺序用 derive_seed(20261101, "residual_la_v21", "V0", "init"|"train", 0)；不看 dev 与任何确认集；锁定结果用于 RB1：LoRA→RB1-07，T0→RB1-01，R_res+KernelMatrix→RB1-02..06，concat→RB1-08、RB1-09；共 2.4–3.2M tokens，超出 plan2 §6.4 的 2M 部分记入 §14.3 储备
- contract: draft, not sealed (research/experiments/E-0020.md)

E-0021 RB1：残差旁路低维因果 linear attention 首轮筛选（plan2 R1 八组 + C_up 核对）

- question: 保留原 attention 的 F 线上，四层 r32 的块输出残差旁路 R_res + 真实因果 KernelMatrix（S1）是否带来任务收益（对 Base、同参数无侧支历史的 T0、近预算同四层 LoRA）？同一旁路里 S2/S4 是否有超出幅度与总步长解释的初步信号？在同一 KernelMatrix 状态上，FFN gate/up 条件融合（C_dual、C_up）是否改善写回？
- intervention: 九个训练 arm，seed 0：RB1-01 R_res+T0 S1；RB1-02 R_res+KernelMatrix S1；RB1-03 同 S2；RB1-04 同 S4；RB1-05 S4 且 query 固定为第 1 步；RB1-06 S4 且 η=0.25/4；RB1-07 同四层 all-7 LoRA rank7；RB1-08 C_dual+KernelMatrix S1；RB1-09 C_up+KernelMatrix S1（plan2 无此组，用于核对 V1 的 C*=C_up）。共同：四层 [6,13,20,27]、r32、qknorm_pre_rope、inclusive 先写一次再读、η=0.25 与 g=1 固定、readout 为循环增量、出口零初始化；Base 只评估。训练后对 RB1-02、S_F* 对应的 arm、RB1-08、RB1-09 做冻结侧支历史干预
- comparator: Base（只评估，评估代码不变时沿用 ATT-0024-1/EV-0022）；RB1-01（同参数、无侧支历史）；RB1-07（630,784 参数，比 R_res 少 4.94%）；循环对照 RB1-05/06；concat 对照 RB1-02
- metric: dev（每任务 1,000 条）上 PIQA、BoolQ 准确率与宏平均（主）及答案 NLL；25/50/75/100% 的 dev 曲线；冻结干预下的 dev 宏平均变化、当前 token 读权重与读权重的 n_eff 和熵（定义同 M 线）；描述性配对分层 bootstrap 95% 区间（10,000 次，seed 20261101，各 arm 共用索

引）；stored/trainable/有梯度参数；每 token 读写次数与原块调用数；峰值训练显存；prefill 吞吐、decode p50/p95；新增状态字节（M+c FP32 r32 为 4,224 字节/层/序列）；node·h

- scope: F 线、冻结 Qwen3-0.6B-Base、四层 r32、qknorm_pre_rope、KernelMatrix（不换 SoftmaxSlot）；V1 切分、模板与候选打分，max_length 512，按 E-0003 排除不可编码的 train 行；单 seed 筛选，不作确认；所有 arm 相同的样本预算与训练顺序 derive_seed(20261101, "residual_la_v21", "R1", "train", 0)：累计非 padding 输入 tokens 首次达到 2M 的完整步（或 3 epochs，先到为准）；LR 来自 RBc；最终 checkpoint；接口参数不等，不宣称参数效率；T0 只有 S1，不能排除 S4 的逐 token 深度解释；任何确认集都不开封

- contract: draft, not sealed (research/experiments/E-0021.md)

E-0022 V3：T2 链终点上的循环矩阵（共同起点阶段 + plan §9.2 十组，单 seed 筛选）

- question: 在单独编码写入协议与 D5 配方下，先经共同起点阶段使 T2 hop1 达到 H3，再从同一起点训练十组：累加循环 S2/S4 在 T2 hops2–3 上是否优于独立训练的 S1，且该差不被固定 query 的幅度、匹配预算的更宽 S1 与同源 tokenwise 核解释？

- intervention: 起点阶段：从 D5 seed 0 adapter（ATT-0159-1）按 D5 续训配方在 T2 hop1 上续训全部侧支参数 6,000 步（同 D5 每段步数），取最终 checkpoint 作共同起点。十组（plan §9.2）：V3-01 累加 S1；V3-02 S2；V3-03 S4；V3-04 S4 且 η=0.25/4；V3-05 S4 且 query 固定为第 1 步；V3-06 同源 tokenwise 核 S4；V3-07 更宽 S1（r=152，每个问题 token 的 MAC 与 V3-03 匹配）；V3-08 S4 解除 Rq/Ro 跨步共享；V3-09 同核锚定 S4；V3-10 S4 完整状态读出。各组从共同起点在 T2 train hops1–3 上续训 6,000 步，数据顺序与样本相同。

- comparator: V3-01（独立训练 S1）；V3-05（幅度）；V3-07（预算）；V3-06（无记忆的逐 token 深度）；V3-04、V3-08、V3-09、V3-10 只作描述；E-0011 的先验与结构捷径基线；各 arm 内的 zero、shuffled、permuted 对照

- metric: T2 dev 上 hops2–3 的 EM（hop2、hop3 等权平均，主）、各 hop 的 EM 与答案 NLL；S_eval∈{1,2,4,8} 扫描（V3-08 只 1/2/4）；每步状态范数、读出范数、相邻步 query cosine、目标 key 权重与 n_eff、准确率随 s 的变化；S_M* arm 的中间状态干预；按 hop 分层的配对 bootstrap 95% 区间（描述）；参数、每 token 读次数、实测 step 时间与 node·h

- scope: M 线、单独编码写入协议（H-0005@2 的范围）；冻结 Qwen3-0.6B-Base，第 14 层，direct_hidden，SoftmaxSlot，F_out，r128；D5 续训配方（LR 3e-3、AdamW 重置、warmup 300 步后 cosine、每步 8 个 episode、只用最终 checkpoint）；T2 只用已发布的 train/dev（hops1–3），hops4/6/8 不开封；单 seed 筛选，不作确认；各组样本预算相同，共用 LR，不逐组校准

- contract: E-0022@2, sealed as DOC-0211@2 (research/experiments/E-0022.md)

4 Results

下列表格给出各实验可用 attempt 的描述性均值，不是统计检验。E-0001 与 E-0004 是 diagnostic 证据：V0 修正后 85/85 项、E-0004 130/130 项检查通过。E-0003 是单 seed 的 screening 证据：Base 的 dev 宏平均为 70.15%，各训练组在 77.05%–79.20% 之间。大部分提升来自 BoolQ（Base 68.0%，训练组 80.8%–83.6%）；PIQA 只从 72.3% 升到 73.0%–74.9%。按预先固定的规则，F*=F_out（77.80%），C*=C_up（77.70%）。最好的 concat 组 C_dual（78.15%）只比 F* 高 0.35 pp（描述

区间 [−0.60, 1.35] ），记为 concat 未胜。C* 比 4 层与 28 层 LoRA 分别低 1.50 与 1.35 pp；C_dual 对线性退化组只高 0.15 pp [−0.70, 1.00]。E-0005 与 E-0007 是 diagnostic 证据。Base 在 dev 上的 M 线与 F 线 EM 分别为 1.1% 与 1.3%，先验基线为 1.5%。在 calibration 上，上下文事实数 E 为 1、2、4、8、16 时，F 线 EM 依次为 67.0%、6.9%、2.2%、1.4%、1.4%；$E \geq 2$ 时，91%–96% 的回答是问题主语；E=16 时，在支持中的名字（去掉主语）里选对的比例为 14.2%，随机为 3.7%。E-0007 有一项实现检查按原规则未通过：BF16 下 batch 1 与 batch 8 的逐条 NLL 平均绝对差为 0.088，门槛为 0.05。FP32 对照把该差异降到 1e-5 量级，因此判为 BF16 数值敏感。

V2 的结果如下。E-0008 的 F 线正对照通过：事实在上下文中训练后，calibration EM 为 99.95%（训练前 1.90%）。E-0010 十组的 dev 正常 EM 为 0.8%–2.1%，清零记忆为 1.1%，打乱记忆为 1.1%–1.9%；正常 EM 对两者中较大者的差最大为 +0.6 pp。所有组的平均目标权重都低于 0.125，也低于均匀的 1/16。正常与打乱记忆的 NLL 几乎相同（4.42–4.61），清零记忆为 9.25。把读取换成目标的已学 value（value oracle）后，EM 为 1.1%–2.3%。按锁定规则没有合格组，H-0003 记为 challenged（screening）；H-0004 的读算子比较中，两种算子都只在 0/4 格占优，记为 inconclusive。E-0012 的六项正对照为 96.6%–100%。在分隔符位置，冻结特征解码尾实体的准确率为 99.6%（direct_hidden）与 92.9%（qknorm_pre_rope）。以目标事实为标签的双线性寻址探针在最后一个问题位置、秩 128–1024 下的 16 选 1 top-1 为 82.2%–90.2%，秩增大没有一致的提升。训练后，每组在全部 1,000 个 dev episode 的答案位置都把第一个写入的槽位排在最前。独立 ridge 探针下，读出向量的答案准确率为 1.8%–2.3%，已学目标 value 为 16.6%–23.0%。逐层 softmax 探针中，训练后 value 投影解码尾实体的准确率（17.0%–37.1%）不高于随机初始化投影（32.2%–38.5%）。融合增量与第 14 层残差范数之比的组均值为 18.7–608（V2-09 为 1.1e6）。

记忆路线的修复结果如下，都是 screening 或 diagnostic 证据。E-0014 中只有 R1 达到它的门槛：在 E=1 dev 上正常记忆比清零与打乱记忆高 82.6 pp。没有一组在 E=16 上达到 H3，R2 也没有达到 E=4 的门槛；R3、R4、R6 重现了 V2 的槽位 0 塌缩（D-0013）。有地址监督的 R5 在最后问题位置的目标权重为 0.518，EM 优势只有 4.6 pp，oracle_value 只有 10.7%（EV-0084）：读对了，读出也解码不好。E-0015 在 calibration 上把这一点拆开（EV-0087）：R1 的上下文编码 value 在 E=16 上只对第一条事实可用（oracle_value 第 0 条 83.4%，第 1 条及以后 1.6%），上下文中第 1–3 条事实的 value 的 L2 范数（103.1）约为第 0 条（43.5）的两倍；每条事实单独编码后，同一解码器在 E=16 上的 oracle_value 为 86.4%（第 1 条及以后 86.7%）。

E-0016 的三条路线中，只有单独编码写入加 E=1 预训练的 D5 达到 H3（EV-0095）：正常 EM 42.8%，清零 1.1%，打乱 1.7%（优势 41.1 pp），NLL 2.514 低于两个对照，全位置目标权重 0.162，最后位置 0.571。从零训练的单独编码写入 D4 读取均匀，优势 0.3 pp（EV-0096）。解码器先行在 teacher-forced 单槽读取下学成（D1，+65.5 pp，用了目标索引），但之后只用答案 CE 续训时没有学到寻址（D2a +3.2 pp，D2b +3.0 pp）；槽位边际似然 D3 停在答案先验解（+0.4 pp）。H-0010 与 H-0011 记为 challenged（screening，D-0022、D-0021）。E-0017 用 seed 1、2 复现 D5，优势为 47.8 与 46.2 pp，连同 seed 0 共 3/3 达标；全位置目标权重 0.158–0.162，最后位置最常见的槽位只占 7.6%–8.0%，没有塌缩到固定槽位。H-0012 记为 supported（screening，D-0023）。同一批 adapter 换回上下文编码写入时，EM 只有约 5%。

E-0018 把三个 D5 adapter 不经训练用到 T2（EV-0107–EV-0109）。S=1 的输出与已审核的评估器逐条一致。三个 seed 在一跳问题上都没有达到 H3：EM 优势 0.9、1.8、3.0 pp，全位置目标权重 0.036–0.038；把读取换成目标 value 后，EM 为 75.7%–80.2%。所以解码器迁移到了 T2，寻址没有迁移。不训练时把 S_{eval} 提到 2、4、8，2–3 跳的 EM 不变（2.7%–4.5%），NLL 上升。E-0022 因此先在 T2 上训练共同起点，该阶段正在运行。

V5 缩减版的官方 validation 只开封一次（ATT-0179-1，EV-0110；PIQA 1,838 条，BoolQ 3,270 条）。C_up、F_out 与 LoRA 的三 seed 平均宏平均准确率分别为 77.48%、77.25%、77.90%。主比较 C_up 减 LoRA 为 −0.42 pp，95% 区间 [−1.02, +0.20] pp；三个 seed 的差都为负（−0.28、−0.40、−0.58 pp），PIQA −0.74 pp，BoolQ −0.09 pp。上界低于 +1 pp，按封存规则 H-0008 记为 challenged（confirmatory，D-0028）。这个区间只反映给定三个训练 seed 时的数据不确定性；它含 0，所以既不宣称等价，也不宣称 LoRA 更好。次要比较 C_up 减 F_out 为 +0.23 pp（Holm

p=0.34), F_out 减 LoRA 为 −0.65 pp (Holm p=0.069), 校正后都不拒绝。WikiText 上相对 Base (NLL 2.647) 的 NLL 增量为 C_up +0.200、F_out +0.238、LoRA +0.337。

RB0 (E-0019) 的 397 项检查在 PBS 中全部通过 (ATT-0188-1)。逐 token decode 与 batch 从 2 扩到 3 的两项 BF16 比较超出 0.50002 (0.8203、0.8574), 未改动的 reference 给出相同的值，按 D-0027 只作记录，对应的 FP32 检查通过。计时覆盖 21 个配置，每个配置测 100 个稳定步；RB0 共用 0.122 node·h。代码审核 RV-0060 通过，留下四条 low 发现，正在交付前修复。

Table 1: E-0001 V0 实现与不变量检查: descriptive means over usable attempts

arm	seeds	attempts	bf16_atol	check.src/test_core.py::test_kernel_matrix_eq
base_calibration	173	1 ended/success	0.5 (n=1)	-
hardware_probe	0	1 ended/failed, 1 ended/success	-	-
invariant_checks	42	2 ended/failed, 2 ended/success	-	1.176e-08 (n=2)

Per-attempt records: rl run show E-0001.

Table 2: E-0002 V1 任务切片与家族学习率校准: descriptive means over usable attempts

arm	seeds	attempts	C_dual.calibration.boolq.accuracy	C_dual.calibra
base-eval	0	1 ended/cancelled, 1 ended/success	-	-
cal-cdual-1e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-cdual-3e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-fout-1e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-fout-3e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-lora-1e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-lora-3e-4	0	1 ended/cancelled, 1 ended/success	-	-
release-schedule-estimate	0	1 ended/success	-	-
selection-boundary-audit	0	1 ended/success	-	-
selection-release-audit	0	1 ended/failed	-	-
smoke	0	1 ended/success	0.6562 (n=1)	32 (n=1)
smoke-corrected	0	1 ended/success	0.75 (n=1)	32 (n=1)
smoke-reloadfix	0	1 ended/success	0.75 (n=1)	32 (n=1)
smoke-reviewfix	0	1 ended/success	0.75 (n=1)	32 (n=1)

Per-attempt records: rl run show E-0002.

Table 3: E-0003 V1 FFN 融合接口筛选（八组）: descriptive means over usable attempts

arm	seeds	attempts	C_dual.dev.boolq.accuracy	C_dual.dev.boolq.example
Base	0	1 ended/success	-	-
V1-01	0	1 ended/success	-	-
V1-02	0	1 ended/success	-	-
V1-03	0	1 ended/success	-	-
V1-04	0	1 ended/success	-	-
V1-05	0	1 ended/success	-	-
V1-06	0	1 ended/success	-	-
V1-07	0	1 ended/success	-	-
V1-08	0	1 ended/success	-	-
analysis-regression	0	1 ended/success	-	-
base-dev-comparison	0	1 ended/success	-	-
formal-analysis	0	1 ended/success	-	-
smoke-eight-arms	0	1 ended/failed, 1 ended/success	0.625 (n=1)	32 (n=1)

Per-attempt records: rl run show E-0003.

Table 4: E-0004 记忆路径就绪检查（RV-0002 跟进）: descriptive means over usable attempts

arm	seeds	attempts	check.src/test_core.py::test_kernel_matrix_equals_slot_reference_
readiness	42	1 ended/success	1.176e-08 (n=1)

Per-attempt records: rl run show E-0004.

Table 5: E-0005 M 线合成任务生成器与 Base 上下限（concat_loop_v2）: descriptive means over usable attempts

arm	seeds	attempts	F.elapsed_s	F.em	F.examples	F.nll	M
base-dev-MF	-	1 ended/success	4.633 (n=1)	0.013 (n=1)	1000 (n=1)	5.706 (n=1)	4.
base-dev-smoke	-	1 ended/success	0.1641 (n=1)	0 (n=1)	32 (n=1)	5.918 (n=1)	0.
data-generation	-	1 ended/success	-	-	-	-	-
data-smoke	-	1 ended/success	-	-	-	-	-
legacy-inventory	-	1 ended/success	-	-	-	-	-

Per-attempt records: rl run show E-0005.

Table 6: E-0006 V2 M 线训练与诊断流水线（实现与冒烟）: descriptive means over usable attempts

arm	seeds	attempts	arms_completed	check.src/test_core.py::te
eight-arm-E0005-integration	2100042	1 ended/success	-	-
eight-arm-copy32	2100042	1 ended/success	8 (n=1)	-
eight-arm-timing	2100042	1 ended/success	8 (n=1)	-
invariant-checks	2100042	1 ended/success	-	1.176e-08 (n=1)
shuffled-diagnostic-regression	2100042	1 ended/failed, 1 ended/success	-	-

Per-attempt records: rl run show E-0006.

Table 7: E-0007 F 线 Base 难度诊断 (concat_loop_v2) : descriptive means over usable attempts

arm	seeds	attempts	F.E1.em	F.E1.nll	F.
diagnostic-smoke32	20261029	1 ended/failed	-	-	-
diagnostic-smoke32-fixed	20261029	1 ended/success	0.625 (n=1)	1.519 (n=1)	0.
fp32-calibration256-and-saved-bf16-audit	173	1 ended/success	-	-	-
full-calibration2048	20261029	1 ended/success	0.6699 (n=1)	1.422 (n=1)	0.
postmerge-regression	20261029	1 ended/success	-	-	-
scoring-difference-audit	20261029	1 ended/success	-	-	-

Per-attempt records: rl run show E-0007.

Table 8: E-0008 V2 F 线训练后正对照 (事实在上下文中) : descriptive means over usable attempts

arm	seeds	attempts	calibration.all.em	calibration.all.examples	calibratio
fline-4m	0	1 ended/success	0.9995 (n=1)	2048 (n=1)	0.01557 (
fline-cal-0.0001	0	1 ended/success	0.9941 (n=1)	2048 (n=1)	0.02509 (
fline-cal-0.0003	0	1 ended/success	0.998 (n=1)	2048 (n=1)	0.009097
fline-cal-0.001	0	1 ended/success	0.9995 (n=1)	2048 (n=1)	0.001602
fline-cal-0.003	0	1 ended/success	0.998 (n=1)	2048 (n=1)	0.0107 (n
fline-output-audit	0	1 ended/success	-	-	-
fline-review-regression	0	1 ended/success	0.0625 (n=1)	32 (n=1)	3.515 (n=
fline-smoke	0	1 ended/failed, 1 ended/success	0.0625 (n=1)	32 (n=1)	3.515 (n=

Per-attempt records: rl run show E-0008.

Table 9: E-0009 V2 记忆家族学习率校准与正式 M 线训练程序: descriptive means over usable attempts

arm	seeds	attempts	200000.budget	200000.steps	200000.t
final-grid-analysis	0	1 ended/success	-	-	-
formal-path-checks	0	1 ended/success	-	-	-
initial-grid-analysis	0	1 ended/success	-	-	-
kernel_matrix-lr0.0003	0	1 ended/success	-	-	-
kernel_matrix-lr0.001	0	1 ended/success	-	-	-
kernel_matrix-lr0.003	0	1 ended/success	-	-	-
kernel_matrix-lr0.01-extension	0	1 ended/success	-	-	-
reporting-max-regression-cost-report	0	1 ended/success	-	-	-
schedule-preflight	0	1 ended/success	2e+05 (n=1)	300 (n=1)	2.002e+0
soft_slot-lr0.0003	0	1 ended/success	-	-	-
soft_slot-lr0.001	0	1 ended/success	-	-	-
soft_slot-lr0.003	0	1 ended/success	-	-	-
soft_slot-lr0.01-extension	0	1 ended/success	-	-	-
twelve-arm-32step-smoke	0	1 ended/success	-	-	-

Per-attempt records: rl run show E-0009.

Table 10: E-0010 V2 M 线单跳记忆读取：2×2×2 析因与 C_dual 探针: descriptive means over usable attempts

arm	seeds	attempts	H3.eligible	H3.qualified	H4.kernel_wins	H4.soft_win
V2-01	0	1 ended/success	-	-	-	-
V2-02	0	1 ended/success	-	-	-	-
V2-03	0	1 ended/success	-	-	-	-
V2-04	0	1 ended/success	-	-	-	-
V2-05	0	1 ended/success	-	-	-	-
V2-06	0	1 ended/success	-	-	-	-
V2-07	0	1 ended/success	-	-	-	-
V2-08	0	1 ended/success	-	-	-	-
V2-09	0	1 ended/success	-	-	-	-
V2-10	0	1 ended/success	-	-	-	-
analysis-regression	0	1 ended/success	-	-	-	-
factorial-analysis	0	1 ended/success	0 (n=1)	0 (n=1)	0 (n=1)	0 (n=1)

Per-attempt records: rl run show E-0010.

Table 11: E-0011 V3 T2 链终点任务生成器、登记与难度检查: descriptive means over usable attempts

arm	seeds	attempts	F.all.em	F.all.examples	F.all.nll	F.all
F-LoRA-pilot	0	1 ended/success	-	-	-	-
base-dev	173	1 ended/success	0.009 (n=1)	1000 (n=1)	5.568 (n=1)	0.011
base-smoke	173	1 ended/success	0.03333 (n=1)	60 (n=1)	5.771 (n=1)	0.033
full-data	20261029	1 ended/success	-	-	-	-
generator-smoke	20261029	1 ended/success	-	-	-	-
generator-smoke-r2	20261029	1 ended/success	-	-	-	-
pilot-selector-smoke	0	1 ended/success	-	-	-	-

Per-attempt records: rl run show E-0011.

Table 12: E-0012 V2 记忆失败层次定位（表示探针与寻址几何）: descriptive means over usable attempts

arm	seeds	attempts	L3_s	address_s	dev_question_tokens	elap
L0-L4-artifact-audit	0	1 ended/success	-	-	7000 (n=1)	0.77
L0-L4-cost	0	1 ended/success	555.5 (n=1)	111.3 (n=1)	-	57.9
L0-L4-full	0	1 ended/success	-	-	-	945
L0-L4-smoke	0	1 ended/success	-	-	-	25.0

Per-attempt records: rl run show E-0012.

Table 13: E-0013 V5 缩减版：tokenwise 简单接口对强 LoRA 的实用确认（PIQA+BoolQ 官方 validation）: descriptive means over usable attempts

arm	seeds	attempts	Base.elapsed_s	Base.validation.
V5-C	0,1,2,3	8 ended/success	-	-
V5-F	0,1,2,3	7 ended/success	-	-
V5-L	0,1,2,3	8 ended/success	-	-
base	0	1 ended/success	-	-
confirmation-lock-preflight	0	1 ended/success	-	-
confirmation-locked-analysis-audit	-	1 ended/success	-	-
locked-confirmation-base-and-nine-final	-	1 ended/success	375.7 (n=1)	0.6667 (n=1)
mig-profile-guard	-	1 ended/success	-	-
sealed-eval-dev-rehearsal	0	1 ended/success	-	-
sealed-eval-provenance-recovery-regression	0	1 ended/success	-	-
sealed-eval-regression	0	1 ended/success	-	-
v5-cost-tabulation	1	1 ended/success	-	-
v5-final-latency	1	1 ended/success	-	-
v5-final-lr-lock	0	1 ended/success	-	-
v5-grid-report-regression	0	1 ended/success	-	-
v5-initial-grid-report	0	1 ended/success	-	-
v5-latency-map-regression	0	1 ended/success	-	-
v5-latency-smoke	0	1 ended/cancelled	-	-
v5-slice-report-regression	-	1 ended/success	-	-
v5-three-arm-smoke	0	1 ended/failed, 1 ended/success	-	-

Per-attempt records: rl run show E-0013.

Table 14: E-0014 记忆路线修复筛选：事实数、课程、先验吸收、寻址监督与去中心化余弦寻址: descriptive means over usable attempts

arm	seeds	attempts	R1.em_margin	R1.target_weight	R2.em_margin	R2.targ
R1	0	1 ended/success	-	-	-	-
R2	0	1 ended/success	-	-	-	-
R3	0	1 ended/success	-	-	-	-
R4	0	1 ended/success	-	-	-	-
R5	0	1 ended/success	-	-	-	-
R6	0	1 ended/success	-	-	-	-
analysis	0	1 ended/success	-0.007 (n=1)	0.06122 (n=1)	0.012 (n=1)	0.09126
checks	0	1 ended/success	-	-	-	-
checks-rv0040	0	1 ended/success	-	-	-	-
data	0	1 ended/success	-	-	-	-

Per-attempt records: rl run show E-0014.

Table 15: E-0015 记忆路线诊断：value/解码侧与寻址侧的分离（E-0014 checkpoint 上只做评估）: descriptive means over usable attempts

arm	seeds	attempts	analyze_exit	mig_instance_hours	node_hours	pytest_ex
analysis	0	1 ended/success	0 (n=1)	-	-	-
checks	0	4 ended/success	-	-	-	0 (n=2)
diag-R1	0	1 ended/success	-	0.1726 (n=1)	0.04316 (n=1)	-
diag-R2	0	1 ended/success	-	0.1779 (n=1)	0.04448 (n=1)	-
diag-R5	0	1 ended/success	-	0.1755 (n=1)	0.04387 (n=1)	-
diag-V2-03	0	2 ended/success	-	0.1544 (n=1)	0.0386 (n=1)	-
reload-diag	0	1 ended/success	-	-	-	-

Per-attempt records: rl run show E-0015.

Table 16: E-0016 记忆路线训练筛选：解码器先行、槽位边际似然、单独编码写入（顾问轮次 1 的三条路线）: descriptive means over usable attempts

arm	seeds	attempts	D1.curve_seconds	D1.evaluation_second
D1	0	1 ended/success	-	-
D1-D2a-D2b-D3-final-analysis	0	1 ended/success	-	-
D1-D3-interim-analysis	0	1 ended/success	-	-
D2a	0	1 ended/success	-	-
D2b	0	1 ended/success	-	-
D3	0	1 ended/success	-	-
D4	0	1 ended/failed, 1 ended/success	-	-
D5	0	1 ended/success	-	-
checks	0	3 ended/success	-	-
protocols-default-regression-smoke	0	1 ended/success	2.927 (n=1)	3.664 (n=1)
shared-hooks-smoke	0	1 ended/success	3.179 (n=1)	3.945 (n=1)
smoke	0	1 ended/failed, 1 ended/success	-	-

Per-attempt records: rl run show E-0016.

Table 17: E-0017 记忆路线：E=1 预训练加单独编码写入（D5 配方）的 seed 复现: descriptive means over usable attempts

arm	seeds	attempts	arms_completed	compare_exit	dump_new_exit	dump_old_exi
C1	1	1 ended/success (1 not usable: spec mismatch)	-	-	-	-
C2	2	1 ended/success (1 not usable: spec mismatch)	-	-	-	-
P1-A	1	1 ended/success	1 (n=1)	-	-	-
P1-B	1	1 ended/success (1 not usable: spec mismatch)	-	-	-	-
P2-A	2	1 ended/success	1 (n=1)	-	-	-
P2-B	2	1 ended/success (1 not usable: spec mismatch)	-	-	-	-
checks	0	4 ended/success	-	0 (n=2)	0 (n=2)	0 (n=2)

Per-attempt records: rl run show E-0017.

Table 18: E-0018 V3 前置：单独编码写入下 T2 链终点任务的读取迁移、无训练深度扫描与循环实现检查: descriptive means over usable attempts

arm	seeds	attempts	eval_smoke_exit	mig_instance_hours	node_hours
D5@ATT-0159-1	0	2 ended/success	-	0.05046 (n=2)	0.01261 (n=2)
D5@ATT-0172-1	1	1 ended/success	-	0.1068 (n=1)	0.0267 (n=1)
D5@ATT-0173-1	2	1 ended/success	-	0.07832 (n=1)	0.01958 (n=1)
E0-tests	-	3 ended/success	0 (n=1)	-	-

Per-attempt records: rl run show E-0018.

Table 19: E-0019 RB0：残差旁路线实现与计时（因果前缀 KernelMatrix、T0、四层 R_res/concat、侧支历史干预）: descriptive means over usable attempts

arm	seeds	attempts	RB1_core_node_h	benchmark.exit_code	cases.co
RB0-D0026-probe	0	1 ended/failed	-	-	-
RB0-D0027-probe	0	1 ended/success	-	0 (n=1)	-
RB0-RB5-cost-layouts	0	1 ended/success	-	-	11 (n=1
RB0-core	0	1 ended/success	-	-	-
RB0-cost-report	0	1 ended/success	2.444 (n=1)	-	-
RB0-full-timing	0	1 ended/success	-	0 (n=1)	-
RB0-model-probe	0	1 ended/failed	-	-	-

Per-attempt records: rl run show E-0019.

E-0020 RBc：残差旁路线四个训练家族的学习率校准（LoRA4 r7、T0、R_res+KernelMatrix、C_dual+KernelMatrix
: no runs registered.
E-0021 RB1：残差旁路低维因果 linear attention 首轮筛选（plan2 R1 八组 + C_up 核对）: no runs registered.

Table 20: E-0022 V3：T2 链终点上的循环矩阵（共同起点阶段 + plan §9.2 十组，单 seed 筛选）: descriptive means over usable attempts

arm	seeds	attempts	arm_smoke_failures	intervene_smoke_exit	pytest_exit	sta
V3-S0	-	1 queued/unknown	-	-	-	-
tests-smoke	-	3 ended/success	0 (n=2)	0 (n=2)	0 (n=2)	0 (

Per-attempt records: rl run show E-0022.

Evidence (level, validity):

- EV-0001 [diagnostic, valid] Allocated environment probe passed on mg1098 GH200 MIG 2g.24gb; driver615.71.09; outbound HTTPS available. No adapter correctness evidence yet.; scope: Execution environment only [code review: ATT-0001-2 not needed: Environment discovery only; no model or core operator execution]
- EV-0002 [diagnostic, valid] 54/60 checks passed; six failures all at suffix-causality comparison including two padded rows. Cache comparisons before assertion passed. Pinned eager masking uses finite minimum then softmax, so padded query outputs need Base diagnosis; no hypothesis conclusion.; scope: Implementation checks with pending padding-test diagnosis [code review: ATT-0003-1 not needed: Deterministic E0 invariant checks before code review; no training or costly run]

- EV-0003 [diagnostic, valid] 61/61 V0 tests passed on GH200 MIG 2g.24gb, fixed FP32 and Base-calibrated BF16 tolerances. Zero exits match Base; two-step grads and frozen backbone checks pass; kernel output/QKV gradient and constrained-fusion identities, causality on valid tokens, cache, padding/reset, ownership, counts, M-line access and export/reload pass. Base-only diagnostic confirms previous failures were padded query logits (13.443359375 vs valid-prefix 0). Code review RV-0001 pending; no training or mechanism claim. (supports H-0001@1) [code review: ATT-0004-1 not needed: Deterministic E0 checks and Base padding diagnosis; unchanged production code and tolerances]

- EV-0004 [diagnostic, valid] RV-0001 fixes: 79/85 pass; six new positive-control failures measure rank32 effects only .359-.781 and swapped support effect .383, below detection thresholds based on unchanged BF16 calibration. Strengthen only nonzero test exits to demonstrate sensitivity before recheck; no production initialization change.; scope: Positive-control fixture adequacy [code review: ATT-0005-1 not needed: Deterministic checks of RV-0001 fixes before recheck; no training]

- EV-0005 [diagnostic, valid] 85/85 expanded V0 checks pass after RV-0001 fixes: real mutation controls rejected, delimiter support mask/capacity and stale-cache/missing-memory guards pass, masked/reset/empty kernel gradient identities pass, observed read accounting and configured key epsilon verified, actual fusion hook gradient identity passes. Nonzero fixture effects1.84375-6.890625 and swapped-owner effects3.53125-3.6484375 exceed unchanged BF16 thresholds. No training or mechanism claim; recheck pending. (supports H-0001@1) [code review: ATT-0006-1 not needed: Deterministic review-fix checks with stronger positive-control fixture; fixed production init and tolerances]

- EV-0006 [diagnostic, valid] V1 initial smoke:96 invariant tests passed and three20-step train/eval/reload paths passed;11586 train tokens. Training loss reduction subsequently superseded by coordinator MSG-0025; smoke establishes engineering compatibility/timing only, not LR or interface evidence. [code review: ATT-0007-1 not needed: deterministic regressions and 20-step three-family smoke before code review]

- EV-0007 [diagnostic, valid] Corrected equal-example V1 smoke:98 invariant tests passed including loss reduction and failed-LR continuation;three5-step training/evaluation/reload paths passed with exact reload equality.2943tokens,116s,.032222MIGh. Engineering diagnostic only; no LR selection or interface comparison. [code review: ATT-0008-1 not needed: corrected loss and divergence invariant regression plus5-step three-family smoke before review]

- EV-0008 [diagnostic, valid] RV0003fix smoke:104 invariant tests passed, including nongold NaN/Inf evaluation, retained failed-LR usage/continuation, Base nonfinite and finite reload mismatch errors, strict smoke acceptance.3families completed5steps and exact reload,2943tokens,114s,.031667 No calibration or interface claim. [code review: ATT-0009-1 not needed: RV-0003 failure-path fixes deterministic checks and5step smoke before recheck]

- EV-0009 [diagnostic, valid] Reload classification fix:106checks passed, including finite initial evaluation then actual NaN/Inf candidate guard during reload -> partial engineering error retaining usage without failed-LR flags. Existing initial-evaluation failure and finite reload mismatch behaviors pass.3families complete5steps/exact reload;2943tokens,119s,.033056MIGh=.008264nodeh. [code review: ATT-0010-1 not needed: RV-0004 reload error classification regression before coordinator-required new review round]

- EV-0010 [diagnostic, valid] base-eval complete finite calibration NLL 2.3312495373933837, calibration accuracies PIQA 0.72802734375 BoolQ 0.689453125; dev PIQA 0.723 BoolQ 0.68; exact reload passed. Base no training. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0011-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0011 [diagnostic, valid] cal-lora-1e-4 complete finite calibration NLL 1.6253322842401003, calibration accuracies PIQA 0.7255859375 BoolQ 0.78955078125; dev PIQA 0.723 BoolQ 0.766; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0012-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0012 [diagnostic, valid] cal-lora-3e-4 complete finite calibration NLL 1.6021037854622986, calibration accuracies PIQA 0.728515625 BoolQ 0.80419921875; dev PIQA 0.733 BoolQ 0.78; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0013-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0013 [diagnostic, valid] cal-fout-1e-4 complete finite calibration NLL 1.6600830964101538, calibration accuracies PIQA 0.71923828125 BoolQ 0.7958984375; dev PIQA 0.723 BoolQ 0.777; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0014-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0014 [diagnostic, valid] cal-fout-3e-4 complete finite calibration NLL 1.6157966926032628, calibration accuracies PIQA 0.72119140625 BoolQ 0.79833984375; dev PIQA 0.725 BoolQ 0.78; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0015-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0015 [diagnostic, valid] cal-cdual-1e-4 complete finite calibration NLL 1.6392950516932028, calibration accuracies PIQA 0.72607421875 BoolQ 0.7919921875; dev PIQA 0.722 BoolQ 0.767; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0016-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0016 [diagnostic, valid] cal-cdual-3e-4 complete finite calibration NLL 1.6024291552247223, calibration accuracies PIQA 0.72705078125 BoolQ 0.798828125; dev PIQA 0.728 BoolQ 0.777; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0017-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0017 [diagnostic, valid] Deterministic E0002 audit selects3e-4 allfamilies; relative NLL gaps1.4499%,2.7408%,2.3006%; slice gate passes. E3 full train audit finds exactly1PIQA row id5528 candidates537/537tokens (.007654% of13065), BoolQ0/6379. Not in calibration prefix. E3 filter authorized MSG0058, production implementation deferred next task. No H0002 interface claim. [code review: ATT-0019-1 not needed: deterministic selection audit and identification of overlength E3 examples; unchanged data/encoding rules]

- EV-0018 [diagnostic, valid] Diagnostic-only post-shuffle/pre-interleave exclusion of PIQA5528 under MSG0058 yields E3 schedule13077steps,2000040tokens,26154examples,epoch index1;

dev steps3270/6539/9808. Production filter not implemented; next task must implement/review and verify same schedule. Costs are family-based projections,28layer LoRA only conservative unmeasured bound. [code review: ATT-0020-1 not needed: cost/schedule estimate only under coordinator MSG0058; production filter and review deferred next task]

- EV-0019 [diagnostic, invalid] ATT-0021-1: 116 passed, one new audit fixture incorrectly expected success for 100 percent excluded train rows; actual guard correctly rejected >1 percent. No arm trained. Fixture repaired to exactly 1 percent without changing production guard. PBS21s, 0.005833333333333333 MIGh / 0.0014583333333333333 nodeh.; scope: Engineering regression fixture failure, not scientific evidence [code review: ATT-0021-1 not needed: Pre-review regression and twenty-step smoke for all eight arms]

- EV-0020 [diagnostic, valid] 126 tests passed; all eight arms match Base at initialization and reload exactly after20steps; parameter counts match contract, intermediate evaluation preserves training trajectory in regression, identical full 13077step/2000040token schedule with one excluded PIQA5528 row and points3270/6539/9808. LoRA28 measured steady .435291s/step. Cost .122778 MIGh=.0306945nodeh,30896inputtokens; not interface-effectiveness evidence.; scope: Implementation readiness only, eight twenty-step smokes on 32 dev examples/task [code review: ATT-0022-1 not needed: Pre-review regression and twenty-step smoke after fixture fix; revised E3 analysis]

- EV-0021 [diagnostic, valid] 145tests passed with0failures/errors/skips on e55eea3, including actual ATT0022 smoke rejection, duplicate/aliased paths, incorrect arm/config/schedule/data/order/paramet membership, and complete correctlymapped fixture acceptance. Existing paired bootstrap/selection arithmetic and V0/V1 suite preserved. PBS28s=.007778MIGh=.0019445nodeh; no trainingtokens.; scope: RV0007 formal-input validation regression only [code review: ATT-0023-1 not needed: Regression of RV-0007 validation fix before recheck]

- EV-0022 [diagnostic, valid] Rerun Base full1000/task dev and reload passed. PBS comparison validated formal Base input and exact equality of all2000 prediction records including scores/NLL, dev metric dictionary and split map againstATT0011-2. PIQA.723 BoolQ.680 macro.7015 devNLL2.3374779898760703. Base307s=.085278MIGh=.0213195nodeh; comparator1s=.000278MIGh=.0000695nodeh; no trainingtokens.; scope: Full untrained Base dev rerun and exact E-0002 comparison [code review: ATT-0024-1 covered by RV-0008 pass; ATT-0033-1 not needed: Cheap deterministic equality comparison using reviewed Base reader; no training or selection]

- EV-0023 [diagnostic, valid] E-0004 deterministic readiness checks: 130/130 V0, V1, and added state cases pass. Chunk3 FP32 state max_abs_err r32=0 and r128=3.814697265625e-6 across all three backends, inclusive/strict, S=1/4; chunk1 diagnostics r32=0.17662876844406128 and r128=0.2529714107513428. Capacity boundary 2 accepted and overflow at 1 rejected for all backends. Per-component state corruption controls rejected. Fixed tolerances unchanged. Code review pending; no training or mechanism claim. (supports H-0001@1) [code review: ATT-0034-1 not needed: Deterministic E0 readiness checks; review requested after validation per task contract]

- EV-0024 [diagnostic, valid] Read-only probe succeeded on mg1098/PBS 3760163.opbs: 169 run registry rows, 129 distinct referenced sample paths with zero locally available; nine committed sample archives contain episode/query IDs and predictions but no support facts or question text. Content-level overlap remains unverified. Pinned Qwen tokenizer confirms

semicolon ID 26 and space-semicolon ID 2587. Measured PBS 9s = 0.0025 MIG-instance-h = 0.000625 node-h; usage imported from executor observation.; scope: Legacy public assets at a255e814740ff85ab3b4ec4836c88f0a8f802b2d only; no claim about H-0003 or content-level overlap. [code review: ATT-0035-1 not needed: Read-only data schema and tokenizer smoke; no model evaluation]

- EV-0025 [diagnostic, valid] 123 core/model/M-line tests passed in 46.37 seconds on registered GH200 MIG job. Eight-arm M-line Base identity, two-step gradient startup, owner reorder and snapshot isolation, offline controls/oracles and hand-computed read metrics passed. Extended smoke and code review pending.; scope: Pipeline correctness only; no support or challenge to H-0003 [code review: ATT-0036-1 not needed: Deterministic protocol checks and two-step gradient smoke before code review; no costly run]

- EV-0026 [diagnostic, valid] Eight PBS regression tests passed; 80 smoke identities and canonical contents unique; independent regeneration SHA256 matched; 92 seen and 89 unseen single-leading-token names. Legacy pinned source/registry checks establish zero labeled fact-set+question overlap by disjoint relation construction (MSG-0093), not by enumerating legacy episode content. Full dataset generation and Base evaluation pending.; scope: Generator and metric code smoke only; no Base results or H-0003 claim. [code review: ATT-0037-1 not needed: Deterministic generator and metric regression smoke before code review; no Base model evaluation]

- EV-0027 [diagnostic, valid] Eight two-step copy-fixture arms completed, 672 training input tokens, application elapsed 9.26s. Stable step about 0.128-0.172s, 244-328 tokens/s; initialization max_abs=0. Two-step positive-control normal-vs-zero/shuffled criterion not yet met; planned longer smoke follows.; scope: Timing only; two updates are not the positive-control completion test [code review: ATT-0038-1 not needed: Two-step eight-arm timing smoke after 123 passing invariant checks, before review]

- EV-0028 [diagnostic, valid] All eight arms completed 32 smoke updates, all have normal-memory answer NLL below zero and shuffled; 10752 input tokens, 42.53s application elapsed, 303-317 input tokens/s. Validates learnability on tiny copy positive control only.; scope: Two-example copy fixture only; not evidence supporting H-0003 or single-hop generalization [code review: ATT-0040-1 not needed: Task-authorized 32-step tiny copy-fixture smoke before review; measured pilot 9.26s, expected under one minute, no formal training]

- EV-0029 [diagnostic, valid] Full V2 dataset generated twice with matching hashes and 21432 unique episode identities and semantic contents: train16384/calibration2048/dev1000/sealed confirmation2000. 8 tests pass, delimiter and single-token answer checks pass. Train has1381843 input tokens per epoch, 4145529 in three epochs, covering4M per arm. Constant prior dev EM0.015. Legacy overlap0 by authorized disjoint-relation construction proof, not empirical episode enumeration. RV-0010 code review passes with zero findings.; scope: Full generator/data contract validation; no Base or memory-mechanism claim. [code review: ATT-0039-1 not needed: Deterministic data materialization after 8 passing PBS invariants; approximately 30s; no model evaluation]

- EV-0030 [diagnostic, valid] Exact commit 94fb78b integration passed 124 regression tests, eight four-update E16 smoke arms, checkpoint roundtrip and independent train/calibration probe path. 5248 smoke forwarded input tokens; 672 protocol-gradient tokens separately reported. 800-959 tokens/s, training-only V2 extrapolation 2.424 node-hours; two-backend

two-LR calibration 0.0588-0.0653 node-hours. Tiny sample is not scientific validation.; scope: E-0005 provisional interface and timing; excludes shuffled address fields pending RV-0011; no H-0003 conclusion [code review: ATT-0041-1 not needed: Coordinator MSG-0106 authorizes provisional E-0005 integration smoke while RV-0010/RV-0011 pending. Four updates per eight arms, plus split probe and regression checks; no formal training.]

- EV-0031 [diagnostic, valid] Reviewed frozen Base evaluator ran with finite full-vocabulary metrics. M32 EM0 NLL9.32865; F32 EM0 NLL5.91752. Measured model load0.849s, M0.287s, F0.164s; used for full-dev walltime sizing, without changing data or protocol.; scope: 32-example dev-prefix timing smoke only; not a task-suitability estimate. [code review: ATT-0042-1 covered by RV-0010 pass]

- EV-0032 [diagnostic, valid] Full dev fixed-protocol Base: M EM0.011 NLL9.2494026874; F EM0.013 NLL5.7063963076; train-selected constant prior EM0.015. M<=prior+0.05 passes; F>=prior+0.20 fails (required0.215). Per E-0005 outcome_actions this is task-difficulty information for V2 design, not a reason to tune the protocol or a memory-mechanism refutation. Review RV-0010 passed; metrics/predictions complete and finite. Qwen3-0.6B-Base pinned revision, BF16 eager, no training.; scope: Frozen Base dev1000 M/F bounds only. Does not support or challenge H-0003 memory-read hypothesis; no confirmation evaluation. [code review: ATT-0043-1 covered by RV-0010 pass]

- EV-0033 [diagnostic, invalid] Five of six focused checks passed; SoftSlot regression hit exact NLL equality using fused mean CE vs evaluator none-then-mean CE, despite identical logits. Difference 4.77e-7. Corrected test to use defined evaluator reduction without changing production code or tolerance.; scope: Test-reference reduction mismatch; not mechanism evidence [code review: ATT-0044-1 not needed: Focused deterministic regression for RV-0011 medium diagnostic fix before recheck; no training]

- EV-0034 [diagnostic, valid] Six focused checks passed; both backends omit shuffled target-dependent address metrics and preserve entropy/neff. Distinct facts and differing targets with owners[1,0,1] produce identical diagnostic/unhooked logits and defined NLL/EM. Original-target read errors agree within1.27e-7 overall and4.47e-8 shuffled. Normal/permuted address metrics unchanged. PBS3760300.opbs exit0,12s.; scope: RV-0011 diagnostic correction only, no training [code review: ATT-0045-1 not needed: Focused deterministic RV-0011 regression rerun with matching CE reduction; no training]

- EV-0035 [diagnostic, valid] Post-merge regression on commit 3ccde7d: 155 tests passed, zero failures/errors/skips, 46.759s pytest, 49s PBS GH200 MIG; 0.013611 MIG h = 0.00340275 node h. No hypothesis inference. [code review: ATT-0046-1 not needed: post-merge regression]

- EV-0036 [diagnostic, valid] Initial diagnostic smoke: 13 tests passed, then strict E16 original-path NLL check failed. Independent PBS audit reproduced CPU versus CUDA CE reduction difference on identical logits (max 0.00006485 first M batch), zero top1 disagreements in M/F smoke. This is implementation diagnostic only; smoke scientific summaries not produced. Fix preserves original CUDA batch scoring without tolerance changes. [code review: ATT-0047-1 not needed: diagnostic Base evaluation, seconds; reviewed with outputs before delivery; ATT-0048-1 not needed: bounded diagnostic of failed smoke outputs]

- EV-0037 [diagnostic, valid] Fixed 32-calibration smoke: 13 tests passed; diagnostic E16 exactly matches original batch8 predictions and NLL. Prefix-only batch1/8 F top1 agreement 0.96875

and NLL MAD 0.10860 miss stability thresholds; this is timing/smoke evidence, full2048 comparison still required. Application6.635s. [code review: ATT-0049-1 not needed: diagnostic Base evaluation, seconds; reviewed with outputs before delivery]

- EV-0038 [diagnostic, valid] Full calibration2048: diagnostic E16 exactly matches original batch8 EM/NLL. Implementation gate fails: F batch1/8 NLL MAD0.088144 exceeds0.05 (top1 agreement99.2676% passes99%); M passes both. E1 F NLL1.421515 below M9.233152 passes. Per E0007@2 first outcome, investigate implementation and do not use F results for V2 design yet. Descriptive full-vocab EM M0.8301%, F E16 1.4160%, F E1 66.9922%. No H0003 inference; independent code/output review pending. [code review: ATT-0050-1 not needed: diagnostic Base evaluation, seconds; reviewed with outputs before delivery]
- EV-0039 [diagnostic, valid] MSG0135 follow-up after strict BF16 gate failure: true FP32/eager first256 calibration, TF32 off, batch1/8 top1 agreement100% M/F; NLL MAD M3.98e-6 F9.56e-6 versus same-prefix BF16 M0.01325 F0.10252. Supports BF16 numerical batch/padding sensitivity rather than an observed scoring/masking defect in tested path. Full BF16 F absDeltaNLL correlation padding +0.2240, prompt length -0.2165;347 unpadded rows zero difference. Saved12288 records/6 groups audited. Strict original @2 failure retained; MSG0135 diagnostic interpretation allows weak E16 retrieval branch (candidate-b14.209%<50%). No H0003 inference; review pending. [code review: ATT-0051-1 not needed: coordinator MSG0135 authorized bounded FP32 diagnosis; reviewed with outputs before delivery]
- EV-0040 [screening, valid] V1-01 F_out: dev PIQA 0.738, BoolQ 0.818, macroNLL 1.5767698148424294. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0025-1 covered by RV-0008 pass]
- EV-0041 [screening, valid] V1-02 F_pre: dev PIQA 0.73, BoolQ 0.811, macroNLL 1.567339824867386. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0026-1 covered by RV-0008 pass]
- EV-0042 [screening, valid] V1-03 C_gate: dev PIQA 0.734, BoolQ 0.808, macroNLL 1.5777984531278557. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0027-1 covered by RV-0008 pass]
- EV-0043 [screening, valid] V1-04 C_up: dev PIQA 0.736, BoolQ 0.818, macroNLL 1.5652449947787976. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0028-1 covered by RV-0008 pass]
- EV-0044 [screening, valid] V1-05 C_dual: dev PIQA 0.739, BoolQ 0.824, macroNLL 1.5619847145143861. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0029-1 covered by RV-0008 pass]

- EV-0045 [screening, valid] V1-06 linear: dev PIQA 0.735, BoolQ 0.825, macroNLL 1.5582456389768808. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0030-1 covered by RV-0008 pass]

- EV-0046 [screening, valid] V1-07 LoRA4: dev PIQA 0.748, BoolQ 0.836, macroNLL 1.5496020224121514. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0031-1 covered by RV-0008 pass]

- EV-0047 [screening, valid] V1-08 LoRA28: dev PIQA 0.749, BoolQ 0.832, macroNLL 1.5535729848258581. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0032-1 covered by RV-0008 pass]

- EV-0048 [screening, valid] Reviewed PBS analysis selects Fstar=F_out77.80%, Cstar=C_up77.70%. Best concat Cdual78.15% exceeds Fstar by0.35pp<1pp, hence concat_not_win. Cdual exceeds Cgate by1.05pp (fixed flag true), but nonlinear-vs-linear difference only0.15pp95CI[-.70,1.00]. Cstar trails LoRA4 by1.50pp95CI[-2.40,-.60] and LoRA28 by1.35pp[-2.50,-.20]. All8complete/exactn schedule2,000,040tokens. 10000paired stratified bootstrap seed20261029 descriptive only. Challenges broad H0002 superiority in this setting; no hypothesis-state change or V2 conclusion. Full tables and limitations in T-0003-results.md. (challenges H-0002@1); scope: E-0003@2 tokenwise F-line proxy, seed0, full fixed dev1000/task, final checkpoint; screening only [code review: ATT-0025-1, ATT-0026-1, ATT-0027-1, ATT-0028-1, ATT-0029-1, ATT-0030-1, ATT-0031-1, ATT-0032-1, ATT-0052-1 covered by RV-0008 pass]

- EV-0049 [diagnostic, valid] Verified shared V2 schedule on sealed train: order hash 0d4b55ed8655d91cf0d0532b 0.2M=>300 steps,200170 tokens,warmup15; 4M=>6000 steps,4000470 tokens,warmup300. Engineering schedule only, no hypothesis evidence. [code review: ATT-0053-1 not needed: Deterministic schedule preflight only, no training]

- EV-0050 [diagnostic, valid] 146 broad core/model and V2 checks passed in60.43s, source f7c4d2b, including twelve-arm zero-exit/protocol, exact checkpoint reload, fixed donor composition and microbatch mean-CE equivalence. No scientific conclusion. [code review: ATT-0055-1 not needed: Deterministic regression tests before review and calibration]

- EV-0051 [diagnostic, valid] F-line smoke:11 tests passed9.39s;four actual training steps2681tokens, fresh checkpoint reload32calibration+32dev exact per-row match; independent shared sample-order hash and 300/6000step counts match researcher-2. Smoke is not positive-control evidence. Earlier ATT0054-1 cwd failure cost3s and retry24s imported; combined0.001875nodeh. [code review: ATT-0054-2 not needed: Correct cwd for same four-step smoke]

- EV-0052 [diagnostic, valid] All12 arms32step smoke complete; zero-exit logits exact Base, final checkpoint dev predictions identical after reload, controls/curves finite; formal sample schedule independently matches F line. Timing only, not H0003 evidence. [code review: ATT-0056-1 not needed: Twelve-arm bounded smoke before code review;146 regression checks passed]

- EV-0053 [diagnostic, valid] Three300step200170token F calibrations finite/exact reload on2048calibration. LR1e-4/3e-4/1e-3 NLL0.02508849/0.00909726/0.00160201 EM99.414/99.805/99.951percent. Best at upper endpoint requires one3e-3 extension. Calibration signals context task learnability; formal4M and dev remain pending, no H0003 inference. [code review: ATT-0058-1, ATT-0059-1, ATT-0060-1 not needed: E0008 permits 200k calibration after tests and smoke;20tests and real4step smoke passed ATT0057;formal4M awaits review]

- EV-0054 [diagnostic, valid] Reporting-only RV0017 max aggregation fix passed focused regression,1 selected test; no model/metric-selection behavior changed. PBS-generated12-arm forecast1467-1544s per6000step arm including full final eval/curves/reload, requests2201-2316s. Forecast only. [code review: ATT-0068-1 not needed: Focused low-severity RV0017 reporting fix regression and arithmetic timing forecast]

- EV-0055 [diagnostic, valid] Required single upper endpoint3e-3 completed300steps200170tokens,calibrationNI reload. Reviewed PBS selector used calibration only across four candidates and finalized1e-3 (NLL0.00160201). No further extension;formal4M pending. [code review: ATT-0067-1 covered by RV-0018 pass]

- EV-0056 [diagnostic, valid] Six initial backend x LR calibration runs completed successfully,300steps/200170tokens each, all manifests validated. Selection pending registered arithmetic analysis ATT0070-1. Diagnostic LR calibration only; does not support or challenge H0003. [code review: ATT-0061-1, ATT-0062-1, ATT-0063-1, ATT-0064-1, ATT-0065-1, ATT-0066-1 not needed: E0009 explicitly allows calibration after tests/smoke while review RV0017 is pending;146tests and12arm smoke passed; rerun affected calibration if review changes path]

- EV-0057 [diagnostic, valid] Initial grid finite and verified300steps/200170tokens/reload. Kernel NLL5.138646/4.704182/4.627133 and soft5.130321/4.682995/4.627491 for LR3e-4/1e-3/3e-3. Both select upper endpoint0.003; sealed E0009 rule triggers single extension0.01. No H0003 conclusion. [code review: ATT-0070-1 not needed: Deterministic arithmetic analysis of already reviewed training results; no model execution]

- EV-0058 [diagnostic, invalid] T-0010 generator smoke: 9 PBS invariant tests pass; 480 rows independently reproduce with 480 unique identities and contents, zero raw and normalized V2 overlap, verified legacy relation-label disjointness and passing smoke shortcut gates. Not a full dataset or hypothesis test.; scope: E-0011 implementation smoke only; no model evaluation; no H-0005 support/challenge [code review: ATT-0073-1 not needed: deterministic smoke and invariant checks before code review; ATT-0060-1 not needed: E0008 permits 200k calibration after tests and smoke;20tests and real4step smoke passed ATT0057;formal4M awaits review]

- EV-0059 [diagnostic, valid] T-0010 generator smoke: 9 PBS invariant tests pass; 480 rows independently reproduce with 480 unique identities and contents, zero raw and normalized V2 overlap, verified legacy relation-label disjointness and passing smoke shortcut gates. Corrects the result-link recording error in invalid EV-0058.; scope: E-0011 implementation smoke only; no model evaluation; no H-0005 support/challenge [code review: ATT-0073-1 not needed: deterministic smoke and invariant checks before code review]

- EV-0060 [diagnostic, valid] Both prescribed LR0.01 endpoint extensions completed with passing artifacts at300steps/200170tokens. Final selection uses all8 candidates in registered ATT0074-1; no further extensions prescribed. Diagnostic only. [code review: ATT-0071-1, ATT-0072-1 covered by RV-0017 issues, RV-0019 pass]

- EV-0061 [diagnostic, valid] Final sealed-rule choice after single endpoint extension: KernelMatrix LR0.01 normal calibration NLL4.584628655; SoftmaxSlot LR0.003 NLL4.627490530. SoftLR0.01 rawNLL4.606242569 differs by<0.5%, so smallerLR wins. All8 candidates finite. Diagnostic calibration only; no H0003 claim. [code review: ATT-0074-1 not needed: Deterministic final calibration selection using reviewed selector; no model execution]

- EV-0062 [diagnostic, valid] Final reviewed F-line seed0 LoRA6000steps4000470tokens: calibration2047/2048 EM99.95117% vs prior1.90430% passes +20pp gate;dev1000/1000. Nonself2020/2021 and987/987,zero nonselfsubjectcopy. Exact full checkpoint reload;PBS savedoutput audit matches all summaries/counts and curve checkpoints. Single cal error NLL31.875/rank486 explains increased mean NLL0.0155656. Diagnostic context-task learnability only;neither supports nor challenges H0003. Joint E0010 release requires E0009 finite LRs and coordinator decision. [code review: ATT-0069-1 covered by RV-0018 pass; ATT-0076-1 not needed: Readonly deterministic saved-output audit; no inference or training]

- EV-0063 [diagnostic, valid] E-0011@2 rotation repair: 10 PBS tests pass, inspecting actual terminal edges in all 12 hop/order/noise combinations and the expected 83/84 dev cell sizes; all six hop-by-order ignore-relation scores equal 0.5 in calibration and dev smoke. 480 records reproduce byte-identically with zero V2 overlap and passing shortcut gates.; scope: RV-0020 regression repair; 60 rows per slice, implementation evidence only; no model evaluation or H-0005 support/challenge [code review: ATT-0075-1 not needed: deterministic smoke and regression checks for RV-0020 fix before re-check]

- EV-0064 [diagnostic, valid] Full E-0011@2 T2 preparation passes 10 tests and all data gates: train 18432, calibration 2048, dev 1000, sealed confirmation 2000 and four OOD slices of 1000. All 27480 identities/semantic contents are unique and independently reproduce; zero raw/normalized V2 overlap and legacy label disjointness rechecked. Three training epochs cover 5527605 prompt tokens. No model evaluation yet.; scope: V3 data engineering and structural shortcut gates only; no support/challenge of H-0005 [code review: ATT-0077-1 covered by RV-0022 pass]

- EV-0065 [diagnostic, valid] Base evaluator runtime smoke completes on 60 released V3 dev episodes with finite full-vocabulary scores and hop/order strata, M EM 0 and F EM 0.0333333, before full dev evaluation. Smoke is timing/engineering evidence only.; scope: E-0011 Base evaluator runtime and startup validation, not full difficulty result [code review: ATT-0079-1 covered by RV-0022 pass]

- EV-0066 [diagnostic, pending] T-0011 analysis implementation:21 synthetic PBS regressions passed in2.06s,0 failures/errors/skips. Covers H3/H4 threshold and probe rules, strict V3 near-ties, exact shared bootstrap draws, factorial normalization and formal-artifact corruption rejection. RV-0023 pass0 findings. PBS3762004.opbs on mg1098 exit0,3s walltime=0.000833 MIG-h=0.00020825 node-h. No formal training efficacy or hypothesis conclusion tested.; scope: Source9145309f16231d52bb121118779c986e875951e5; analysis-only synthetic checks; full ten-arm CLI and10000 real-artifact replicates remain pending [code review: ATT-0078-1 not needed: Low-cost synthetic analysis regression before analysis review; no training or scientific decision run]

- EV-0067 [diagnostic, valid] Full 1000-episode V3 Base dev: M EM 0.0, NLL 9.5821453819; F EM 0.009, NLL 5.5676069493, subject-copy 0.079; train-selected dev prior 0.011. M is within prior+5pp overall and for every hop. Per-hop/order outputs retained. Difficulty diagnostic

only, no hypothesis support/challenge.; scope: E-0011@2 frozen M/F Base on released dev only; full-vocabulary metrics [code review: ATT-0090-1 covered by RV-0022 pass]

- EV-0068 [diagnostic, valid] Integrated V3 F-line selector smoke: all 26 PBS tests pass, including unchanged V2 defaults/failure boundaries plus V3 release pins, phase-order seeds, sealing and stratum joins. Real V3 LoRA completes four steps/3158 prompt tokens at LR1e-3 and exact fresh-Base adapter reload on 32 calibration/dev rows. Engineering readiness only, not the full 0.2M pilot.; scope: E-0011 pilot selector and checkpoint smoke; no H-0005 support/challenge [code review: ATT-0091-1 not needed: four-step engineering smoke before scoped review of V3 data selector; common F-line engine previously reviewed RV-0018]

- EV-0069 [diagnostic, valid] Fixed V3 F-line LoRA pilot completes 251 steps and 200281 support+question tokens at LR1e-3. Full2048 calibration: EM0.92529296875, NLL0.3153590976. Hop1 EM0.9297218, hop2 EM0.9604685, hop3 EM0.8856305; hops2-3 pooled EM0.9230769 vs prior0.0205128. Both per-hop and pooled +20pp gates pass. Fresh-Base adapter reload exactly matches all2048 per-episode records and file hashes. No dev or sealed model evaluation in this pilot.; scope: E-0011@2 context-retained T2 learnability under fixed 0.2M pilot; not evidence for H-0005 or permission to start M-line V3 [code review: ATT-0092-1 covered by RV-0022 pass, RV-0024 pass]

- EV-0070 [screening, valid] E-0010@2 locked H3 output: challenged, screening only, seed0 final dev1000. No arm qualifies: max normal-minus-max(zero,shuffled) EM margin is+0.6pp vs required10pp; all target weights are below0.125. Per-arm inputs in V2 order (EM normal/zero/shuffled %, NLL normal/zero/shuffled, target weight):01 [0.9/1.1/1.9;4.427822/9.248186/4.427827;0
[1.7/1.1/1.5;4.448453/9.248186/4.449490;0.059004];03 [1.6/1.1/1.2;4.422075/9.248186/4.423024;0.053784];04
[1.9/1.1/1.3;4.442024/9.249403/4.441777;0.054220];05 [0.8/1.1/1.5;4.427029/9.248186/4.426557;0.058403];06
[1.8/1.1/1.2;4.446569/9.248186/4.447419;0.057738];07 [1.7/1.1/1.1;4.425355/9.248186/4.423455;0.055640];08
[2.0/1.1/1.5;4.445451/9.248186/4.447843;0.058278];09 [2.1/1.1/1.5;4.610641/9.249403/4.609989;0.059095];10
[1.8/1.1/1.9;4.497515/9.248186/4.499404;0.052307]. Qualified0/10; eligible0/10; V3 selection none. Released E-0008 positive control passed, ATT-0069-1. ATT-0093-1/RES-0094 full real-artifact validation and10000 paired bootstrap replicates passed, PBS3762550.opbs on mg1098 exit0/5s. Normal EM0.8-2.1% and value-oracle EM1.1-2.3%: contract qualitative oracle rule points to value-writing/fusion diagnosis, without isolating the cause or ruling out addressing. Confirmation unread. (challenges H-0003@1); scope: E-0010@2 final checkpoints, ten fixed arms including C_dual, dev1000,4M budget/6000steps/4000470tokens each, seed0 screening; not confirmation or a general impossibility claim; coordinator owns conclusion review and hypothesis decision [code review: ATT-0080-1, ATT-0081-1, ATT-0082-1, ATT-0083-1, ATT-0084-1, ATT-0085-1, ATT-0086-1, ATT-0087-1, ATT-0088-1, ATT-0089-1 covered by RV-0019 pass (RV-0023 pending); ATT-0093-1 covered by RV-0023 pass]

- EV-0071 [screening, valid] E-0010@2 locked H4 output inconclusive: SoftmaxSlot better0/4 cells, KernelMatrix better0/4; neither reaches3/4. No supports/challenges flag. Four cell inputs kernel->soft (normal EM%; target weight; n_eff; normalized read error): direct/F_out [0.9->1.6;0.060240->0.053784;9.969244->1.110847;0.165140->2.492829]; direct/C_up [1.7->1.9;0.059004->0.054220;6.274531->1.033493;0.354737->0.488782]; qknorm/F_out [0.8->1.7;0.058403->0.055640;5.618480->2.537960;0.392928->0.289613]; qknorm/C_up [1.8->2.0;0.057738->0.058278;4.044499->5.774581;0.433630->0.245538]. Soft EM is higher in all four cells, but none also has both higher target weight and lower n_eff. C_dual probes excluded. Reader normal-EM factorial effect+0.50pp, paired bootstrap95% interval[-0.05,+1.05]pp; intervals descriptive,10000 draws seed20261029, same

episode indices across arms/controls. This does not establish a useful read operator or explain differences by matrix compression.; scope: H-0004 four feature-by-fusion cells only F_out/C_up, final dev1000, seed0 screening; strict simultaneous rule unchanged; hypothesis status remains coordinator-owned [code review: ATT-0080-1, ATT-0081-1, ATT-0082-1, ATT-0083-1, ATT-0084-1, ATT-0085-1, ATT-0086-1, ATT-0087-1, ATT-0089-1 covered by RV-0019 pass (RV-0023 pending); ATT-0093-1 covered by RV-0023 pass]

- EV-0072 [diagnostic, valid] E-0012 L0-L4 measurements complete and independently audited: sixL0 controls96.6278%-100% (min31663/32768; question2048/2048). L1 delimiter-tail direct32648/32768,qknorm30433/32768; last-question supervised address acrossr128/256/512/1024 direct1804-1847/2048,qknorm1684-1756/2048. L2 trained value-tail direct8151-12148/32768,qknorm5556-8615/32768 versus random12609/32768,10539/32768. L3clean-to-clean341-472/2048,read-to-read37-47/2048,clean-to-read38-62/2048 acrossten checkpoints. L4 every trainedarm last-question argmaxslot0 in1000/1000 dev episodes, target52/1000,same-relation249/1000; key cosine means0.888802-0.999020 with120000pairs/arm; last fusion/residual mean18.6638-1.12365e6 withn1000/arm under stated increment definitions. RV-0027 pass; exact120probe/10L3/14L4 entries; independent counts/baseline/selection/hash/manifest/histogram checks passed; five earlier regressions passed. TotalE0012=1071MIGs=.074375nodeh. Descriptive diagnostic evidence only; no information-absence, causal or route conclusion.; scope: E-0012@1 fixed V2 T1 seed0; train16384 with14336 fit/2048 internal selection, calibration2048, dev1000; L3train4096/calibration2048 ridge1; no confirmation, M-line training, hypothesis update or route selection [code review: ATT-0096-1 covered by RV-0027 pass; ATT-0097-1 not needed: Deterministic independent artifact validator; no extraction, fitting, training, evaluation, split or metric change; measurement core covered by RV-0027]

- EV-0073 [diagnostic, valid] V5 three-arm engineering smoke passed:36 V1/V5 regressions;39 optimizer steps per arm;FP32 trainable counts524288/524288/630784;exact zero-exit Base logits and fresh-Base dev reload for all arms;three epoch-end monitoring points. Full plan19443 rows x3=58329 presentations,29165 steps,4173300 input tokens; same seed0 order across arms. This is implementation evidence only, not hypothesis support.; scope: E-0013@3 code f4b2c19;GH200 MIG2g.24gb on mg1098/PBS3764329;dev32 per task for smoke, no official validation. Stable step seconds C0.1990/F0.1950/L0.4083. Measured181MIGseconds=0.01256944 node-h;launcher failure separate2seconds. [code review: ATT-0098-2 not needed: Low-cost V5 regression and end-to-end smoke before code review; no LR selection or formal training]

- EV-0074 [diagnostic, invalid] 8 focused latency regression tests pass on2250925: synthetic LoRA merge weight update/wrapper removal;null,empty,whitespace,non-string checkpoint path rejection;both checkpoint files required before loading. RV-0035 pass. No full256 timing measured yet.; scope: Engineering only;PBS3764470 mg1100,10MIGseconds=0.00069444 node-h;no task-data or trained checkpoint inference. [code review: ATT-0110-1 not needed: Focused low-cost regression before recheck RV-0034;no full latency benchmark or confirmation data; ATT-0099-1 not needed: Cheap synthetic regression, dev-only software rehearsal and one Base WikiText timing before review; no validation access, training or selection]

- EV-0075 [diagnostic, valid] 8 focused latency regression tests pass on2250925: synthetic LoRA merge weight update/wrapper removal;null,empty,whitespace,non-string checkpoint path rejection;both checkpoint files required before loading. RV-0035 pass. No full256 timing measured yet. Corrects the administrative result-link error in invalid EV-0074.; scope: Engineering only;PBS3764470 mg1100,10MIGseconds=0.00069444 node-h;no task-data or trained check-

point inference. [code review: ATT-0110-1 not needed: Focused low-cost regression before recheck RV-0034;no full latency benchmark or confirmation data]

- EV-0076 [diagnostic, valid] 4 initial-grid report regressions passed on87bbb59:raw-minimum extension before tie choice,incomplete/mismatched budgets rejected,unflagged null NLL rejected,explicit failed LR with partial usage accepted. RV0038 pass. This does not yet execute report on actual grid results.; scope: Synthetic manifest fixtures only;PBS3764578 mg1098 Exit0,10MIGseconds=0.00069444 node-h;no official validation or model inference. [code review: ATT-0114-1 not needed: Focused low-cost regression for RV-0037 low finding before recheck;no selection from actual results or training]

- EV-0077 [diagnostic, valid] All9 initial seed0 LR calibrations completed with finite NLL,matched1101steps/220 Reviewed report requires3e-3 extension for all3arms;initial best1e-3 is provisional,not a final lock. Calibration only;no support/challenge claim.; scope: ATT0117 retains automatic stale review flag:per coordinator MSG0363,executed report/trainer exactly match RV0038/RV0033;stale entries were T0014 files absent from old branch and never imported. Master merge5cad3e restores covered status for all3endpoint registrations;byte-identical checks passed. Initial9 actual5996MIGseconds=0.41638889node-h;report10seconds separate. [code review: ATT-0101-1, ATT-0102-1, ATT-0103-1, ATT-0104-1, ATT-0105-1, ATT-0106-1, ATT-0107-1, ATT-0108-1, ATT-0109-1 covered by RV-0033 pass, RV-0034 issues; ATT-0117-1 stale: research/artifacts/T-0014/sealed-manifest.json changed since RV-0039; research/notes/T-0014-RV-0036-fixes.md changed since RV-0039; research/notes/T-0014-lock-schema.md changed since RV-0039; 8 more]

- EV-0078 [diagnostic, valid] Untrained Base dev1000 per task:PIQA accuracy0.723 NLL2.750281812879043;Bo accuracy0.680 NLL1.9246741668730973;task-mean NLL2.3374779898760703. Fresh Base repeat produced exactly identical per-example scores/predictions and metrics. No official validation read.; scope: Same-device dev repeat onGH200 MIG2g.24gb,mg1100/PBS3764504;344MIGseconds=0.023 h. Diagnostic baseline/noise only. [code review: ATT-0113-1 covered by RV-0033 pass, RV-0034 issues, RV-0035 pass (RV-0036 pending, RV-0037 pending)]

- EV-0079 [diagnostic, valid] Required3e-3 endpoints all finite and paired1101steps/200103tokens;all12 calibration manifests passed reviewed lock CLI. Final LR1e-3 for C,F,L;exact lock committed before any seed1–3 run. LockSHA25670ba5356203afc128cbf3e4c14a7b4474ece54a55d28b8401b852d957069e01 No dev or official-validation selection.; scope: E0013@3;lock source5cad3e,committed config2b5c809. Endpoint2047MIGseconds=0.14215278node-h;lock14seconds separate. Diagnostic selection/provenance only. [code review: ATT-0118-1, ATT-0119-1, ATT-0120-1, ATT-0128-1 covered by RV-0033 pass, RV-0034 issues, RV-0035 pass, RV-0037 issues, RV-0038 pass, RV-0039 pass]

- EV-0080 [screening, valid] R1 (train E=1): value path usable. E=1 dev normal/zero/shuffled correct 836/9/10 → margin +82.6 pp (R1 gate ≥50 pp: pass; normal−zero 95% CI 80.3–85.0 pp); NLL 0.538 vs 9.265/18.229. Addressing parameters get no gradient at E=1, so on E=16 dev: 4/11/6 correct (−0.7 pp), NLL 18.74 vs 9.25/18.90, target weight 0.061, last-position argmax hits target 4.9%; H3 fail. Fusion/residual at last position ≈18.5×.; scope: E-0014@1 seed 0 final checkpoint, M-line T1 hop 1, E=16 dev n=1000 (ATT-0039-1); single-seed screening; CIs are paired-bootstrap descriptive only [code review: ATT-0122-1, ATT-0138-1 covered by RV-0041 pass]

- EV-0081 [screening, valid] R2 (train E=4): E=4 dev correct 82/0/16 → +6.6 pp, NLL 3.687 < 5.955, target weight 0.371 → R2 gate fail (needs ≥10 pp and ≥0.5). E=16 dev 23/11/8 →

+1.2 pp (normal−shuffled CI 0.5–2.6 pp), NLL 6.222 < 6.724, target weight 0.091 → H3 fail. Last-position argmax picks a same-relation fact 99.8% of the time (target 24.2%): addressing keys on relation, not head entity.; scope: E-0014@1 seed 0 final checkpoint, M-line T1 hop 1, E=16 dev n=1000 (ATT-0039-1); single-seed screening; CIs are paired-bootstrap descriptive only [code review: ATT-0123-1, ATT-0138-1 covered by RV-0041 pass]

- EV-0082 [screening, valid] R3 (curriculum E 1→4→8→16 by step): E=16 dev 10/11/12 → −0.2 pp, NLL 4.4315 vs 4.4296 shuffled, target weight 0.052, last-position argmax slot 0 in 100% (n_eff 1.000): the V2-03 collapse recurs; H3 fail. Calibration monitors at segment ends: E=1 EM 15.3% (step 1000), E=4 1.6%, E=8 1.6%, E=16 2.1%. End-of-training E=1 dev 64/9/20 (+4.4 pp), E=4 dev 30/0/10 (+2.0 pp).; scope: E-0014@1 seed 0 final checkpoint, M-line T1 hop 1, E=16 dev n=1000 (ATT-0039-1); single-seed screening; CIs are paired-bootstrap descriptive only [code review: ATT-0124-1, ATT-0138-1 covered by RV-0041 pass]

- EV-0083 [screening, valid] R4 (prior path r32 trained 1000 steps at zero memory then frozen, verified unchanged): the prior path alone gives E=16 dev NLL 4.740 (vs 9.248 zero memory without it). After memory training: 6/11/12 → −0.6 pp, NLL normal 4.4303 = shuffled 4.4303, target weight 0.061, last-position argmax slot 0 100%; H3 fail. Absorbing the answer prior did not stop the slot-0 collapse.; scope: E-0014@1 seed 0 final checkpoint, M-line T1 hop 1, E=16 dev n=1000 (ATT-0039-1); single-seed screening; CIs are paired-bootstrap descriptive only [code review: ATT-0125-1, ATT-0138-1 covered by RV-0041 pass]

- EV-0084 [screening, valid] R5 (with address supervision: +1·(−log w_target) at last question position, extra supervision): E=16 dev 62/11/16 → +4.6 pp (normal−shuffled CI 3.1–6.2 pp), NLL 4.102 < min(9.248, 4.970) pass, token-mean target weight 0.143 ≥ 0.125 pass, EM margin < 10 pp → H3 fail. Last position: target weight 0.518, argmax = target 60.2%, same relation 100%. oracle_value EM only 10.7% (NLL 3.64), oracle_key 6.7%: even a correct read decodes poorly at E=16.; scope: E-0014@1 seed 0 final checkpoint, M-line T1 hop 1, E=16 dev n=1000 (ATT-0039-1); single-seed screening; CIs are paired-bootstrap descriptive only [code review: ATT-0126-1, ATT-0138-1 covered by RV-0041 pass]

- EV-0085 [screening, valid] R6 (centered cosine address, learnable tau init 10): E=16 dev 15/11/13 → +0.2 pp, NLL 4.4205 vs 4.4244 shuffled, target weight 0.060, n_eff 5.73, last-position argmax slot 0 86.3%; H3 fail. Fusion/residual at last position ≈41.7×.; scope: E-0014@1 seed 0 final checkpoint, M-line T1 hop 1, E=16 dev n=1000 (ATT-0039-1); single-seed screening; CIs are paired-bootstrap descriptive only [code review: ATT-0127-1, ATT-0138-1 covered by RV-0041 pass]

- EV-0086 [diagnostic, valid] ATT-0144-1 (V2-03) failed the E-0015 dev reload check (normal 51/1000, zero 12/1000, shuffled 37, oracle_value 11 mismatches; NLL max |diff| 0.14-0.23) because it ran on mg1098_mig3, which nvidia-smi reports as a MIG 1g.24gb device with 26 SMs (the ledger labels it 2g.24gb). ATT-0146-1 on the same slice, same code, evaluator settings identical to E-0010 (batch 16, right padding, episode order sha b52c83a6...): R1 also mismatches (normal 4, zero 12), V2-03 repeats the same 51/12 in-process, and the Base-only zero forward differs from the recorded zero predictions (12/1000). So BF16 kernel choice on the other MIG profile changes results; the checkpoint is not at fault. ATT-0144-2 on a 2g.24gb slice (guarded by nvidia-smi) reproduces all four recorded V2-03 dev controls exactly, as R1/R2/R5 (ATT-0141..0143) did. Bitwise cross-job reproducibility in this project needs 2g.24gb slices.; scope: E-0015 reload check; Miyabi debug-mig slice mg1098_mig3 [code review: ATT-0144-1

stale: src/v2_train.py changed since RV-0041 (RV-0042 pending); ATT-0146-1 not needed: diagnostic of the reload check / tests for the RV-0042 recheck; included in the recheck]

- EV-0087 [diagnostic, valid] R1 calibration EM normal E=1/4/16 = 84.1/1.1/0.5%. Contextual oracle_value 84.1/21.9/7.6%, high only for target fact 0 (E=16: 83.4% t0 vs 1.6% t>=1). Isolated-encoding iso_oracle_value 85.3% (E=4) and 86.4% (E=16; 83.4% t0, 86.7% t>=1); iso_wrong_oracle 1.5% vs answer prior 1.9%; iso_v/iso_kv E=16 2.1/2.5%. Probes: Pv context 74.5/47.9/37.6%, Pv isolated 77.6% (E=16), Pv iso->ctx 7.1% (E=16), Pf context 99.8/97.4/94.9%. Contextual values at fact positions >=1 have ~2x the L2 of position 0 / isolated values (E=16: 43.5 vs 103.1 (1-3), 89.2 (>=4)); frozen delimiter features do not change in L2 (~57). Sweep at E=1: alpha 0.1/0.3 EM 1.0/55.7%, non-final-only 82.4/83.6%. Rules 4 and 5 hold on R1.; scope: E-0015@4 diagnostics, checkpoint R1 (ATT-0122-1 final), calibration E=1/4/16 (2048 rows each); E=16 dev reload bitwise identical [code review: ATT-0141-1 stale: src/v2_train.py changed since RV-0041 (RV-0042 pending); ATT-0148-1 stale: src/v2_train.py changed since RV-0041]
- EV-0088 [diagnostic, valid] R2 calibration EM normal E=1/4/16 = 11.9/9.7/2.8%; oracle_value 11.9/9.5/4.1%; last_only - normal E=4 -1.5 pp; iso_v - normal +1.7 pp (E=4), +1.1 pp (E=16); iso_oracle_value 12.2/10.9% (E=4/16); uniform 3.7% (E=4). Probes: Pv context 71.1/45.9/38.8%, Pv isolated 75.2/75.3% (E=4/16), Pf context 99.8/97.4/94.9%, Pz normal 46.1/21.7/3.5%. Magnitude sweep at E=4: alpha 0.1/0.3 EM 0.9/5.7% (normal), 1.4/5.9% (oracle_value), 8.1/8.9% (non-final only), all below alpha=1. Contextual values keep one dominant coordinate (dim 63) at every position; L2 rises from 70.7 (pos 0) to 88.9 (pos >=4, E=16). Rule 1 does not hold (Pv 45.9% < 80%); rule 2 holds here (Pv < 50%, Pf >= 90% at E=4/16); rules 3, 6, 7 do not.; scope: E-0015@4 diagnostics, checkpoint R2 (ATT-0123-1 final), calibration E=1/4/16 (2048 rows each); E=4 and E=16 dev reload bitwise identical [code review: ATT-0142-1 stale: src/v2_train.py changed since RV-0041 (RV-0042 pending); ATT-0148-1 stale: src/v2_train.py changed since RV-0041]
- EV-0089 [diagnostic, valid] R5 calibration EM normal E=1/4/16 = 6.1/7.6/6.4%; oracle_value 6.1/7.2/10.4%; last_only - normal E=16 -0.9 pp; iso_v - normal -3.2 pp (E=4), -3.3 pp (E=16); iso_oracle_value 4.7/6.0% (E=4/16); uniform 2.1% (E=16). Probes: Pv context 69.4/45.8/45.8%, Pv isolated 74.8/75.1% (E=4/16), Pf context 99.8/97.4/94.9%, Pz normal 44.4/22.2/11.7%. Magnitude sweep at E=16: alpha 0.1/0.3 EM 0.5/3.7% (normal), 0.9/6.0% (oracle_value), 6.2/6.0% (non-final only). Values have L2 ~77-82 with one dominant coordinate (dim 27) at every position, contextual and isolated. Rule 1 does not hold (Pv 45.8% < 80%, oracle_value 10.4%); rule 2 holds here; rules 3, 6, 7 do not.; scope: E-0015@4 diagnostics, checkpoint R5 (ATT-0126-1 final), calibration E=1/4/16 (2048 rows each); E=16 dev reload bitwise identical [code review: ATT-0143-1 stale: src/v2_train.py changed since RV-0041 (RV-0042 pending); ATT-0148-1 stale: src/v2_train.py changed since RV-0041]
- EV-0090 [diagnostic, invalid] V2-03 is at chance under every control: calibration EM 1.0-2.2% for all controls at E=1/4/16 (normal 1.6/1.7/1.5%, oracle_value 1.6/1.7/1.3%, iso_oracle_value 1.7/1.3%), answer prior 1.4-1.9%. Probes: Pv context 69.3/38.4/32.9%, Pv isolated 73.7/74.3% (E=4/16), Pf context 99.8/97.4/94.9%, Pz normal 35.7/5.3/1.4%. Contextual values shrink with fact position (E=16 L2 109.7 at pos 0, 68.3 at 1-3, 44.4 at >=4); isolated values stay at 109.7. Rule 2 holds here (Pv < 50%, Pf >= 90% at E=4/16).; scope: E-0015@4 diagnostics, reference V2-03 (ATT-0082-1 final), calibration E=1/4/16 (2048 rows each); E=16 dev reload bitwise identical on a MIG 2g.24gb slice (ATT-0144-1 on a 1g slice mismatched, see EV-0086) [code review: ATT-0144-2, ATT-0148-1 stale: src/v2_train.py changed since RV-0041]

- EV-0091 [diagnostic, valid] V2-03 is at chance under every control: calibration EM 1.0-2.2% for every non-zero control at E=1/4/16 (normal 1.6/1.7/1.5%, oracle_value 1.6/1.7/1.3%, iso_oracle_value 1.7/1.3%), zero 0.7/0.9/0.8%, answer-prior EM 1.4/1.3/1.9%. Probes: Pv context 69.3/38.4/32.9%, Pv isolated 73.7/74.3% (E=4/16), Pf context 99.8/97.4/94.9%, Pz normal 35.7/5.3/1.4%. Contextual values shrink with fact position (E=16 L2 109.7 at pos 0, 68.3 at 1-3, 44.4 at >=4); isolated values stay at 109.7. Rule 2 holds here (Pv < 50%, Pf >= 90% at E=4/16). Replaces EV-0090.; scope: E-0015@4 diagnostics, reference V2-03 (ATT-0082-1 final), calibration E=1/4/16 (2048 rows each); E=16 dev reload bitwise identical on a MIG 2g.24gb slice (ATT-0144-1 on a 1g slice mismatched, see EV-0086) [code review: ATT-0144-2, ATT-0148-1 stale: src/v2_train.py changed since RV-0041]

- EV-0092 [diagnostic, valid] Hardware audit:two original calibration manifests report1g.24gb(C1e-3 ATT0103 and L1e-4 ATT0107),other14 completed model manifests report2g;old manifest hashes match the unchanged LR lock. New allocated-UUID guard passed live2g on mg1098 and7 regressions including1g masked by unrelated2g. No inference/timing method changed.; scope: D0015 preserves original valid LR rule/lock and9formal runs;two diagnostic2g sensitivity reruns authorized,not yet complete. Guard sourcefbf97b4,ATT0149/PBS3765777 Exit0,2MIGs. No retroactive nvidia-smi evidence claimed for prior jobs. [code review: ATT-0103-1, ATT-0107-1 covered by RV-0033 pass, RV-0034 issues; ATT-0149-1 not needed: Infrastructure fail-fast guard per coordinatorMSG0408;no model path or timing method changed;fixtures and live inventory only]

- EV-0093 [diagnostic, valid] Two completed duplicate calibration attempts imported as plain diagnostic records under D0018/MSG0472. No slice comparison,lock recomputation or LR-selection analysis performed. Original lr-lock.json is final and unchanged.; scope: Same original RunSpecs,seed0,1101steps/2202examples/200103tokens;execution and raw outputs only. Profiles metadata only;no sensitivity caveat and no retraining. [code review: ATT-0103-2, ATT-0107-2 not needed: D0015 slice sensitivity rerun;identical RunSpec/model code reviewed RV0033;no new model path. Tested guard fbf97b4 is launcher-only. Unexecuted T0014 files absent from original snapshot.]

- EV-0094 [diagnostic, valid] All six C/F seed1–3 training runs complete:29165steps,58329presentations,4173300 tokens each;all3epoch dev monitoring,final checkpoint/inventory and exact fresh-Base reload. Original LRlock hash matches;C/F training orders paired within each seed. Three LoRA runs remain in progress.; scope: Training/dev/provenance only,not confirmation evidence. Hardware profiles retained as metadata per D0018/D0019;no rerun or flag based on slice. Official validation not accessed. [code review: ATT-0129-1, ATT-0130-1, ATT-0131-1, ATT-0132-1, ATT-0133-1, ATT-0134-1 covered by RV-0033 pass, RV-0034 issues, RV-0035 pass, RV-0037 issues, RV-0038 pass, RV-0039 pass]

- EV-0095 [screening, valid] D5 passes the E-0010@2 H3 thresholds under the isolated write: EM normal 42.8% vs zero 1.1%, shuffled 1.7% (margin 411/1000 = 41.1 pp); NLL 2.514 < 9.248, 7.977; target weight all positions 0.162 (>= 0.125), last position 0.571. Passes in both strata: target 0 (n=52) EM 44.2, margin 38.5 pp, weight 0.140; target >=1 (n=948) EM 42.7, margin 41.2 pp, weight 0.163. Isolated permuted 2.4, oracle_key 24.5, oracle_value 73.6. Contextual-encoding normal (descriptive) 5.7% (t0 65.4 / t>=1 2.4), NLL 15.05. Curve EM normal 4.3/20.6/35.3/42.8 at 25/50/75/100%. Reload identical; R1 replay 0/1000 mismatches. Single seed screening. (supports H-0012@1); scope: E-0016@1 D5 (R1 ATT-0122-1 adapter-only continuation, 6000 steps, seed 0), isolated-encoding write protocol (new; reported separately

from E-0010 H3), E=16 dev 1000 episodes, code 415cac3 [code review: ATT-0159-1 stale: src/v2_train.py changed since RV-0041, RV-0045]

- EV-0096 [screening, valid] D4 does not meet the H3 thresholds under the isolated write: EM normal 1.4% vs zero 1.1%, shuffled 1.1% (margin 3/1000 = 0.3 pp); NLL normal 4.423 < zero 9.248 but not < shuffled 4.424; target weight all positions 0.058 (< 0.125), last position 0.063 with last-position n_eff 16.0 (uniform over the 16 slots). Fails in both strata (t0 n=52 margin 1.9 pp, weight 0.049; t>=1 n=948 margin 0.2 pp, weight 0.058). oracle_key 1.4, oracle_value 1.4, permuted 1.7. The memory acts as a content-independent bias (normal NLL = shuffled NLL). Train CE plateau ~4.41 from step ~2500; curve EM normal 1.0/1.0/1.8/1.4 at 25/50/75/100%. Contextual normal (descriptive) 2.2%. Reload identical; initial zero-exit = Base (0.0). Single seed screening.; scope: E-0016@1 D4 (from scratch, 6000 steps, seed 0), isolated-encoding write protocol (new; reported separately from E-0010 H3), E=16 dev 1000 episodes, code 415cac3 [code review: ATT-0158-2 stale: src/v2_train.py changed since RV-0041, RV-0045]
- EV-0097 [screening, valid] D1 seed0 supports the stage1 decoder portion of H0010 under explicit target-slot supervision: tf_normal668/1000 vs zero11 and tf_shuffled13,margin65.5pp passes the50pp point gate; tf_wrong8. OLS fitted decline0.052743nat,plateau=false. Standard all-slot normal23/1000; oracle_value668/1000. Complete6000steps,full1000dev and exact same-run reload; all checkpoint/audit checks pass. No supports/challenges edge or hypothesis verdict.; scope: E0016@1 D1 only; supervised teacher-forced stage1,seed0 screening; final-window LR max7.3896%peak,plateau label schedule-conditioned; descriptive intervals conditional on trained seed; no confirmation. [code review: ATT-0154-1 stale: src/test_v2r_diagnostics.py changed since RV-0043; src/v2r_diagnostics.py changed since RV-0043; src/v2r_reload_check.py changed since RV-0043 (RV-0046 pending); ATT-0169-1 stale: src/test_v2r16_isolated.py changed since RV-0046, RV-0048; src/test_v2r_diagnostics.py changed since RV-0043; src/v2_isolated.py changed since RV-0046; 3 more]
- EV-0098 [screening, valid] D2a seed0 addressing-only continuation from supervised D1: normal50/1000,zero11,shuffled18 =>3.2pp margin below10pp; normalNLL5.302961 below both controls; all-question target weight0.069221<0.125. Standard H3 point gate not passed. Last-position target weight0.161943 is descriptive. Full6000 continuation steps,full1000dev/E1,exact same-run reload and final artifact audit pass; decoder A_v/Ro/P remains bitwise equal to D1 and all three addressing tensors changed. No supports/challenges edge or hypothesis verdict.; scope: E0016@1 D2a seed0 screening; total recipe includes supervised D1 prewarm,adapter-only load,fresh optimizer/warmup and derived order; H3 uses token-weighted all-question positions; profile metadata only D0018/19; no confirmation. [code review: ATT-0162-1, ATT-0169-1 stale: src/test_v2r16_isolated.py changed since RV-0046, RV-0048; src/test_v2r_diagnostics.py changed since RV-0043; src/v2_isolated.py changed since RV-0046; 3 more]
- EV-0099 [screening, valid] D2b seed0 all-adapter continuation from supervised D1: normal48/1000,zero11,shuffled18 =>3.0pp margin below10pp; normalNLL4.349222 below both controls; all-question target weight0.086935<0.125. Standard H3 point gate not passed. Last-position target weight0.135142 is descriptive; oracle_value170/1000. Full6000 continuation steps,full1000dev/E1,exact same-run reload and final artifact audit pass. No supports/challenges edge or hypothesis verdict.; scope: E0016@1 D2b seed0 screening; recipe includes supervised D1 prewarm,adapter-only load,fresh optimizer/warmup and derived order; H3 uses token-weighted all-question positions; profile metadata only D0018/19; no confirmation. [code review: ATT-0163-1, ATT-0169-1 stale: src/test_v2r16_isolated.py changed since

RV-0046, RV-0048; src/test_v2r_diagnostics.py changed since RV-0043; src/v2_isolated.py changed since RV-0046; 3 more]

- EV-0100 [screening, valid] D3 seed0 latent-slot marginal training: hard last-only normal15/1000,zero11,shuffle =>0.4pp margin below10pp; normalNLL4.424336 below both controls; last-position target weight0.061145<0.125. Registered hard H3 point gate not passed. Soft normal16/1000,NLL4.425512;marginal normal9/1000,NLL4.424916. Complete6000steps,full1000dev and exact same-run reload; final audit passed. No supports/challenges edge or hypothesis verdict.; scope: E0016@1 D3 seed0 screening; no target-index training input; hard/soft/marginal use the new last-question-position protocol and are distinct from standard E0010 H3. Descriptive paired intervals only; profile metadata D0018/19; no confirmation. [code review: ATT-0155-1 stale: src/test_v2r_diagnostics.py changed since RV-0043; src/v2r_diagnostics.py changed since RV-0043; src/v2r_reload_check.py changed since RV-0043 (RV-0046 pending); ATT-0169-1 stale: src/test_v2r16_isolated.py changed since RV-0046, RV-0048; src/test_v2r_diagnostics.py changed since RV-0043; src/v2_isolated.py changed since RV-0046; 3 more]

- EV-0101 [diagnostic, valid] All nine final V5 training runs complete/passed at E0013@3, LR0.001,29165steps/58329examples/4173300inputtokens each; exact within-job saved checkpoint reload. PBS ATT0174 preflight authenticates all9 inventories/configs/weights and matching per-seed sample orders. Training/dev implementation evidence only.; scope: T0013 training/checkpoint readiness and descriptive dev/cost;original LR lock final;profiles metadata only;no official validation access,no H0008 supports/challenges,no slice comparison [code review: ATT-0129-1, ATT-0130-1, ATT-0131-1, ATT-0132-1, ATT-0133-1, ATT-0134-1, ATT-0135-1, ATT-0136-1, ATT-0137-1 covered by RV-0033 pass, RV-0034 issues, RV-0035 pass, RV-0037 issues, RV-0038 pass, RV-0039 pass]

- EV-0102 [screening, valid] P1 passes the E-0010@2 H3 thresholds under the isolated write: EM normal 49.7% vs zero 1.1%, shuffled 1.9% (margin 239/500 = 47.8 pp); NLL 2.279 < 9.249, 8.583; target weight all positions 0.162, last 0.610; last-position n_eff 1.97, most common last slot rate 0.076 (no fixed-slot collapse). Strata: t0 (n=52) EM 50.0, margin 48.1 pp, weight 0.152; t>=1 (n=948) EM 49.7, margin 47.8 pp, weight 0.162 - both pass. Isolated permuted 2.3, oracle_key 23.2, oracle_value 77.1. Stage A (ATT-0166-1, R1 recipe seed 1): E=1 dev EM normal 82.7 / zero 0.9 / shuffled 1.1 (R1 seed 0: 83.6). Descriptive: E=16 calibration isolated normal 49.5 / zero 0.8 / shuffled 1.6; contextual dev normal 4.9 (t0 67.3 / t>=1 1.5). Curve EM 6.0/22.7/41.1/49.7. Start adapter digest pinned (b5b2aec9...) and matched; replay of stage A's recorded dev predictions 4/1000 mismatches on a 1g slice (recorded only, D-0018/D-0019); reload identical. (supports H-0012@1); scope: E-0017@1, isolated-encoding write protocol (reported separately from E-0010 H3), E=16 dev 1000 episodes, code 7d53d5d (RV-0054), D5 recipe 6000 steps, single run per arm; P1: stage A R1 recipe seed 1 (ATT-0166-1) -> stage B continuation seed 1 (ATT-0172-1); replication arm [code review: ATT-0166-1 not needed: D-0020: RV-0049 found nothing on the stage-A path; v2_train.py/v2_schedule.py byte-identical 64331a4..f01b036; ATT-0172-1 covered by RV-0054 pass]

- EV-0103 [screening, valid] P2 passes the E-0010@2 H3 thresholds under the isolated write: EM normal 48.0% vs zero 1.1%, shuffled 1.8% (margin 231/500 = 46.2 pp); NLL 2.378 < 9.248, 8.241; target weight all positions 0.158, last 0.600; last-position n_eff 2.01, most common last slot rate 0.077. Strata: t0 (n=52) EM 50.0, margin 48.1 pp, weight 0.143; t>=1 (n=948) EM 47.9, margin 46.1 pp, weight 0.159 - both pass. Stage A (ATT-0167-1, R1 recipe seed 2): E=1 dev EM normal 81.1 / zero 0.9 / shuffled 0.9. Start adapter digest pinned (aa3919e2...)

and matched; replay 0/1000 mismatches; reload identical. (supports H-0012@1); scope: E-0017@1, isolated-encoding write protocol (reported separately from E-0010 H3), E=16 dev 1000 episodes, code 7d53d5d (RV-0054), D5 recipe 6000 steps, single run per arm; P2: stage A R1 recipe seed 2 (ATT-0167-1) -> stage B continuation seed 2 (ATT-0173-1); replication arm [code review: ATT-0167-1 not needed: D-0020: RV-0049 found nothing on the stage-A path; v2_train.py/v2_schedule.py byte-identical 64331a4..f01b036; ATT-0173-1 covered by RV-0054 pass]

- EV-0104 [screening, valid] C1 (descriptive) passes the H3 thresholds under the isolated write: EM normal 45.3% vs zero 1.1%, shuffled 2.3% (margin 43/100 = 43.0 pp); NLL 2.406 < 9.249, 8.205; target weight all 0.150, last 0.589. Strata: t0 EM 42.3, margin 34.6 pp, weight 0.131; t>=1 EM 45.5, margin 43.5 pp, weight 0.151 - both pass. Start adapter = trusted R1 digest 9ea30adb...; replay 4/1000 mismatches on a 1g slice (recorded only); reload identical. (supports H-0012@1); scope: E-0017@1, isolated-encoding write protocol (reported separately from E-0010 H3), E=16 dev 1000 episodes, code 7d53d5d (RV-0054), D5 recipe 6000 steps, single run per arm; C1 (descriptive, not in the replication verdict): R1 seed 0 (ATT-0122-1) -> continuation seed 1 [code review: ATT-0170-1 covered by RV-0054 pass]

- EV-0105 [screening, valid] C2 (descriptive) passes the H3 thresholds under the isolated write: EM normal 46.5% vs zero 1.1%, shuffled 2.3% (margin 221/500 = 44.2 pp); NLL 2.394 < 9.248, 8.245; target weight all 0.157, last 0.598. Strata: t0 EM 44.2, margin 38.5 pp, weight 0.152; t>=1 EM 46.6, margin 44.5 pp, weight 0.157 - both pass. Start adapter = trusted R1 digest; replay 0/1000 mismatches; reload identical. (supports H-0012@1); scope: E-0017@1, isolated-encoding write protocol (reported separately from E-0010 H3), E=16 dev 1000 episodes, code 7d53d5d (RV-0054), D5 recipe 6000 steps, single run per arm; C2 (descriptive, not in the replication verdict): R1 seed 0 (ATT-0122-1) -> continuation seed 2 [code review: ATT-0171-1 covered by RV-0054 pass]

- EV-0106 [diagnostic, valid] Final seed1 fixed256-dev latency complete on allocation-verified mg1098 with existing2g timing guard; all5 models have768timings. Independent PBS artifact tabulation verifies all3840 raw timings positive/finite and summary statistics match. Nine-checkpoint provenance and measured costs packaged.; scope: Descriptive cost and engineering readiness only;single fixed GPU/seed1 input workload;merged LoRA timing only. No official validation,no H0008 outcome claim,no slice comparison;original LR lock final. [code review: ATT-0174-1 covered by RV-0033 pass, RV-0034 issues, RV-0035 pass, RV-0037 issues, RV-0038 pass, RV-0039 pass, RV-0047 pass; ATT-0175-1 not needed: Bounded metadata/timing-file validation and descriptive table generation only;no model execution,no new training/eval/loading path,no scientific conclusion;source latency reviewed RV0035 and complete RES0172]

- EV-0107 [diagnostic, valid] D5 seed 0 on T2: S=1 implementation check exact vs reviewed evaluator (4 controls). Hop1 H3 fails: EM normal/zero/shuffled 9/0/6 of 334 (margin 0.9 pp), target weight all 0.036 (<0.125, below 1/16), last 0.054; NLL 7.74<8.81<9.58. oracle_value hop1 EM 75.7% (decoder transfers, address does not). S_eval 2/4/8: hops2-3 EM 3-4.5% unchanged, NLL rises, step-2 weight does not move to hop-2 fact (descriptive).; scope: E-0018@1 E1, isolated-write D5 adapter ATT-0159-1 (seed 0) on T2 dev (1000; hops 1-3), S_eval 1/2/4/8, no training; reviewed code 7f3b5a0 (RV-0057) [code review: ATT-0182-1 covered by RV-0057 pass]

- EV-0108 [diagnostic, valid] D5 seed 1 (P1) on T2: S=1 exact vs reviewed evaluator on T2

and T1 E=16 dev; T1 S=1 vs P1 recorded predictions 0/1000 mismatches, identical NLL (4 controls). Hop1 H3 fails: EM 10/0/4 of 334 (margin 1.8 pp), target weight all 0.037, last 0.056; oracle_value hop1 EM 80.2%. S_eval>1 no gain on hops 2-3; NLL rises with S. Ran on 1g slice (metadata only, D-0018).; scope: E-0018@1 E1, isolated-write D5 adapter ATT-0172-1 (seed 1, P1) on T2 dev (1000; hops 1-3), S_eval 1/2/4/8, no training; reviewed code 7f3b5a0 (RV-0057) [code review: ATT-0183-1 covered by RV-0057 pass]

- EV-0109 [diagnostic, valid] D5 seed 2 on T2: S=1 exact vs reviewed evaluator. Hop1 H3 fails: EM 15/0/5 of 334 (margin 3.0 pp), target weight all 0.038, last 0.069; oracle_value hop1 EM 76.3%. S_eval>1 no gain on hops 2-3 (2.7-3.9%), NLL rises with S, no step-2 shift toward the hop-2 fact (descriptive).; scope: E-0018@1 E1, isolated-write D5 adapter ATT-0173-1 (seed 2) on T2 dev (1000; hops 1-3), S_eval 1/2/4/8, no training; reviewed code 7f3b5a0 (RV-0057) [code review: ATT-0184-1 covered by RV-0057 pass]

- EV-0110 [confirmatory, valid] E0013@3 single authorized official-validation opening: C_up r32 vs all28/all7 rank1 LoRA seed-mean macro difference -0.417652pp,paired stratified10000-bootstrap95CI[-1.020109,+0.197062]pp. Upper bound<+1pp triggers only the sealed challenged edge; interval includes0,so this does not establish equivalence or LoRA superiority. Seed differences[-0.275412,-0.401297,-0.576248]pp;task means PIQA-0.743562pp,BoolQ-0.091743pp. Macro means C77.484174%,F77.250252%,L77.901827%. S1 C-F+0.233922pp,rawCI[-0.239300,+0.708736],HolmP0.3356;S2 F-L-0.651575pp,rawCI[-1.252136,-0.037860],HolmP0.0692;neither rejects afterHolm. WikiText deltaNLL C+0.200122,F+0.238017,L+0.336505 vsBase2.647376. All10 bundles authenticated,single complete receipt,PIQA1838/BoolQ3270,zero exclusions/deviations;exact full locked analysis reproduced from saved predictions. RV0039/RV0056 passed; no methods/lock/source/data/checkpoint changes and no inference rerun. Hypothesis status decision remains coordinator. (challenges H-0008@2); scope: H0008@2,E0013@3;Qwen3-0.6B-Base,F-line attention retained,3 fixed training seeds1/2/3,final three-epoch checkpoints,allLR0.001 selected by sealed short calibration protocol;C/F524288 parameters,L630784. Intervals represent data uncertainty conditional on these trained seeds;no seed resampling. +1pp/per-task-1pp are point gates;challenged is not equivalence. Secondary intervals raw,not Holm simultaneous;S1 budget comparison changes rank and interface together. LoRA conclusion limited to rank1 and this calibration/training recipe. Training/latency costs reused authenticatedT0013 records;LoRA mergeable and merged timing only,confirmation scoring unmerged. No zero-tuning-transfer or parameter-efficiency claim. Validation consumed once;cannot be reused to confirm modified methods. [code review: ATT-0179-1, ATT-0194-1 stale: research/artifacts/T-0020/confirmation-lock.json changed since RV-0056; research/artifacts/T-0020/preflight-allocation.json changed since RV-0056; research/artifacts/T-0020/preflight-result.json changed since RV-0056; 5 more]

5 Discussion

证据支持以下几点。第一，V0 侧分支在所测试的恒等式与隔离项上实现正确（diagnostic，D-0003）。第二，没有侧支历史的 tokenwise 接口没有显示出优势：V1 中 gate/up 条件融合没有超过加法写回 1 pp（screening，D-0004）；V5 在官方 validation 上，C_up 没有超过参数相近的 28 层 rank1 LoRA（confirmatory，D-0028）。第三，记忆路线的结果与这样一种解释相容：V2 的失败包含 value 的上下文迁移。上下文中编码的 value 只对第一条事实可解码；每条事实单独编码、并先在 E=1 上训练后，只用答案 CE、不用地址标签，E=16 的单跳读取在三个 seed 上都达到门槛（screening，D-0023）。在这个设置下单独编码本身不够：从零训练的 D4 失败，解码器先行后只用答案 CE 续

训（D2a、D2b）与槽位边际似然（D3）也没有学到寻址（D-0021、D-0022）。第四，在 T1 句式上学到的寻址没有迁移到 T2，解码器迁移了（diagnostic，E-0018）。

证据不支持以下推断：上下文编码的记忆可用（D5 的 adapter 在上下文编码写入下只有约 5% EM），所以单独编码是须声明的结构化输入条件；D5 的结论适用于 T1 一跳、E=16、V2-03 配置之外；任何关于多次读取（H-0005）、读算子选择性（H-0004）或带侧支历史的残差旁路（H-0013–H-0016）的结论，这些检验还在进行；tokenwise 接口在其他任务、层或训练长度上同样无效。除 V5 外，所有结果都是单 seed 或三个 seed 的筛选级证据。官方 validation 已经用掉，RB 线的确认改用 A2 集（D-0025）。

- D-0001 route:pause — 旧 S-RE 推理资产（writer.pt、sq2 loader 所需文件）不在本文件系统：/work/xg24i002/x10041/LoopLora 目录不存在，只有 GitHub 源码 a255e81 可读。用户 2026-11-01 选择按计划 fallback 跳过冻结 S-RE 诊断（plan §6.1），新方案直接从 Base 实现；不声称旧指标已复现。(reopen if: 用户提供 S-RE 推理资产（writer.pt 及 looplora.llm.sq2.load_inference 所需的归一化 buffers 与配置）的位置，并为该诊断分配 $\leq$1 node·h)

- D-0002 route:continue — V0 实现检查 85/85 通过（ATT-0006-1/RES-0006/EV-0005，GH200 MIG 2g.24gb，容差在适配代码前锁定且未改动）；代码审核 RV-0001 的问题全部修复，RV-0002 通过，两条 low 跟进（FP32 chunk3 状态比较、KernelMatrix 支持容量保护）只约束分块/缓存记忆实验与 V2，不影响 V1 tokenwise 路径。按用户授权的 V0+V1 里程碑进入 V1：先做 E-0002（PIQA/BoolQ 切片与三个家族的学习率校准），再按实测成本生成 E-0003 八组 release。H-0001 的证据状态暂不改为 supported，等 V1 阶段报告时一并做结论审核。V0 实际用量 0.045 MIG instance-h，0 训练 tokens。

- D-0003 route:continue->supported — V0 侧分支实现语义在测试范围内成立（diagnostic，E0）：E-0001 修正后 85/85 项检查通过（RV-0002 pass），E-0004 130/130 项通过（RV-0009 pass）。覆盖陈述中的全部恒等式与隔离项；prefill、分块与逐 token decode 的输出在预先锁定的 BF16 容差内一致；chunk3 FP32 记忆状态在 2e-5 内一致。逐 token（chunk1）记忆状态的 FP32 差异 0.18–0.25 只作诊断记录，不宣称相等；全 padding 的 query 行无语义约定。结论审核 RV-0014 通过、零发现。范围只是实现正确性，不涉及学习效果或机制价值。下一步：实现作为 V1/V2 的基础继续使用；用到逐 token 缓存记忆的实验前，先查清 chunk1 差异的来源。[conclusion review RV-0014 pass (same_family_unverified)]

- D-0004 route:pause->challenged — 按 E-0003@2 运行前固定的规则（RV-0008 代码审核通过；RV-0015 结论审核通过、零发现）：在 Qwen3-0.6B-Base、F 线、四层 r32 tokenwise proxy、预先锁定的家族校准配方（三族均为 3e-4）、seed 0、每任务 1,000 条 dev、最终 checkpoint 的条件下，最佳 concat（C_dual 78.15%）只比 F*（F_out 77.80%）高 0.35 pp，未达到 +1 pp 的筛选标准，记为 concat 未胜；所选 C*（C_up 77.70%）的观测准确率低于两个 LoRA（79.20%、79.05%）；非线性 proxy 对线性退化组只高 0.15 pp [-0.70, 1.00]。因此 H-0002 在该设置下记为 challenged（screening）。这不排除其他训练 seed、接口专属调参或总体上 $\geq$1 pp 的真实效应（最佳 concat 对 F* 的描述区间上限为 1.35 pp），也不否定有记忆的 concat（H-0003）。按契约，V2 保留 C_dual 作为交互探针，融合用 F*=F_out 与 C*=C_up。此决定不释放 V2 预算。(reopen if: V2 中有记忆的 C*（C_up）或 C_dual 交互探针明显优于 F*（F_out），或用户要求用多个 seed 复核 V1 筛选) [conclusion review RV-0015 pass (same_family_unverified)]

- D-0005 stage:continue_probe — V0+V1 里程碑完成，总用量 11.25 MIG-h $\approx$ 2.81 node·h（预算 200）。V1 没有发现 gate/up 融合相对加法写回的优势（H-0002 暂停），但 V2 的问题（有记忆的单跳读取，H-0003）不由无记忆的 V1 proxy 决定；按 E-0003 契约，V2 融合用 F*=F_out 与 C*=C_up，并保留 C_dual 交互探针。E-0007 显示 E=16 的 F 线对 Base 很难（问题主语复制 96%，限定候选 (b) 14.2%，随机 3.7%；E=1 时 EM 67%），所以建议

V2 分两步：先用训练后的 F 线正对照确认任务在上下文中可学，同时做记忆家族学习率校准；正对照达到预先锁定的水平后，再跑八组析因。V2 是否释放、预算多少由用户决定。

- D-0006 route:pause->inconclusive — E-0010@2 locked H4 rule: SoftmaxSlot better in 0/4 cells and KernelMatrix in 0/4, neither reaches 3/4. With no arm producing a usable episode-memory read (EV-0070), a read-operator comparison cannot separate selectivity from absent memory content; soft EM is higher in all four cells by under 1.2 pp but its target weights are lower in three. No further H-0004 runs until memory reads work. (reopen if: An M-line configuration produces a usable episode-memory read (H-0003 gate met, or the single plan 8.4 probe shows a signal), so the two read operators can be compared on working memory.)

- D-0007 route:diagnose->challenged — E-0010@2 locked rule: 0/10 arms qualify (max EM margin +0.6 pp vs 10 pp; every arm-mean target weight below 0.125 and 1/16; normal and shuffled NLL nearly equal) after the E-0008 positive control passed. Screening only: one seed, released configuration and 4M tokens per arm; not a general impossibility result. The learned-value oracle failure points at values and the downstream path without isolating the cause; LR/schedule, implementation and write-position alternatives are neither shown nor excluded. Per plan 8.4 the next step is E-0012 diagnostic measurement (no automatic mapping), then one preregistered and reviewed probe of at most 4M tokens or a reviewed stop for limited expected value. V3 stays closed. Conclusion review RV-0025 (medium finding on the E-0012 draft mapping) resolved in RV-0026 (pass). [conclusion review RV-0026 pass (same_family_unverified)]

- D-0008 route:stop — Expected-value stop of further spending on the new memory route (M line); H-0003 stays challenged (screening). Basis: E-0010 locked gate 0/10 released arms (EV-0070) and the E-0012 diagnostic measurements (EV-0072; L0 controls 96.6-100%). Observations (single seed, diagnostic): at the last question position the highest-weight slot (argmax of mixture weights) is slot 0 in 1000/1000 dev episodes for every trained arm, so its target rate equals the slot-0 base rate (52/1000); reads remain weighted mixtures (E-0010 mean n_eff 4.0-10.0 kernel, 1.0-5.8 soft); arm-mean target margins there are negative (soft -2.1 to -16.8, kernel -1247 to -3501); arm-mean fused-increment to layer-14 residual norm ratios are 18.7-608 (1.1e6 for V2-09). Under the L3 ridge probes (4,096 train episodes, calibration held-out) actual-read accuracy is near the prior (1.8-2.3%; clean-to-read 1.9-3.0%) and learned target values give 16.6-23.0%; in the separate L2 softmax probes trained value projections decode the tail no better than random-init projections (17.0-37.1% vs 32.2-38.5%). Frozen layer-14 features give 92.9-99.6% tail decoding and 82.2-90.2% supervised (target-labelled bilinear) addressing top-1 among 16 facts at the last question position across ranks 128-1024, with no consistent gain with rank. Normal and shuffled NLL are nearly equal (4.42-4.61, near the 4.39-nat train answer-prior entropy; zero memory 9.25), while normal is below permuted in V2-03 (-0.18) and V2-10 (-0.11 nats), so outputs are not fully content-independent. Interpretation (not established): predominantly prior-like adaptation through the memory path without demonstrated useful episode-specific retrieval; an optimization cause under answer-only CE is plausible but not isolated; information absence is not shown. Rationale: ten released arms failed; the diagnostics show coupled difficulties; no permitted rescue has demonstrated benefit and their expected value is judged too low to prioritize over the reduced F-line work. Accepted uncertainty: plan 8.2 temperature/gain calibration with a registered restart or paired continuation, S=2 training and wider value/fusion width remain untested and are not falsified; the single plan 8.4 probe (at most 4M tokens) stays available as a reopen path. Scope: one seed, 4M tokens per arm, answer-only CE from Base, layer 14, visible-delimiter writes, E=16,

released LRs. Conclusion review RV-0028 (two medium findings) resolved in RV-0029 (pass, two low wording corrections applied here). (reopen if: The user approves a change outside the plan 8.4 probes that targets the observed failure (for example fewer facts per episode or a curriculum, a separate path that absorbs the answer prior, or addressing supervision), or a reason appears to spend the single permitted probe (temperature/gain calibration with restart, S=2, or same-layout width; at most 4M tokens).) [conclusion review RV-0029 issues (cross_model_unverified)]

- D-0009 route:pause — V3/V4 test multi-read gains on a fixed memory that a single read can already use (plan 5.1: V3 only after a usable V2 read). No released V2 arm produced a usable single-hop memory read (EV-0070) and the memory route is stopped (D on H-0003, RV-0029). The T2 chain-end data contract and generator (E-0011, T-0010) are ready and kept for reuse. (reopen if: A working single-hop M-line memory read (H-0003 locked gate met, or a reopened memory route shows the gate signal).)

- D-0010 stage:stop — V2 delivers a stop of the memory route as an expected-value decision (D-0008, RV-0029): no released arm met the locked single-hop gate, diagnostics show a fixed top-ranked slot, read-out increments far above the residual scale and near-prior read decodability, while frozen features remain linearly informative. Per plan 5.1 and 14.2 the project continues with V5 reduced to a simple-interface comparison (V1 tokenwise F_out and C_up against a parameter-matched strong LoRA, F line, PIQA+BoolQ, three seeds, sealed validation opened once). Changes that target the observed failure but lie outside the permitted probes (fewer facts per episode or a curriculum, a separate prior-absorbing path, addressing supervision) are listed for the user; changing facts per episode is the user's decision. [conclusion review RV-0029 issues (cross_model_unverified), RV-0030 issues (cross_model_unverified)]

- D-0011 route:expand — Release the reduced V5 as one confirmatory experiment, E-0013@3 (DOC-0128@3), testing H-0008@2. Basis: plan 5.1/14.2 allow V5 to shrink to a simple-interface comparison after the memory route stops (D-0008 stop, D-0009 V3/V4 paused, D-0010 V2 stage stop). The V1 screening evidence EV-0048 (C_up trails LoRA4 r10 by 1.50 pp and LoRA28 r2 by 1.35 pp on dev, seed 0) sets modest expectations, so the experiment is a preregistered estimate rather than an expected win. Design: C_up r32 and F_out r64 (524,288 parameters each) against all-28-layer all-7 rank-1 LoRA (630,784) with seeds 1-3, exactly 3 epochs, the V1 example-mean objective, per-arm LR selection on calibration NLL with one endpoint expansion per side, and the final checkpoint only. The sealed official validation is opened once, after code reviews, a dev rehearsal and a lock manifest. Gates are fixed in E-0013. Intervals are conditional on the three trained seeds, and the +1 pp and per-task -1 pp rules are point-estimate gates. Budget at most 12 node-h (estimate 7.8; project total 4.2 of 200). Conclusion review RV-0031 found two issues (medium objective mismatch, low wording), both fixed in E-0013@2 and H-0008@2. Re-check RV-0032 passed; its low provenance finding was fixed in E-0013@3, which adds a tests relation to H-0008@2.

- D-0012 route:reopen — The D-0008 reopen condition is met: on 2026-11-01 the user approved out-of-plan changes that target the observed failure (fewer facts per episode or a curriculum, a separate prior-absorbing path, addressing supervision), including changes to the sealed data protocol through new registered data versions, and authorized all meaningful attempts without further approval (docs/constraints.md). H-0003 stays challenged at screening level for the released V2 recipe (EV-0070); that result is not rewritten. The reopened route tests modified training recipes as H-0009 (derived from H-0003) in the screening experiment E-0014: six

single-change arms on the V2-03 configuration, judged by the locked E-0010@2 H3 gate on T1 E=16 dev, plus an E=1 value-transport control. The memory route resumes beyond E-0014 only if a recipe meets the gate and replicates with new seeds; V3/V4 (D-0009) stay paused until then.

- D-0013 route:consult — E-0014 outcome: only R1 passed its value gate (E=1 dev +82.6 pp); no E=16 arm met H3; R2 failed its E=4 gate. This is the pre-registered branch ' 只有 R1/R2 达标 -> 试组合，需要时请顾问'. Three single changes (curriculum R3, prior path R4, centered cosine R6) reproduced the V2 slot-0 collapse, so combining them is unlikely to help. Diagnostics point beyond addressing: R2 on E=4 dev has last-position target weight 0.983 (argmax=target 100%) yet 8.2% EM; R5 oracle_value at E=16 is 10.7%; R1 oracle_value drops from 83.6% (E=1) to 5.8% (E=16). Surviving explanations (slow coupled address/value learning, mixed reads at non-final question positions, context shift of support features, oversized fusion exit) are not separated by existing tests, so one advice round (AD-0001, AD-0002) informs the next screening contract. H-0009 stays untested; V3/V4 stay paused (D-0012).

- D-0014 route:expand — Advice round 1 (AD-0001, AD-0002) agrees: continue one bounded round rather than pause. Both checked from per-episode files that one value map does not transfer between 'fact 0' and 'facts with a prefix': R1 oracle_value 45/52 at target position 0 vs 13/948 later; R5 the mirror image, 1/52 vs 106/948. No E=16 arm has trained its decoder on clean reads. Magnitude and clipping do not explain the failure: R1 succeeded at E=1 at about 9x residual with about 99.6% of steps clipped. Correction to EV-0083's last sentence: R4's prior path reached only NLL 4.74 at zero memory vs 4.42 for the collapse solution, so 'prior absorbed yet still collapses' is not shown. Next round: E-0015@4 (eval-only diagnostics on calibration, T-0016) plus E-0016 (training screen, sealed) testing three advice-derived hypotheses: H-0010 decoder first (D1 teacher-forced single-slot, D2a/D2b address learning from answer CE), H-0011 slot-marginal likelihood with last-position read (D3), H-0012 per-fact isolated writes (D4/D5, only if E-0015 rule 5 holds). Budget 0.25 + 0.9 node·h. Rejected candidates: plain R1 warm start with contextual writes, mixed E=1/E=16 batches, longer runs of the old arms, a magnitude cap as a standalone arm.

- D-0015 route:continue — V5 LR 校准的 12 条 run 中有 2 条在 MIG 1g.24gb 上（ATT-0103：C@1e-3，即 C 的选中值；ATT-0107：L@1e-4），其余在 2g.24gb（MSG-0421）。处理如下。(1) 原 lr-lock.json 保留为正式输入：它按登记规则（取 calibration NLL 在最小值 1% 以内的最小 LR）从有效 run 得出；E-0013 没有规定校准的硬件，1g 与 2g 的差别是 BF16 舍入级的数值差异，不是计算错误。(2) 敏感性检查（diagnostic）：用相同代码、配置与 seed 0，在 guard 确认的 2g 上重跑这两条，报告各自 calibration NLL 的差；用已审核的 lock CLI 在全 2g 的 12 条 manifest 上重算 lock，输出到单独文件，不覆盖 lr-lock.json。(3) 预先规定：重算结果与原 lock 相同，记为 LR 选择对切片稳健；若不同（只可能发生在 C：2g 下 C@1e-3 的 NLL 比 C@3e-3 的 1.6025 高出 1% 以上，即约大于 1.6185），也保留原 lock 与九个正式 run，不重训、不加 run，并在 V5 分析与 H-0008 结论中把 C 的 LR 选择列为对硬件数值敏感的限制。(4) 正式 run 不因训练所在切片重跑，每个 run 的硬件 profile 按最终 manifest 记录；validation 开封评估与延迟测量都在 guard 确认的 2g 上进行，九个 checkpoint 与 Base 的确认评估用同一种切片。

- D-0016 route:continue — Coverage exception for the E-0016 formal runs ATT-0154-1 (D1) and ATT-0155-1 (D3), and for D2a/D2b later run from the same reviewed source 6da33b3. run register marks them stale on src/test_v2r_diagnostics.py, src/v2r_diagnostics.py and src/v2r_reload_check.py. Those are E-0015 files that RV-0046 (the T-0018 review of the

same experiment) also lists. All three paths are absent from commit 6da33b3: the coordinator checked with git ls-tree and git grep that the T-0017 sources neither contain nor import them. So the runs cannot execute them. RV-0045 (pass, 0 findings) reviewed every path these runs execute, and ATT-0152-1 verified the final source. Decision: run without a re-check. A model re-check of files that the run does not contain cannot change what the run computes. The stale label stays on the attempts with this reason. It does not apply to T-0018 runs, which do execute the isolated-write code under RV-0046.

- D-0017 route:continue — Coverage of T-0018's D4 (ATT-0158-1) and D5 (ATT-0159-1) at 415cac3. run register marks them stale on src/v2_train.py. In that commit the file is master 8cccbb1 plus T-0017's train() hook. researcher-3 checked that the hook is byte-identical to the one RV-0045 reviewed. The rest of the file differs from 6da33b3 by T-0016's evaluate() change, which RV-0042 and RV-0043 reviewed and ATT-0140-1 regression-tested bitwise. The two hunks merged automatically and sit in different functions. All shared v2r16_* files are byte-identical to 6da33b3. Unlike D-0016, this file is executed, and no reviewer has seen this combination. So, instead of a model re-check, a deterministic check comes first. In one guarded 2g PBS job at 415cac3 (no training), run the full test suites of T-0016, T-0017 and T-0018: test_v2r_diagnostics, test_v2r16_analysis, test_v2r16_objective, test_v2r16_schedule, test_v2r16_memory, test_v2r16_isolated, plus T-0017's exact default-identity check. If everything passes, D4/D5 run at 415cac3 under this decision without a re-check, and the stale label stays with this reason. If anything fails, stop and report.

- D-0018 route:continue — User instruction 2026-11-01 (about 23:10 JST): **不要考虑 BF 的舍入这一点误差**. This supersedes D-0015 items (2) and (3). The V5 slice-sensitivity comparison and the all-2g lock recomputation are dropped. The two finished reruns (ATT-0103-2, ATT-0107-2) are kept only as plain diagnostic records. The original lr-lock.json is final, with no sensitivity caveat. Project-wide: BF16 rounding differences between MIG 1g.24gb and 2g.24gb slices are not treated as an error source. Nothing is rerun, analysed or flagged because of the slice it ran on, and the profile is recorded as metadata only. A cross-job bitwise mismatch that a different slice explains is not a failure. Existing 2g guards stay where they are already wired in, because they cost nothing and prevent false exact-match alarms. No new slice checks are added.

- D-0019 route:continue — Follow-up to D-0018. D4 (ATT-0158-1, PBS 3766228) failed in 1 s on its 2g guard. PBS placed it on mg1109_mig3, which is a 1g.24gb slice. Slot 3 on these nodes is 1g, so a fail-fast guard rejects a large share of placements. Under D-0018 that buys nothing: the remaining E-0016 reload checks run in the same job, so they use the same slice. Rule from now on: MIG guards record the profile and never fail the job. The one exception is a job that measures time for comparison (latency or throughput), which keeps fail-fast on 2g. A job that already failed only on the guard is resubmitted as a new attempt with the guard in record-only mode. Jobs already queued are not cancelled.

- D-0020 route:continue — E-0017 stage A (R1 recipe at seeds 1 and 2) may run at f01b036 before the stage-B review round closes. RV-0049 reviewed the stage-A seed path at 64331a4 (v2_train.py --seed: initialization and the appended-seed order rule; v2_schedule.py) and raised no finding on it: it answered that seed 0 stays exact and that the positive-seed derivation is a reasonable, documented reading of the contract. Its two findings are in v2r16_isolated.py (stage B) and src/test_e0017_seeds.py. The coordinator checked with git diff that only those two files changed between 64331a4 and f01b036, and with git grep that v2_train.py does not

import v2r16_isolated.py, so the stage-A runs execute only code that RV-0049 saw unchanged and judged sound. run register may label the stage-A attempts as not covered because the RV-0049 round still has a medium finding open; that label stays with this reason. Stage B (C1, C2 and the P continuations) still waits for a passing review of the stage-B fix and of the start-adapter hash pinning.

- D-0021 route:pause->challenged — H-0011 被挑战，证据级别 screening，范围限于从零训练（没有解码器预热）：单 seed 0，E=16 上下文编码写入，V2-03 配置，6000 步。D3 硬读取 15/11/10，+0.4 pp，最后位置目标权重 0.061。NLL 只比 shuffled 低 0.004 nat，是答案先验解。失败方式（RV-0052）：逐槽解码器没有学成事实相关的映射，oracle_value 17/1000，NLL 4.421，和 normal 差不多；所有非零记忆的对照都在 4.42–4.43；边际目标几乎不给寻址信号。最后位置权重接近均匀：n_eff 7.9，argmax 77.5% 落在槽位 0。训练 CE 在 2001–2500 步为 4.53，2501–3000 步为 4.52，结束时 4.41，当时学习率仍高（RV-0053 的更正）。这次没有检验' 有可用解码器时边际似然能否发现寻址'。按 E-0016 的 outcome_actions（D1 达标，D2a、D2b、D3 都未达标）暂停。记忆路线经 H-0012/E-0017 继续。RV-0053 通过。(reopen if: (a) 要在已有可用解码器（如 D1 checkpoint ATT-0154-1）上检验边际似然能否发现寻址；(b) E-0017 没有复现 D5，需要另一条不用标签的路线；(c) 后续阶段在单独编码写入下需要换训练目标。) [conclusion review RV-0052 issues, medium open (same_family_independent_session), RV-0053 pass (same_family_independent_session)]

- D-0022 route:pause->challenged — H-0010 是两段的合取，记为 challenged，证据级别 screening：单 seed 0，E=16 上下文编码写入，V2-03 配置，每段 6000 步。阶段 1 成立：D1 用 teacher-forced 单槽读取，用了目标索引，属额外监督；668/11/13，+65.5 pp，达到 50 pp（EV-0097）。阶段 2 不成立：只用答案 CE 续训后，D2a 只训寻址 50/11/18，+3.2 pp，全位置目标权重 0.069（EV-0098）；D2b 全部训练 48/11/18，+3.0 pp，权重 0.087，oracle_value 从 668 降到 170（EV-0099）。读法（RV-0051）：只能说明干净读取下有解码容量，而这个答案 CE 配方没能发现寻址；不是唯一的因果识别，也不说寻址学不会。按 E-0016 的 outcome_actions 暂停，不在上下文编码写入下用监督寻址开展 V3 机制研究。不用标签的 D5（单独编码写入）已在 seed 0 达到 H3（EV-0095），记忆路线经 H-0012/E-0017 继续。RV-0051 通过，0 个问题。(reopen if: E-0017 没有复现 D5，而 V3 机制研究需要一条声明额外监督的寻址路线；或者要在单独编码写入下检验解码器先行。) [conclusion review RV-0051 pass (same_family_unverified)]

- D-0023 route:continue->supported — H-0012 记为 supported，证据级别 screening，只限 R1 续训这一支（D5 配方）：先按 E-0014 R1 配方在 E=1 上训练 6000 步，再在 E=16 上用单独编码写入续训 6000 步，只用答案 CE，不用地址标签。按 E-0017 的预先规则，P1、P2 都达到 E-0010@2 的 H3（单独编码写入协议，统计量与 E-0010 分开报告），连同 seed 0 共 3/3。三者的差分别是 41.1、47.8、46.2 pp（EV-0095、EV-0102、EV-0103），NLL 低于 zero 与 shuffled，全位置目标权重 0.158–0.162，两个目标位置分层都达标。最后位置没有塌缩到固定槽位：最常见的最后位置槽位占 7.6–8.0%（RV-0055 的更正）。描述性的 C1、C2 也达标（EV-0104、EV-0105）。' 从零训练' 这一支不成立：D4 失败，读取均匀（EV-0096）。同一批 adapter 在上下文编码写入下只有约 5%，第 1 条及以后约 1.4–2.4%（RV-0055 的更正）；所以单独编码是绕开上下文迁移的、须声明的结构化支持输入条件，不是让上下文编码的 value 变得可用。四条结果的 arm 标签不一致只是命名问题（RV-0055 同意），见各证据的 validity 说明。范围：M 线 T1 hop 1，E=16，V2-03 配置，每个 seed 一个 run，不读 confirmation。RV-0055 通过，只有一个 low（描述范围），已在本文更正。动作：按 E-0017/E-0016 的 outcome_actions，用这个配方与协议重开 V3/V4 路线（H-0005）。[conclusion review RV-0055 issues (cross_model_unverified)]

- D-0024 route:reopen — 按 E-0016 与 E-0017 的 outcome_actions 重开 V3/V4 路线（D-0023）：在单独编码写入的 D5 配方下，单跳读取在 3/3 seed 达到 H3，满足 plan §8.4 进入循环的条件，只限这个协议与配方。H-0005 的范围已改为这个协议、配方与 T2 链终点任务（E-0011）。先做 V3 的前置诊断：D5 adapter 在 T2 hop1 上能否迁移，以及不训练时扫描 S_eval 1/2/4/8；然后再定 V3 的训练矩阵。

- D-0025 route:continue — plan2 作为 RB 线加入（用户 2026-11-02）。新增 H-0013..H-0018 与契约 E-0019（RB0）、E-0020（RBc）、E-0021（RB1），由 Opus 5.5 subagent 先在记录副本中建出，协调者审核后在主项目重放（research/notes/plan2-commands.sh，提取见 research/notes/plan2-extraction.md）。协调者对待定问题的决定如下。Q1 RB 线确认集选方案 A2，而不是 subagent 推荐的 A1：从 V1 train 中每任务切出 1,000 条，按 derive_seed(20261101,'residual_la_v21','RB5','confirma 分层抽取，登记 id 清单与 SHA256 后封存；RB 线的 train 为 V1 train 去掉这些行；calibration 与 dev 不变。理由是 A1 的行已被 E-0002 与 E-0013 的 LR 选择用过，A2 的行从未影响任何选择；官方 validation 将由 E-0013 开封，之后不再作确认。这符合' 改方法后的确认用新数据'，没有改变确认集规则。Q2 保留 RB1-09（C_up）。Q3 采用 RB1 Design 中的推进规则 G0/GR/GH/GL/GC 与规则 11：GR 只是释放 RB5 的资源门槛，主线主张仍要求确认集上 +1 pp。Q4 RBc 用三点网格加一次端点扩展。Q5 F 线干预的操作定义按 RB0 Design 第 4 项，由 RB0 的代码审核核对。Q6 T-0013 已合并（f0d9283）。Q7 KernelMatrix 版的 D5 配方与 H-0004 的重开，放到 M 线 V3/V4 设计时决定。Q8 RB0 现在开始；RB5 最晚 11-18 释放。Q9 RB 线的固定字段与确认集安排写入 docs/constraints.md，标为协调者默认。

- D-0026 route:modify — RB0 检查 ATT-0186-1 三项 BF16 比较失败的处理：拓扑改用逐位恒等（零容差）并逐层检查 FFN 输入；cache 保留 BF16 锁定容差并照实记录 chunk1 超出，另加 FP32 决定性检查与状态清空负对照。不放宽任何锁定容差。全文见 rationale-file。

- D-0027 route:modify — D-0026 的处理推广为一条通则（ATT-0187-1：395/397 通过；剩下两项是 cache 重排测试中 order=[1,0,0] 把 batch 从 2 扩到 3，BF16 logits 差 0.857421875 > 0.50002，prefix 与未改动的 reference 在同一位置给出相同值）。通则：BF16 比较若跨越 GEMM 形状的改变（逐 token decode、batch 扩展），且超出量在未改动的 reference 上完全相同，就只作诊断，照实记录，0.50002 不改；决定性门槛是同一比较在 FP32（关闭 TF32）下按 E-0019 的 FP32 门槛（逐张量 $\max|a-r| \le 1e\text{-}5 + 1e\text{-}4 \cdot \max|r|$）。形状不变的比较仍按 BF16 0.50002 判定。对本项：契约的 "batch 重排" 用同 B 的真置换判定（BF16 0.50002，输出须随行移动）；B2→B3 扩展另做 FP32 的扩展与所有权检查（每个扩展行等于其来源行）；BF16 下 prefix 与 reference 的差仍须 ≤ 0.50002。FP32 检查不过 -> 按真实问题修复后重测，不进入训练。

- D-0028 route:stop->challenged — H-0008@2 记为 challenged（confirmatory，按 E-0013@3 运行前封存的规则），停止这条路线。PIQA+BoolQ 官方 validation 只开封一次（ATT-0179-1，EV-0110）：F 线四层 [6,13,20,27] 的 tokenwise C_up（r32，524,288 参数）对全 28 层 all-7 LoRA rank1（630,784 参数），3 个 seed 平均的宏平均差 −0.42 pp，配对按任务分层 bootstrap 95% 区间 [−1.02, +0.20] pp（只反映给定这三个训练 seed 时的数据不确定性）；上界 < +1 pp，按封存规则记 challenges。三个 seed 的差都为负（−0.28、−0.40、−0.58 pp）；每任务 PIQA −0.74、BoolQ −0.09 pp。区间含 0，不宣称等价，也不宣称 LoRA 更好。次要比较 S1（C−F +0.23 pp，Holm p=0.34）与 S2（F−L −0.65 pp，Holm p=0.069）在 Holm 校正后都不拒绝。WikiText ΔNLL：C +0.20、F +0.24、L +0.34。范围：V1 切分、模板与评估，每 run 正好 3 epochs，短校准 LR（三族都锁定为 1e-3），LoRA 结论只针对 rank1，不含侧支历史与多次读取。官方 validation 已用掉，H-0008 不能在该集上再确认，所以 stop。父假设 H-0006 仍为 paused/untested；带因果侧支历史的版本由 RB 线的 H-0013/H-0014 在 D-0025 的 A2 确认集上检验，不复用 V5 的 checkpoint。来源（按 RV-0061 更正）：a09c318

封存了确认锁；03de68f 加入 preflight 结果与 allocation、说明与开封 launcher；RV-0056 覆盖两者。ATT-0179-1/ATT-0194-1 的代码审核标签 stale 来自这些任务分支输入不在 2b5c809 中，以及未使用的切片检查路径，不是执行源码的漂移。[conclusion review RV-0061 issues (cross_model_unverified)]

- D-0029 stage:stop — V5 缩减版（H-0008）阶段结束：无侧支历史的 tokenwise C_up 接口在官方 validation 上没有超过参数相近的 LoRA28 r1，按封存规则记 challenged 并停止（路线决定 D-0028，结论审核 RV-0061）。官方 validation 已用掉；带因果侧支历史的实用价值问题转由 RB 线（H-0013/H-0014，A2 确认集）检验。[conclusion review RV-0061 issues (cross_model_unverified)]

6 Conclusion

在锁定的配置与预算内，没有侧支历史的 tokenwise gate/up 接口在 V1 与 V5 中都没有显示出优于加法写回或参数相近的 LoRA；V5 的确认比较按封存规则记为 challenged，这条路线停止。只保留低维记忆的单跳读取在 V2 配方下失败；改成每条事实单独编码写入、并先在 E=1 上训练后，三个 seed 都达到门槛，但这依赖声明的结构化支持输入，学到的寻址也没有迁移到 T2 句式。接下来两条线并行：V3 在 T2 上训练共同起点后，按运行前写定的规则比较 S=1/2/4 的循环读取（E-0022）；RB 线修好 RB0 审核的四条 low 发现后，做学习率校准（E-0020）与首轮筛选（E-0021），在 A2 集上检验带侧支历史的残差旁路是否有实用价值。截至 2026-11-02，项目共用约 12.6 node·h（上限 200）。

A Records

- records at data_version 2808; plan DOC-0001@1 (docs/plan.md); constraints docs/constraints.md
- experiments: E-0001@1 (DOC-0003@1), E-0002@1 (DOC-0019@1), E-0003@2 (DOC-0040@2), E-0004@1 (DOC-0047@1), E-0005@1 (DOC-0048@1), E-0006@1 (DOC-0053@1), E-0007@2 (DOC-0063@2), E-0008@1 (DOC-0086@1), E-0009@1 (DOC-0087@1), E-0010@2 (DOC-0083@2), E-0011@2 (DOC-0097@2), E-0012@1 (DOC-0117@1), E-0013@3 (DOC-0128@3), E-0014@1 (DOC-0135@1), E-0015@4 (DOC-0157@4), E-0016@1 (DOC-0160@1), E-0017@1 (DOC-0180@1), E-0018@1 (DOC-0202@1), E-0019@1 (DOC-0203@1), E-0020@0 (draft), E-0021@0 (draft), E-0022@2 (DOC-0211@2)
- attempts: 212 (8 ended/cancelled, 13 ended/failed, 190 ended/success, 1 queued/unknown)
- reviews and advice: AD-0001 answered, AD-0002 answered, RV-0001 answered issues, RV-0002 answered pass, RV-0003 answered issues, RV-0004 answered issues, RV-0005 answered pass, RV-0006 answered pass, RV-0007 answered issues, RV-0008 answered pass, RV-0009 answered pass, RV-0010 answered pass, RV-0011 answered issues, RV-0012 answered pass, RV-0013 answered pass, RV-0014 answered pass, RV-0015 answered pass, RV-0016 answered issues, RV-0017 answered issues, RV-0018 answered pass, RV-0019 answered pass, RV-0020 answered issues, RV-0021 answered pass, RV-0022 answered pass, RV-0023 answered pass, RV-0024 answered pass, RV-0025 answered issues, RV-0026 answered pass, RV-0027 answered pass, RV-0028 answered issues, RV-0029 answered issues, RV-0030 answered issues, RV-0031 answered issues, RV-0032 answered issues, RV-0033 answered pass, RV-0034 answered issues, RV-0035 answered pass, RV-0036 answered issues, RV-0037 answered issues, RV-0038 answered pass, RV-0039 answered pass, RV-0040 answered blocking, RV-0041 answered pass,

RV-0042 answered blocking, RV-0043 answered pass, RV-0044 answered issues, RV-0045 answered pass, RV-0046 answered issues, RV-0047 answered pass, RV-0048 answered pass, RV-0049 answered issues, RV-0050 answered issues, RV-0051 answered pass, RV-0052 answered issues, RV-0053 answered pass, RV-0054 answered pass, RV-0055 answered issues, RV-0056 answered pass, RV-0057 answered pass, RV-0058 answered issues, RV-0059 answered pass, RV-0060 answered pass, RV-0061 answered issues

- resource use: 0.00583333 mig_instance_hours compute on GH200-MIG-2g.24gb (measured); 5.81063 node_h compute on GH200 MIG 2g.24gb (measured); 0.000138889 node_h compute on GH200 MIG 2g.24gb (guard verified) (measured); 6.94444e-05 node_h compute on GH200 MIG allocation; profile not measured (artifact tabulation only) (measured); 0.00263889 node_h compute on GH200 MIG allocation;profile metadata only (measured); 0.000694444 node_h compute on GH200 MIG2g.24gb (measured); 0.119236 node_h compute on GH200 MIG2g.24gb verified timing guard (measured); 6.94444e-05 node_h compute on Miyabi-G GH200 MIG allocation;artifact-only CPU tabulation;profile not inspected (measured); 26.5781 mig_instance_hours gpu_mig_instance on GH200 MIG 2g.24gb (measured); 1.2221e+06 input_tokens training on - (measured)